Lesson **40 minutes · concept**

ICT Level 3 - Complete Lesson Plans

Chapter 1 - Advanced Programming Concepts

`(..) => ..`

Key Vocabulary

Function: Reusable block of code that performs one specific task perfectly each time
Definition: Writing function code once so it can be called later anywhere
Call: Telling program to run the function by writing its name with brackets
DRY: Do Not Repeat Yourself principle encouraging reuse of code efficiently

Engage (5 min)

Teacher Students, imagine you are given a task to draw a house 20 times on the board. Would you draw every single line from scratch each time, or would you create a template and reuse it? Today we learn functions, which are like reusable templates in programming. Think of a function like a recipe card. You write the recipe once, and whenever you want to make that dish, you just follow the card. You do not have to remember every step each time.

Students A recipe card? So instead of writing the same code again and again, I write it once and call it whenever I need? That would save so much time! Is it like a shortcut?

Explore (10 min)

Teacher Let us try something together. I will write a small program without functions first. Look at this code that prints the school morning prayer three times. Now I will show you how to write the same thing using a function. You can see that the code inside the function is written only once but executed three times. Try to guess what will happen when I call the function name. Discuss with your partner and write down your prediction before I run it.

Students Oh I see! The print statements are inside the function but the function name is called three times. So the computer goes to the function, runs the code, comes back, and does it again. My prediction is it will print the prayer three times just like before.

Explain (10 min)

Teacher A function in Python is a named block of code that performs a specific task. You define a function using the keyword def followed by the function name and parentheses. The code inside the function is indented. To execute the function you call it by writing its name followed by parentheses. The function definition tells the computer what to do and the function call tells the computer when to do it. Functions follow the DRY principle which stands for Do Not Repeat yourself. In KREIS we use functions to organize our programs just like how our school day is organized into periods each with a specific purpose.

Students So def creates the function and then we use the name with brackets to run it. The indentation shows which lines belong to the function. This is much cleaner than writing the same code everywhere. I can see how this helps when programs get longer.

Elaborate (10 min)

Teacher Now let us apply this to something from our daily life at KREIS. Write a function called morning_assembly that prints three steps: stand in line, sing prayer, listen to announcements. Call this function three times to represent Monday, Tuesday, and Wednesday assemblies. Then challenge yourself: write a second function called lunch_routine that prints the steps of our lunch time. Call both functions and observe the output. Notice how each function has its own task and they work independently. Modify: change one parameter/block and predict outcome before running.

Students I wrote the morning_assembly function and called it three times. Then I created lunch_routine with wash hands, collect food, eat quietly, return plate. When I call both functions the output shows assembly steps then lunch steps. The functions are like different routines in our school day.

Evaluate (5 min)

Teacher I will give you a short program that has repeated code. Your task is to identify the repeated part and convert it into a function. Then call the function the correct number of times. I will check if your function definition is correct, if the indentation is proper, and if your function calls produce the expected output. You have ten minutes to complete this. Who can explain why using a function here is better than the original code? Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students I converted the repeated print statements into a function called greet_student. I called it four times for four students. Using a function is better because if I need to change the greeting I only change it in one place instead of four places.

Cross-Curricular Connections

Mathematics: Mathematics: Functions in math take an input and produce an output just like f of x equals x plus 2. Python functions work the same way conceptually.
Science: Science: A scientific experiment follows a procedure that can be repeated. A function is like a repeatable procedure in code.
Language: Language Arts: A function is like a paragraph that can be referenced by a topic sentence. You write the paragraph once and refer to it whenever needed.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Python IDLE
- Textbook Chapter 1 Page 1
- KREIS programming worksheet handout

Differentiation

Approaching: Class 8 Foundation: Focus on understanding what a function is and writing functions with no parameters. Use step-by-step worksheets with fill-in-the-blank code. Teacher guides through each line of the function definition.
On Level: Class 9 Application: Write functions with simple print statements and call them multiple times. Students work in pairs to create functions for school routines. Minimal teacher intervention expected.
Advanced: Class 10 Mastery: Write functions and explain the DRY principle. Compare function usage with and

without functions. Begin thinking about parameter passing which will be covered in the next lesson.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Observation and worksheet check during class activity
CT Skill Assessed: Decomposition: Breaking a repeated task into a reusable function

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a function called study_time that prints three activities: read textbook, solve problems, review notes. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Parameter: Variable listed in function definition receiving value when called correctly for flexibility
Argument: Actual value you pass to function when you call it for execution
Input: Data given to function so it works with different values
Signature: First line showing function name and its expected parameters clearly together

Engage (5 min)

Teacher Yesterday we learned how to write functions. But notice that our functions always did exactly the same thing. Today we learn parameters which let us customize what a function does each time we call it. Think of a water tap. The tap is the function. The handle is the parameter. When you turn the handle to different positions you get different amounts of water. The tap stays the same but the result changes based on what you give it.

Students So parameters are like inputs to the function? If I write a function to greet a student I can pass different names as parameters and it will greet each student differently? That makes the function much more useful!

Explore (10 min)

Teacher Look at this function I wrote yesterday: def say_hello which just prints Hello Student. Now I will add a parameter: def say_hello name and inside it prints Hello plus the name. When I call say_hello with Arjun it prints Hello Arjun. When I call it with Priya it prints Hello Priya. Try changing the name inside the parentheses and predict what will print. Write three different calls with three different names from our class.

Students I tried say_hello with Ravi and it printed Hello Ravi. Then I tried with KREIS and it printed Hello KREIS. The function works with any text I put in the parentheses. It is like filling in a blank in a sentence.

Explain (10 min)

Teacher A parameter is a variable listed in the function definition. It is a placeholder that holds the value you pass to the function. An argument is the actual value you send when you call the function. So name is the parameter and Arjun is the argument. You can have multiple parameters separated by commas. For example def add a b takes two parameters and you call it with add 5 3. The function receives 5 in a and 3 in b. Parameters make functions flexible because the same function can work with different values. This is like how our school timetable has the same periods but different subjects each day.

Students So parameter is the name in the definition and argument is the value we actually pass. Multiple parameters let the function handle more complex tasks. The function definition is the blueprint and the function call with arguments is the actual building.

Elaborate (10 min)

Teacher Create a function called calculate_area that takes length and width as parameters and prints the area. Test it with the dimensions of your classroom, your textbook, and your slate. Then write a function called travel_time that takes distance and speed as parameters and calculates the time taken. Use it to find how long it takes to walk from the dormitory to the school building. Discuss with your partner: what happens if you call a function with fewer arguments than parameters?

Students My calculate_area function works with any two numbers. I tested it with classroom dimensions 8 meters by 6 meters and got 48 square meters. My textbook is 25 centimeters by 20 centimeters and the area is 500 square centimeters. When I tried calling the function with only one argument Python gave an error saying one positional argument is missing.

Evaluate (5 min)

Teacher I will give you a function definition with two parameters. You must write the correct function call with appropriate arguments. Then I will give you a function call and you must write the function definition that would make that call work. Pay attention to the order of arguments because they must match the order of parameters. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students I wrote the function definition correctly. I noticed that the order matters because def greet name age is different from def greet age name. The first argument goes to the first parameter and so on.

Cross-Curricular Connections

Mathematics: Mathematics: Parameters are like variables in algebra. f of x equals 2x plus 3 has x as a parameter. Python functions work identically with different input values.
Science: Science: In a science experiment the independent variable is like a parameter you control. Different values of the parameter produce different results just like different arguments produce different function outputs.
Language: Language Arts: Parameters are like blanks in a mad-libs story. The story template stays the same but different words produce different funny stories each time.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:

- Python IDLE
- Textbook Chapter 1 Page 3
- KREIS parameter practice worksheet

Differentiation

Approaching: Class 8 Foundation: Work with single parameter functions only. Use predefined function definitions and practice writing the correct calls. Provide a matching activity where students match function definitions to correct calls.
On Level: Class 9 Application: Write functions with two parameters and test with different values. Predict output before running code. Work in small groups to create parameterized functions for real KREIS scenarios.
Advanced: Class 10 Mastery: Write functions with multiple parameters and explore default parameter values. Investigate what happens with wrong number of arguments and different data types. Begin connecting parameters to the next lesson on return values.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Quiz with function definition and call matching questions
CT Skill Assessed: Abstraction: Using parameters to generalize a specific task into a reusable pattern that works with different inputs

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a function called bmi_calculator that takes weight in kilograms and height in meters as parameters and prints the BMI. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Return: Sending result back from function to the place that called it
Value: Result produced by function after doing its calculation correctly for use
Caller: Code that calls function and waits to receive returned result back
None: Special value meaning function returned nothing useful for caller to use

Engage (5 min)

Teacher Students imagine you ask your friend to calculate the total cost of items in the tuck shop. There are two ways your friend can respond. First way: your friend shouts out the total to everyone. Second way: your friend quietly tells you the total and you decide what to do with it. The first way is like printing inside a function. The second way is like returning a value. Today we learn the return statement which lets a function send a result back to whoever called it.

Students So return is like getting an answer back from the function instead of just displaying it? That means I can use the answer in other calculations. Like if the function gives me the total cost I can then check if I have enough pocket money to pay for it.

Explore (10 min)

Teacher Let me show you two versions of a function. First version: def add a b which prints a plus b. Second version: def add a b which returns a plus b. When I call the first version it shows the result but I cannot save it. When I call the second version with result equals add 5 3 the result variable now holds the value 8. I can use this 8 in another calculation like result times 2. Try both versions and write down the differences you observe.

Students The print version shows 8 on screen but result is empty. The return version does not show anything but result holds 8. I can then print result and get 8 or do result plus 10 to get 18. Return is much more useful because I can reuse the value.

Explain (10 min)

Teacher The return statement ends the function execution and sends a value back to the caller. The returned value can be stored in a variable for later use. A function can return numbers strings booleans or even lists. If a function has no return statement it automatically returns None. You should use return when the function is calculating something that other parts of your program need. You should use print when you just want to display information to the user. Most useful functions return values rather than printing them directly.

Students So return gives the value back to the code that called the function. Print just shows it on screen. I should use return when I need the result later and print when I just want to display something. Functions that return values are like calculators and functions that print are like announcements.

Elaborate (10 min)

Teacher Write a function called convert_temperature that takes a temperature in Celsius and returns the value in Fahrenheit using the formula F equals C times 9 divided by 5 plus 32. Store the result and print it with a message. Then write a function called calculate_grade that takes a percentage and returns the grade A B C or D based on KREIS grading rules. Test both functions with at least three different values each. Challenge: can you write a function that returns two values at once? Modify: change one parameter/block and predict outcome before running.

Students My convert_temperature function takes 100 Celsius and returns 212 Fahrenheit. My calculate_grade function takes 85 and returns A, takes 65 and returns B, and takes 45 and returns C. For the challenge I learned that I can return two values by writing return min_value max_value and Python puts them in a tuple automatically.

Evaluate (5 min)

Teacher I will give you a problem description. You must write a function that takes the right parameters and returns the correct result. Then I will give you some test values and you must tell me what the function will return without running the code. This tests whether you understand how return values work. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students I can predict the return value by following the formula step by step. For example if the function returns a times 2 plus 1 and I call it with 5 the answer is 11. Tracing through the function line by line helps me get the right answer.

Cross-Curricular Connections

Mathematics: Mathematics: Return values are exactly like mathematical functions. f of x equals 3x plus 7 returns a value for any input x. Computing functions work the same way.
Science: Science: A thermometer returns a temperature reading. It does not just display it. Similarly a computing function returns a value that can be used in further analysis.
Language: Language Arts: A return value is like the moral of a story. The story processes events and delivers a conclusion. The conclusion is the return value of the narrative.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Python IDLE
- Textbook Chapter 1 Page 5
- KREIS return value practice sheet

Differentiation

Approaching: Class 8 Foundation: Focus on functions that return a single simple value. Use tracing worksheets where students follow the function step by step to predict the return value. Teacher provides the function and students trace the output.
On Level: Class 9 Application: Write functions that return values and use those values in calculations. Compare print versus return approaches. Students work independently on a set of practice problems with increasing complexity.
Advanced: Class 10 Mastery: Write functions returning multiple values using tuples. Explore how return values enable modular program design. Begin connecting return values to data structures which will be covered in Chapter 2.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Trace table exercises and prediction tasks without running code
CT Skill Assessed: Abstraction: Returning computed results enables modular design where functions can be combined to build complex programs from simple parts

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a function called calculate_average that takes three test scores as parameters and returns the average. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Array: Ordered list storing many values of same type under one name
Index: Number showing position of element starting from zero in array quickly
Element: Single value stored at one position inside array for access later
Size: Total number of elements currently stored inside the array altogether

Engage (5 min)

Teacher Students imagine you have ten pencils in your pencil box. You can hold all ten in one box instead of carrying ten separate boxes. Today we learn about lists in Python which work like that pencil box. A list lets you store many values under one name. Instead of creating ten separate variables for ten students marks you put all ten marks in one list. This makes your program much easier to manage.

Students So a list is like a container that holds many items? I can store all my subject marks in one list instead of creating separate variables for each subject. That saves a lot of time and makes the code shorter.

Explore (10 min)

Teacher Let me create a list of five KREIS houses: houses equals opening square bracket Arya Bharati Chandrika Drona Ekalavya closing square bracket. Now I will print the first house using houses square bracket zero. Notice we start counting from zero not one. Try to predict what houses square bracket 2 will give you. Then try to access houses square bracket 5 and see what happens. Discuss with your partner why Python starts counting from zero.

Students houses square bracket 2 gives Chandrika because we count zero one two. houses square bracket 5 gives an error because we only have five houses numbered zero to four. Starting from zero must be because computers count from zero. It is like floors in some buildings where the ground floor is zero.

Explain (10 min)

Teacher A list in Python is created using square brackets with values separated by commas. Each element has an index starting from zero. The first element is at index zero the second at index one and so on. You access an element using the list name followed by the index in square brackets. You can change an element by assigning a new value to its index position. Lists can contain different data types including numbers strings and even other lists. The length of a list tells you how many elements it contains. Indexing from zero is universal in programming and you must always remember this.

Students So square brackets create the list and square brackets with an index access an element. Zero based indexing means the first element is always at position zero. I can also change any element by assigning a new value to its position. Lists are very flexible containers for storing groups of related data.

Elaborate (10 min)

Teacher Create a list called attendance that holds the attendance status present or absent for all students in your group for today. Then create a list called subjects that holds the names of all your subjects today. Access and print the third subject. Change the second attendance record from absent to present. Now create a list called marks that holds five test scores and find which score is the highest by comparing elements.

Students My attendance list has present for six students and absent for one. My subjects list has Mathematics Science English Kannada and Social Studies. The third subject at index two is English. I changed Ravi attendance from absent to present. For finding the highest mark I compared each element and found that 92 at index three is the highest.

Evaluate (5 min)

Teacher I will give you a list and ask you questions about specific elements using index numbers. You must also write code to create a list from a description and access certain elements. I will check if you use correct zero-based indexing and if you can modify elements correctly. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students I used correct indexing. I noticed that practice is needed to remember that the third element is at index two not index three. Writing the index number above each element helps me keep track.

Cross-Curricular Connections

Mathematics: Mathematics: Lists are like sets in mathematics but ordered and changeable. Indexing is like array notation in mathematics where a sub 1 a sub 2 represents elements.
Science: Science: Data collected in experiments is often stored in tables which are like lists of lists. Each row in a data table is a list of measurements.
Language: Language Arts: A list is like a shopping list where each item has a position. The order matters and you can add remove or change items as needed.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Python IDLE
- Textbook Chapter 1 Page 7
- KREIS list practice worksheet

Differentiation

Approaching: Class 8 Foundation: Focus on creating simple lists and accessing elements by index. Use visual diagrams showing indices above list elements. Practice with single line list creation and access tasks with step-by-step guidance.
On Level: Class 9 Application: Create lists and perform basic operations including access modification and length checking. Work on real KREIS data scenarios like attendance records and test scores. Predict output before running code.
Advanced: Class 10 Mastery: Create nested lists and explore advanced indexing including negative indices. Begin connecting list concepts to arrays in mathematics. Prepare for list methods in the next lesson.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Practical coding exercise creating and accessing list elements
CT Skill Assessed: Pattern Recognition: Understanding how indexed access provides systematic way to retrieve specific data from a collection

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a list called weekly_menu that holds the lunch menu for Monday through Friday. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Traverse: Visiting each element of array one by one in order sequentially
Insert: Adding new element at specific position inside existing array carefully
Delete: Removing element from array and shifting others to fill gap
Search: Looking through array to find element matching desired value quickly

Engage (5 min)

Teacher Yesterday we created lists and accessed elements. But what happens when you need to add a new student to your attendance list or remove a student who has left? Today we learn list operations which let us add and remove elements. Think of your list as a train with compartments. You can add new compartments at the end or insert one in the middle or remove one from anywhere.

Students So lists are not fixed. I can change their size by adding or removing elements. That makes them much more useful than regular variables. I can build a list step by step as I collect data.

Explore (10 min)

Teacher I have a list called fruits with banana orange and apple. Let me add mango at the end using fruits dot append mango. Now the list has four items. Let me insert grape at position one using fruits dot insert 1 grape. Now grape is between banana and orange. Let me remove orange using fruits dot remove orange. Try each operation and print the list after each change to see what happens.

Students After append the mango appears at the end. After insert the grape pushes the other items to make room. After remove the orange is gone and the remaining items shift to fill the gap. The list automatically adjusts its size.

Explain (10 min)

Teacher The append method adds an element to the end of the list. The insert method adds an element at a specific index position. The remove method deletes the first occurrence of a specified value. The pop method removes and returns the element at a given index or the last element if no index is given. The len function returns the number of elements in the list. The in operator checks if a value exists in the list and returns True or False. These operations make lists dynamic meaning they can grow and shrink as needed during program execution.

Students So append adds to the end insert adds at a position remove deletes by value and pop deletes by position. Len tells me how many items are in the list and in checks if something is there. These operations let me manage the list like a living collection that changes as needed.

Elaborate (10 min)

Teacher Create a program that manages a KREIS class roster. Start with an empty list. Add five student names using append. Insert a new student at position two. Remove a student who transferred to another school. Print the length of the list after each operation. Check if a specific student is in the list using the in operator. Finally print the complete updated roster. Challenge: use pop to remove the last student and store their name in a variable.

Students My roster started empty. I added Arjun Priya Ravi Kavya and Meena. Then I inserted Sanjay at position two. After that I removed Kavya because she transferred. The list length changed from 5 to 6 to 5. Sanjay is in the list returns True. I popped Meena and stored her name. The final roster has five students.

Evaluate (5 min)

Teacher I will give you a sequence of operations to perform on a list. You must write the code for each operation and predict the state of the list after each step. Then I will give you a final list state and ask what operations could have produced it. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students I can track the list state after each operation by writing it down. The key insight is that insert shifts elements to the right and remove shifts elements to the left to fill gaps.

Cross-Curricular Connections

Mathematics: Mathematics: List operations are like set operations in mathematics. Adding elements is like union and removing is like difference. The len function gives the cardinality.
Science: Science: In laboratory work you add and remove samples from a collection. List operations model this process of managing physical collections programmatically.
Language: Language Arts: Managing a list is like editing a paragraph. You insert new sentences remove unnecessary ones and rearrange the order to improve clarity.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Python IDLE
- Textbook Chapter 1 Page 9
- KREIS list operations cheat sheet

Differentiation

Approaching: Class 8 Foundation: Practice append and remove operations with teacher guided step-by-step instructions. Use physical cards or tokens to model list operations before coding. Focus on understanding what each operation does visually.
On Level: Class 9 Application: Perform all basic list operations independently. Create a student management program that uses multiple operations. Predict list state after a sequence of operations before running code.
Advanced: Class 10 Mastery: Combine multiple list operations to solve complex problems. Explore edge cases like removing from an empty list or inserting at invalid positions. Begin thinking about efficiency of operations.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Step-by-step list state tracking exercise
CT Skill Assessed: Algorithmic Thinking: Planning a sequence of operations to achieve a desired final state of the list

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a program that creates a to-do list for your evening study. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

String: Sequence of characters like name or sentence inside quotes together
Concatenation: Joining two strings together end to end to make one longer string
Substring: Part of a string taken from middle like first three letters
Length: Number of characters including spaces inside a string counted together

Engage (5 min)

Teacher Students you use strings every day when you write your name read signs and send messages. Today we learn how to manipulate strings in Python. Imagine you have a rubber band. You can stretch it join two rubber bands together or cut a piece from it. Strings work similarly in programming. You can join two strings stretch a string by repeating it or cut out a piece using slicing.

Students So strings are not just text. They are like objects I can work with. I can combine them change their case and even cut pieces out. That means I can process text data in my programs just like I process numbers.

Explore (10 min)

Teacher Let me join two strings: first_name equals KREIS and last_name equals Academy. Full_name equals first_name plus space plus last_name. Now I have KREIS Academy. Let me repeat a string: dash equals star times 20 gives me twenty stars. Let me change case: hello dot upper gives HELLO and HELLO dot lower gives hello. Try these operations and create your own joined and repeated strings.

Students I joined my first name and last name with a space between them. I repeated the character equals sign twenty times to make a line. upper changed everything to capital letters and lower changed everything to small letters. These methods are very useful for formatting text.

Explain (10 min)

Teacher String concatenation uses the plus operator to join two strings. String repetition uses the multiply operator to repeat a string multiple times. The upper method converts all characters to uppercase. The lower method converts to lowercase. The strip method removes leading and trailing spaces. The replace method substitutes one substring with another. String slicing uses the syntax string start colon end to extract a portion. The start index is included but the end index is excluded. Slicing does not modify the original string but returns a new one.

Students Plus joins strings multiply repeats them and dot methods transform them. upper and lower change case strip removes spaces and replace swaps text. Slicing with colon extracts a piece. Strings are immutable which means the original string never changes. Every operation creates a new string.

Elaborate (10 min)

Teacher Create a program that formats a KREIS student ID card. Use string concatenation to combine the student name school name and class into a formatted string. Use upper to make the school name stand out. Use strip to clean up any extra spaces in student names. Use replace to change a placeholder name to the actual student name. Use slicing to extract the first three letters of the student name as initials. Print the complete formatted ID card.

Students My ID card shows KREIS ACADEMY in uppercase at the top. The student name is Priya Sharma and her class is Class 9 Section A. I used strip to remove extra spaces from the name input. I used replace to swap placeholder with Priya Sharma. The initials are the first three letters P from Priya. The formatted output looks professional and clean.

Evaluate (5 min)

Teacher I will give you a string and ask you to perform specific operations on it. You must write the correct method calls and predict the result. I will also give you a desired output and you must figure out what string operations produced it. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students I can trace through string operations step by step. For example if the string is hello world then strip removes spaces dot upper makes it HELLO WORLD and replace changes O to zero giving HELLO W0RLD. Practicing these operations builds confidence.

Cross-Curricular Connections

Mathematics: Mathematics: String indexing uses the same zero-based system as list indexing. Slicing is like selecting a range of numbers in mathematics.

Science: Science: String operations are used to process experimental data labels and unit conversions. Cleaning data often requires stripping whitespace and standardizing case.
Language: Language Arts: String operations are like editing text. Concatenation is joining sentences slicing is extracting quotes and replace is find-and-replace in word processing.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Python IDLE
- Textbook Chapter 1 Page 11
- KREIS string methods reference card

Differentiation

Approaching: Class 8 Foundation: Focus on concatenation upper lower and strip methods. Use predefined strings and practice one method at a time. Provide worksheets where students match method calls to their outputs.
On Level: Class 9 Application: Combine multiple string operations to format text output. Work on formatting KREIS documents like certificates and ID cards. Practice slicing to extract substrings independently.
Advanced: Class 10 Mastery: Use string methods in combination with list operations for complex text processing. Explore edge cases in slicing and method chaining. Prepare for file I/O where string operations are essential.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: String manipulation challenge with prediction and coding tasks
CT Skill Assessed: Pattern Recognition: Identifying which string operations produce desired text transformations and combining them effectively

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a program that takes your full name and formats it in three ways: FULL NAME in all capitals, full name in all lowercase, and First M. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Nested Loop: Loop placed inside another loop running inner many times per outer
Outer Loop: First loop controlling how many times inner repeats completely each cycle

Inner Loop: Second loop running fully each time outer takes one step forward
Iteration: Single execution of loop body before checking condition again for next

Engage (5 min)

Teacher Students in our multiplication table we have rows and columns. Each row is a multiple and each column is a multiplier. To fill the entire table you need a loop inside a loop. Today we learn nested loops which are loops within loops. Think of it like this: the outer loop picks which row you are on and the inner loop goes through all the columns in that row before moving to the next row.

Students So the outer loop runs once for each row and the inner loop runs completely for each iteration of the outer loop. That means if the outer loop runs 5 times and the inner loop runs 3 times the total iterations are 15. The inner loop finishes everything before the outer loop moves forward.

Explore (10 min)

Teacher Let me write a nested loop. The outer loop uses i from 1 to 3 and the inner loop uses j from 1 to 3. Inside the inner loop I print i times j. Watch what happens: when i equals 1 the inner loop runs three times printing 1 times 1 equals 1 then 1 times 2 equals 2 then 1 times 3 equals 3. Then i becomes 2 and the inner loop runs again. Draw a table on paper showing which values of i and j produce which outputs.

Students I drew the table. When i is 1 the outputs are 1 2 3. When i is 2 the outputs are 2 4 6. When i is 3 the outputs are 3 6 9. The inner loop runs three times for each value of the outer loop. Total print statements are 9 which is 3 times 3.

Explain (10 min)

Teacher A nested loop is a loop inside another loop. The outer loop controls the number of times the inner loop repeats entirely. For each single iteration of the outer loop the inner loop runs through all its iterations. The total number of iterations is the outer count multiplied by the inner count. Nested loops are essential for processing 2D data like grids tables and matrices. They are also used for pattern printing and coordinate systems. Be careful with nested loops because they can become slow if both loops have large ranges.

Students So the outer loop is like the teacher assigning rows and the inner loop is like students filling each row. The inner loop must finish completely before the outer loop moves on. The total work is outer count times inner count. I need to be careful with large ranges because nested loops do a lot of work.

Elaborate (10 min)

Teacher Create a program that prints a multiplication table from 1 times 1 to 5 times 5 using nested loops. Format the output so numbers align in columns. Then write a program that prints a right triangle of stars where row 1 has 1 star row 2 has 2 stars and so on up to 5 rows. Challenge: print an inverted triangle where row 1 has 5 stars and row 5 has 1 star.

Students My multiplication table prints neatly in columns. The triangle program uses the outer loop for rows and the inner loop prints j stars for row j. For the inverted triangle I print 6 minus j stars for each row. The nested loop pattern is the same but the calculation inside changes the shape.

Evaluate (5 min)

Teacher I will give you a nested loop program and ask you to trace through it completely. You must fill in a table showing the values of i and j at each step and the output produced. Then I will give you a pattern and you must write the nested loop code that produces it. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Tracing nested loops requires patience. I write down each value of i then write all values of j for that i before moving to the next i. This systematic approach helps me avoid mistakes.

Cross-Curricular Connections

Mathematics: Mathematics: Nested loops directly relate to multiplication tables and matrix operations. The concept of rows and columns is fundamental in mathematics.
Science: Science: Data tables in science have rows for experiments and columns for measurements. Nested loops process these two-dimensional data structures.
Language: Language Arts: A nested structure is like a story within a story. The outer narrative contains chapters and each chapter contains paragraphs. Both follow a hierarchical pattern.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Python IDLE
- Textbook Chapter 1 Page 13
- KREIS pattern printing worksheets

Differentiation

Approaching: Class 8 Foundation: Trace simple nested loops with small ranges like 1 to 3. Use provided templates where students fill in the loop bodies. Focus on understanding how the inner loop completes before the outer loop advances.
On Level: Class 9 Application: Write nested loops independently to create multiplication tables and star patterns. Predict output before running code. Experiment with changing loop ranges and observe how patterns change.
Advanced: Class 10 Mastery: Create complex grid patterns and multi-dimensional data processing programs. Analyze the time complexity of nested loops. Connect to real-world applications in data processing.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Trace table completion and pattern creation coding task
CT Skill Assessed: Algorithmic Thinking: Breaking down a grid pattern into nested sequential steps that can be executed by the computer

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a nested loop program that prints the table of 7 from 7 times 1 equals 7 to 7 times 10 equals 70. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Exception: Error happening during program run that would stop execution if unhandled
Try: Block where you attempt code that might cause an error safely
Except: Block running alternative code when try block fails with error gracefully
Finally: Block always running whether error happened or not for cleanup

Engage (5 min)

Teacher Students what happens when you try to divide a number by zero in mathematics. The teacher says it is undefined. In Python the program crashes with an error. But crashing is not good. Today we learn exception handling which lets us catch errors and handle them gracefully. Think of it like this: if you trip while walking you can either fall down or catch yourself on a railing. Exception handling is the railing that catches your program before it crashes.

Students So instead of the program stopping with an ugly error message I can catch the error and show a friendly message or try a different approach. That makes my programs more robust and user-friendly.

Explore (10 min)

Teacher Let me try dividing 10 by 0 in Python. Watch what happens. The program gives a ZeroDivisionError. Now let me wrap it in a try block: try result equals 10 divided by 0 except ZeroDivisionError print cannot divide by zero. Now the program does not crash. It prints a friendly message instead. Try other errors like converting hello to a number using int and see what exception occurs.

Students The ZeroDivisionError was caught and my message printed. When I tried int of hello I got a ValueError. I can catch that too with except ValueError. The try block is like a safety net that catches problems and lets the program continue.

Explain (10 min)

Teacher An exception is an error that occurs during program execution. Common exceptions include ZeroDivisionError for dividing by zero TypeError for wrong data types and ValueError for invalid values. The try except block lets you attempt risky code in the try section and handle errors in the except section. If no error occurs the except section is skipped. You can have multiple except blocks for different error types. The else block runs only if no exception occurred. The finally block always runs whether an exception occurred or not. Exception handling makes programs robust and user-friendly.

Students Try contains code that might fail. Except catches specific errors. Else runs when there is no error. Finally runs always. Multiple except blocks handle different error types. This structure lets me write programs that do not crash unexpectedly and provide helpful feedback to users.

Elaborate (10 min)

Teacher Write a calculator program that takes two numbers from the user and performs division. Handle ZeroDivisionError when the second number is zero. Handle ValueError when the user enters text instead of numbers. Provide clear messages for each type of error. Add an else block that prints the result only when the division succeeds. Add a finally block that prints Thank you for using the calculator. Modify: change one parameter/block and predict outcome before running.

Students My calculator catches division by zero and shows Please enter a non-zero divisor. It catches invalid input and shows Please enter valid numbers. When the calculation works the else block prints the result. The finally block always prints the thank you message. The program never crashes no matter what the user enters.

Evaluate (5 min)

Teacher I will give you programs with potential errors. You must identify what type of exception might occur and write the correct try except block to handle it. You must also write meaningful error messages that help the user understand what went wrong and how to fix it. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students I can identify the error type by thinking about what could go wrong. Division by zero is ZeroDivisionError. Converting text to number is ValueError. Adding a string and number is TypeError. Each requires a specific except block.

Cross-Curricular Connections

Mathematics: Mathematics: Division by zero is undefined in mathematics. Exception handling in computing formally addresses this mathematical concept with program logic.
Science: Science: Experiments can fail due to equipment malfunction or wrong measurements. Exception handling in programs models how scientists handle unexpected results by having backup plans.
Language: Language Arts: Exception handling is like using safe language when delivering bad news. Instead of crashing you provide a thoughtful response that guides the listener to a solution.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Python IDLE
- Textbook Chapter 1 Page 15
- KREIS exception handling reference

Differentiation

Approaching: Class 8 Foundation: Focus on try except blocks for ZeroDivisionError and ValueError only. Use templates where students fill in the except blocks. Practice with simple scenarios like division and number conversion.
On Level: Class 9 Application: Handle multiple exception types in one program. Write meaningful error messages and use else and finally blocks. Create a robust input validation program independently.
Advanced: Class 10 Mastery: Implement comprehensive exception handling in complex programs. Explore custom error messages and nested try except blocks. Connect exception handling to file I/O and database operations in later chapters.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Error identification and handling code writing exercise
CT Skill Assessed: Evaluation: Analyzing potential failure points and implementing defensive strategies to handle them gracefully

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a program that asks for the student age and converts it to an integer. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

File: Collection of data stored on disk with a name like notes.txt
Read: Getting data from file into program for processing or display on screen
Write: Saving data from program into file for future use permanently
Path: Address showing where file is stored like folder inside folder

Engage (5 min)

Teacher Students when you write a program and run it the variables exist only while the program runs. When you close the program everything is lost. But what if you want to save your data like marks attendance or notes? Today we learn file handling which lets your programs save data to files and read data from files. Think of it like writing in a notebook. Your thoughts stay in the notebook even after you stop writing.

Students So file handling lets my program save information permanently. I can write student marks to a file and read them back later even after restarting the computer. This is how real applications store data.

Explore (10 min)

Teacher Let me create a file and write something to it. I will use with open test dot txt in write mode as file then file dot write hello KREIS. When I run this a file called test dot txt is created with the text inside. Now let me read it back: with open test dot txt in read mode as file then print file dot read. Watch the text appear on screen. Try creating your own file with your name and reading it.

Students I created a file called my_name dot txt and wrote my name inside. When I read it back my name appeared on screen. The with statement automatically closed the file when I was done. The file still exists on the computer even after the program ended.

Explain (10 min)

Teacher File handling in Python uses the open function with a filename and mode. The write mode w creates a new file or overwrites an existing one. The read mode r opens an existing file for reading. The append mode a adds data to the end of an existing file. The with statement automatically closes the file when the block ends which is the recommended approach. The write method adds text to the file. The read method returns the entire file content as a string. The readline method reads one line at a time. The writelines method writes a list of strings to the file. Always close files properly to avoid data loss.

Students Open creates or accesses a file. Write mode overwrites read mode accesses and append mode adds to the end. The with statement is the safest way because it auto-closes. Write adds text read gets text and readline gets one line. Proper file handling prevents data loss and corruption.

Elaborate (10 min)

Teacher Write a program that saves a student report card to a file. Create variables for student name class and marks in three subjects. Write all this data to a file called report dot txt in a readable format. Then write a program that reads this file and calculates the total and average marks. Save the results to a new file called results dot txt. Challenge: use append mode to add a new student record to an existing file.

Students My report card file has the student name class and marks formatted nicely. My results program reads the marks calculates total as 235 and average as 78.33 and saves it. For the challenge I appended a second student record without deleting the first one. The append mode preserved the existing data.

Evaluate (5 min)

Teacher I will give you a scenario where a program needs to save and load data. You must write the complete file handling code including opening writing reading and closing files. I will check if you use the with statement correctly and handle possible errors like file not found. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students I used try except with FileNotFoundError in case the file does not exist yet. The with statement ensures the file closes properly. Writing data in a consistent format makes it easier to read back later.

Cross-Curricular Connections

Mathematics: Mathematics: File handling is like recording data in a log book. Mathematical experiments generate data that needs to be stored and retrieved for analysis.
Science: Science: Scientists store experimental results in files for later analysis. File handling in computing models this real-world data persistence need.
Language: Language Arts: Writing to a file is like writing in a diary. The diary preserves your entries and you can read them back later. Both serve as persistent storage.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Python IDLE
- Textbook Chapter 1 Page 17
- KREIS file handling examples sheet

Differentiation

Approaching: Class 8 Foundation: Focus on basic write and read operations using the with statement. Use predefined file operations and trace through the code. Create simple text files with teacher guided step-by-step instructions.
On Level: Class 9 Application: Write and read files independently. Create report card programs that save and load data. Handle file not found errors gracefully. Work on real data persistence scenarios.
Advanced: Class 10 Mastery: Implement complete file-based data management systems. Use append mode for record keeping and explore reading files line by line. Connect file handling to database concepts in Chapter 3.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Practical file handling task creating reading and writing files
CT Skill Assessed: Abstraction: Understanding how persistent storage abstracts the complexity of data preservation beyond program execution

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a program that saves your daily study log to a file. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Module: File containing related functions that can be reused in many programs
Import: Bringing module code into current program to use its functions
Reuse: Using same code in many places without writing again for efficiency
Cohesion: How well parts inside one module work together for single purpose

Engage (5 min)

Teacher Students imagine you are building a house. You would not try to build everything yourself. You would have separate teams for plumbing electrical work and painting. Each team is an expert in their area. Modular programming works the same way. Instead of writing one long program you break it into smaller modules or functions each responsible for one task. This makes your code organized testable and reusable.

Students So modular programming is like dividing work among specialists. Each function becomes an expert in one task. I can reuse the same function in different programs and if there is a bug I only need to fix it in one place.

Explore (10 min)

Teacher Let me show you two versions of a program. First version is one long main function with everything mixed together. Second version has separate functions: get_input process_data and display_output. The second version is much easier to read and understand. Now let me show you the math module: import math then print math dot sqrt 16 gives 4.0. Try importing the random module and using random dot randint to get a number between 1 and 10.

Students The modular version is much cleaner. Each function has a clear name that describes what it does. The math module gave me square root and the random module gave me random numbers. I can use these built-in modules without writing the math myself.

Explain (10 min)

Teacher Modular programming means organizing code into separate self-contained units. Each unit handles one specific task. This follows the Single Responsibility Principle. Python modules are files containing functions and variables that can be imported into other programs. The import statement loads a module. You can import specific functions with from module import function. Built-in modules include math for mathematical functions random for random numbers and datetime for date and time operations. Modular code is easier to debug test and maintain. It also enables teamwork as different people can work on different modules.

Students Modules are like toolboxes. Each toolbox contains tools for a specific job. I import the toolbox and use its tools. The math module has mathematical tools. The random module has random number tools. Breaking programs into functions makes each part testable and reusable.

Elaborate (10 min)

Teacher Create a KREIS student management program using modular design. Write separate functions for add_student view_students search_student and save_to_file. Then create a main menu that lets the user choose which function to run. Import the datetime module to timestamp each student record. Import the random module to generate unique student IDs. Each function should be short and focused on one task.

Students My program has four functions each handling one task. The main menu loops and lets the user choose an option. add_student collects data view_students displays all records search_student finds a specific student and save_to_file writes everything to disk. The datetime module adds timestamps and random generates IDs like STU4521.

Evaluate (5 min)

Teacher I will give you a monolithic program with everything in one function. You must identify the logical sections and break them into separate functions. Then write the main program that calls these functions in the correct order. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students I identified four main tasks in the program and created a function for each. The main program becomes a simple sequence of function calls. Each function is short and focused making the code much easier to understand and modify.

Cross-Curricular Connections

Mathematics: Mathematics: The math module provides functions for square root power and trigonometry. These directly connect to mathematical concepts students learn in mathematics class.
Science: Science: Modular programming mirrors the scientific method where experiments are broken into controlled variables procedures and measurements. Each module handles one aspect of the experiment.
Language: Language Arts: Modular writing separates introduction body and conclusion. Each section has a specific purpose. Similarly modular code separates input processing and output into distinct functions.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Python IDLE
- Textbook Chapter 1 Page 19
- KREIS modular programming examples

Differentiation

Approaching: Class 8 Foundation: Focus on understanding what a module is and using built-in modules with import. Practice breaking simple programs into two or three functions. Use teacher provided templates for modular structure.
On Level: Class 9 Application: Write modular programs with multiple functions and import built-in modules independently. Create a menu-driven program using modular design. Test each function separately.
Advanced: Class 10 Mastery: Design complex modular systems with clear separation of concerns. Explore creating custom modules and understand module namespaces. Connect modular design to software engineering principles.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Code refactoring exercise turning a monolithic program modular
CT Skill Assessed: Decomposition: Breaking a complex problem into manageable independent modules that can be developed and tested separately

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a program that calculates the area of a circle using the math module for the value of pi. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Recursion: Function calling itself to solve smaller version of same problem cleverly
Base Case: Simple situation where recursion stops and returns answer directly without further calls
Recursive Call: Function calling itself with smaller input to move toward base case gradually
Stack: Memory storing each waiting function call until base case reached for completion

Engage (5 min)

Teacher Students imagine you are standing between two mirrors facing each other. You see your reflection repeated infinitely. Recursion in programming is similar but with a twist. A recursive function calls itself but each time it gets closer to a stopping point. Think of it like Russian nesting dolls. Each doll contains a smaller version of itself until you reach the smallest solid doll. That smallest doll is the base case that stops the recursion.

Students So a recursive function calls itself but it must have a stopping condition. Without the stopping condition it would call itself forever like the mirrors. The base case is like the smallest doll that does not open. I can see how this could simplify some problems.

Explore (10 min)

Teacher Let me write a simple recursive function. def countdown n: if n equals 0 print liftoff else print n then countdown n minus 1. When I call countdown 5 it prints 5 4 3 2 1 liftoff. Trace through this on paper. Write down each call and what happens. What would happen if I forgot the if statement. What would happen if I did not change n in each call.

Students I traced it. countdown 5 prints 5 then calls countdown 4. countdown 4 prints 4 then calls countdown 3. This continues until countdown 0 prints liftoff and stops. Without the if statement it would call itself forever causing a stack overflow. Without changing n the base case would never be reached.

Explain (10 min)

Teacher Recursion is when a function calls itself. Every recursive function needs two parts. The base case is the condition that stops the recursion. Without it the function would call itself infinitely. The recursive case is where the function calls itself with a modified parameter that moves toward the base case. The call stack keeps track of all active function calls. Each recursive call adds a new frame to the stack. When the base case is reached the frames start returning values back up the stack. Recursion is useful for problems that have a self-similar structure like counting tree branches or calculating factorials.

Students Base case stops recursion recursive case moves toward base case and the call stack manages everything. Each call waits for the next call to finish. The base case is crucial. Without it Python gives a maximum recursion depth error. Recursion simplifies certain problems by breaking them into smaller versions of the same problem.

Elaborate (10 min)

Teacher Write a recursive function to calculate factorial of a number. Remember that n factorial equals n times n minus 1 factorial and 0 factorial equals 1. Test with factorial of 5 factorial of 0 and factorial of 1. Then write a recursive function to calculate the sum of numbers from 1 to n. Challenge: write a recursive function to reverse a string by taking the last character plus the reverse of the remaining characters. Modify: change one parameter/block and predict outcome before running.

Students My factorial function returns 120 for factorial of 5 because 5 times 4 times 3 times 2 times 1 equals 120. factorial of 0 returns 1 which is the base case. My sum function returns 15 for sum of 1 to 5. For reversing hello it takes o plus reverse of hell which continues until the string is empty returning olleh.

Evaluate (5 min)

Teacher I will give you a recursive function and ask you to trace through it completely. You must show each function call on the stack and the return values as they propagate back. You must also identify the base case and the recursive case. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Tracing recursion requires patience. I draw the call stack showing each level. The base case returns first then each level above uses that return value. Understanding the stack helps me predict recursive function output.

Cross-Curricular Connections

Mathematics: Mathematics: Factorial and Fibonacci sequences are mathematical concepts that naturally use recursion. The mathematical definition directly translates to recursive code.
Science: Science: Natural phenomena like tree branching and river deltas follow recursive patterns. Fractal geometry in nature is based on self-similar recursive structures.
Language: Language Arts: Recursion is like a story within a story within a story. Each level contains a smaller version of the same narrative structure until reaching the innermost tale.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Python IDLE
- Textbook Chapter 1 Page 21
- KREIS recursion tracing worksheets

Differentiation

Approaching: Class 8 Foundation: Focus on understanding the concept of a function calling itself. Trace simple recursion examples step by step on paper. Use countdown and simple addition recursion with heavy teacher guidance.
On Level: Class 9 Application: Write recursive functions for factorial and sum. Trace through recursive calls independently. Understand the base case requirement and predict when recursion will stop.
Advanced: Class 10 Mastery: Implement recursive solutions for complex problems including string reversal and power calculation. Analyze the stack depth and compare recursive versus iterative approaches.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Recursive trace table and base case identification exercise
CT Skill Assessed: Pattern Recognition: Identifying self-similar subproblems that can be solved by the same function applied to smaller inputs

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a recursive function called sum_digits that takes a number and returns the sum of its digits. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Programming: Writing instructions so computer performs tasks step by step
Code: Written instructions that computer can understand and follow exactly
Instruction: Single command telling computer to do one specific action
Bug: Mistake in code that makes program behave incorrectly or stop

Engage (5 min)

Teacher Students over the past weeks you have learned many programming tools: functions parameters return values lists loops file handling and more. Today you will combine all these skills into a complete project. This is like building a house where you use bricks cement wood and paint together. Each material alone is simple but combined they create something amazing. Your project will demonstrate how all these concepts work together.

Students I am excited to use everything I learned in one big project. I need to plan which functions I need and how they will work together. This feels like being a real programmer who builds complete applications.

Explore (10 min)

Teacher Let us plan a KREIS Library Management System together. First we identify the main tasks: add a book issue a book return a book display all books and search for a book. Each task becomes a function. We use a list to store book records. We use file handling to save the data. We use loops for the menu. We use string operations for formatting. Let me sketch the program structure on the board and you suggest what each function should do.

Students The add_book function should collect title author and availability. The issue_book function should mark a book as issued. The return_book function should mark it available again. The display function should loop through all books. The search function should find a book by title. We need a main menu loop to let the user choose which operation to perform.

Explain (10 min)

Teacher A programming project requires planning before coding. Start by identifying all the functions your program needs. Write pseudocode for each function describing what it does step by step. Decide what data structures you need such as lists or dictionaries. Plan how functions will communicate through parameters and return values. Write one function at a time and test it before moving to the next. Use comments to explain complex logic. Debug by checking each function individually then testing the complete program. Documentation helps others understand your code and helps you maintain it later.

Students Planning saves time because I know what to build before I start coding. Testing each function separately catches errors early. Comments explain my thinking. The project structure shows how simple functions combine to create a useful application.

Elaborate (10 min)

Teacher Build the complete KREIS Library Management System. Include at least five functions each handling one operation. Use a list of dictionaries to store book records with title author and availability status. Implement a menu loop that runs until the user chooses to exit. Save all book data to a file when the program ends and load it when the program starts. Add error handling for invalid menu choices and missing books. Format the output neatly using string methods. Modify: change one parameter/block and predict outcome before running.

Students My library system has six functions: add_book issue_book return_book display_books search_book and load_data. I use a list of dictionaries where each dictionary represents a book. The menu loops until the user chooses exit. I save data to library_data dot txt using json format. Error handling catches invalid input and missing books. The output is neatly formatted in columns.

Evaluate (5 min)

Teacher I will evaluate your project based on these criteria: correct use of functions with parameters and return values proper use of lists and file handling clear program structure with a working menu error handling that prevents crashes and neat formatted output. Present your project to the class explaining your design choices and how each function works. Predict: before running, students write their prediction of output and compare with actual result (PRIMM Predict). Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students My project meets all criteria. I used six separate functions with proper parameters. Lists store the data and file handling saves it. The menu loop works correctly. Error handling catches all invalid inputs. My presentation explained how I planned each function before coding.

Cross-Curricular Connections

Mathematics: Mathematics: Library management involves counting sorting and searching which are mathematical operations. Calculating overdue fines uses arithmetic operations.
Science: Science: Project management in science follows the same plan implement test and document cycle. A programming project mirrors the scientific method of hypothesis experimentation and analysis.
Language: Language Arts: Documentation and presentation of the project develop communication skills. Explaining code to others is like explaining a story plot clearly.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Python IDLE
- Textbook Chapter 1 review pages
- KREIS project planning template

Differentiation

Approaching: Class 8 Foundation: Complete a simplified project with three functions: add display and search. Use predefined data structures and follow the teacher provided template. Focus on understanding how functions work together.
On Level: Class 9 Application: Build the complete project with five functions and file handling. Work independently or in pairs. Implement error handling and test all functions thoroughly before submission.
Advanced: Class 10 Mastery: Extend the project with additional features like sorting books by title calculating overdue fines and generating reports. Add comprehensive error handling and documentation. Present and defend design choices to the class.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Project submission with code review and class presentation
CT Skill Assessed: Creation: Combining multiple programming concepts into a complete functional application that solves a real-world problem

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Build a simple Student Grade Tracker project. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 2 - Data Structures

`(..) => ..`

Key Vocabulary

List: Ordered collection holding many values that can change during program run
Index: Position number starting zero indicating where element lives in list currently
Slice: Part of list taken as new smaller list like first three items
Mutability: Ability to change list contents after creation without making new list

Engage (5 min)

> **Teacher** Students yesterday we learned basic list operations like append and remove. Today we go deeper into lists. Imagine you have a row of 30 students in assembly. You want to take a photograph of just students 5 through 15. That is like slicing a list. You take a portion of the list without changing the original. Advanced lists also let you create new lists from existing ones using powerful expressions.

Students Slicing a list means taking a portion of it. I can get students 5 through 15 without removing anyone from the original row. List comprehension sounds like a shorter way to create lists. I am curious about how that works.

Explore (10 min)

Teacher Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me create a list of numbers 0 through 9. Now slice it: numbers 2 colon 7 gives me elements at indices 2 through 6. numbers colon colon 2 gives every other element. numbers colon colon minus 1 reverses the list. Now list comprehension: square_list equals bracket x times x for x in range 10 bracket. This creates 0 1 4 9 16 through 81 in one line. Try different slices and comprehension expressions.

Students The slice with step 2 skips every other element. Reversing with minus 1 step is clever. The list comprehension created squares of 0 through 9 in one line instead of a for loop. I can add conditions too like even numbers only. This is much faster to write.

Explain (10 min)

Teacher List slicing uses the syntax list start colon stop colon step. Start is where slicing begins inclusive. Stop is where it ends exclusive. Step determines the increment between elements. Omitting start begins at the beginning. Omitting stop goes to the end. Step of 2 takes every other element. Negative step reverses the direction. List comprehension creates a new list from an expression in one line using the pattern bracket expression for item in iterable bracket. You can add if conditions to filter elements. List copying is important because assigning one list to another creates a reference not a copy. Use copy method or slicing to create independent copies.

Students Slicing extracts portions with start stop and step. Comprehension builds lists concisely with expression for item in iterable. Copying requires care because simple assignment creates a reference. The copy method and slicing create actual copies that can be modified independently.

Elaborate (10 min)

Teacher Create a list of marks for 20 students. Use slicing to extract marks of the first 5 students the last 5 students and every third student. Use list comprehension to create a list of pass marks where marks above 50 are marked as pass and below as fail. Use comprehension with a condition to create a list of only the failing marks. Copy the original list and modify the copy to show what happens with reference versus actual copies.

Students My slices give first 5 marks from index 0 to 4 and last 5 from index 15 to 19. Every third student uses step 3. My pass_fail comprehension creates PASS or FAIL for each mark. My failing marks list uses if mark less than 50. The copy was modified independently proving it is a real copy not a reference.

Evaluate (5 min)

Teacher I will give you a list and ask you to write slicing expressions to extract specific portions. Then I will give you a description and ask you to write the list comprehension that produces it. You must also explain the difference between reference and copy assignment. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Slicing notation becomes natural with practice. The key is remembering that stop is exclusive. List comprehension is a powerful tool that replaces multi-line loops with a single expressive line.

Cross-Curricular Connections

Mathematics: Mathematics: Slicing is like selecting a subset from a sequence. List comprehension is like set-builder notation in mathematics where you define a set by a rule.
Science: Science: Scientists often need to sample data from a larger dataset. Slicing provides a programmatic way to extract representative samples for analysis.
Language: Language Arts: Slicing a list is like extracting a paragraph from a longer essay. You select a specific portion while keeping the original intact.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:

- Python IDLE
- Textbook Chapter 2 Page 1
- KREIS advanced list exercises

Differentiation

Approaching: Class 8 Foundation: Focus on basic slicing with start and stop only. Use visual index diagrams to understand slicing. Practice list comprehension with simple expressions like x times 2 for x in range 5.
On Level: Class 9 Application: Use slicing with step parameters and negative indices. Write list comprehensions with conditions. Understand reference versus copy through practical experiments.
Advanced: Class 10 Mastery: Combine slicing and comprehension to solve complex problems. Explore nested list comprehension. Analyze when to use comprehension versus traditional loops for clarity and efficiency.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Slicing and comprehension coding challenges with prediction tasks
CT Skill Assessed: Pattern Recognition: Identifying patterns in data to extract subsets and create filtered lists using slicing and comprehension

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a list of numbers from 1 to 30. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Method: Function belonging to list called with dot like list.append for action
Append: Adding item to end of list without changing other elements quickly
Pop: Removing and returning last item from list for use elsewhere immediately
Sort: Arranging list elements in order like smallest to largest automatically quickly

Engage (5 min)

Teacher Students in our KREIS school we line up in order of height for assembly. The teacher arranges us from shortest to tallest. Today we learn list methods that sort and organize data just like the teacher arranges the line. Python has built-in methods that can sort reverse count and find elements in a list.

Students So Python can sort lists automatically. I do not have to write sorting code from scratch. The sort method will arrange my marks from lowest to highest and I can reverse it to go from highest to lowest.

Explore (10 min)

Teacher Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me create a list of 8 student marks: 78 92 45 88 63 95 72 81. Now I call marks dot sort and print the list. The marks are now in order from 45 to 95. Now I call marks dot reverse and they go from 95 to 45. Let me count how many students scored above 80 using a comprehension then use marks dot count to find how many got exactly 72. Try these methods and observe the changes.

Students sort rearranged the list in ascending order. reverse flipped it to descending. count told me how many times 72 appears in the list. index found the position of a specific mark. These methods modify the original list directly.

Explain (10 min)

Teacher The sort method arranges elements in ascending order by default. Pass reverse equals true to sort in descending order. sort modifies the original list in place and returns None. The reverse method flips the list order. The count method returns how many times a value appears. The index method returns the position of the first occurrence of a value. The extend method adds all elements from another list to the end. The clear method removes all elements leaving an empty list. These methods are essential for data manipulation and organization.

Students Sort orders the list and reverse flips it. count counts occurrences and index finds position. extend merges lists and clear empties them. All these methods except clear modify the list in place. They are like tools for organizing and managing data efficiently.

Elaborate (10 min)

Teacher Create a program that manages KREIS exam results. Store marks for 10 students in a list. Sort them to find the highest and lowest scores. Count how many students scored above 80. Find the position of the median score. Extend the list with 5 new marks from another test. Reverse the sorted list to show results from highest to lowest. Finally clear the list and print its length to confirm it is empty.

Students My sorted list shows lowest 35 and highest 98. Eight students scored above 80. The median is at index 4 in the sorted list. I extended with 5 more marks from the science test. Reversed shows 98 at the beginning. Clear made the list empty and length is 0. The program handles all data management tasks.

Evaluate (5 min)

Teacher I will give you a list of data and a series of operations to perform. You must write the correct method calls and predict the output. Then I will give you a desired final state and ask which methods could produce it. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Knowing which method to use for each task requires practice. Sort for ordering count for frequency index for position and extend for merging are the key tools.

Cross-Curricular Connections

Mathematics: Mathematics: Sorting is fundamental in statistics for finding median and quartiles. Count relates to frequency in data analysis. Index relates to position in sequences.
Science: Science: Laboratory data often needs sorting to identify trends. Counting occurrences helps determine experimental frequency of results.
Language: Language Arts: Sorting words alphabetically is a language skill. Counting word frequency helps analyze writing style and vocabulary usage.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Python IDLE
- Textbook Chapter 2 Page 3
- KREIS list methods quick reference

Differentiation

Approaching: Class 8 Foundation: Focus on sort reverse and count methods with simple lists. Use provided data and step-by-step instructions. Practice one method at a time with clear examples.
On Level: Class 9 Application: Use all list methods independently to manage real data. Combine multiple methods to solve problems. Predict output before running code.
Advanced: Class 10 Mastery: Use list methods in complex data processing pipelines. Compare sort versus sorted function. Analyze when to use each method for optimal results.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Method application exercise with data manipulation tasks
CT Skill Assessed: Algorithmic Thinking: Selecting appropriate methods to transform data from one state to another efficiently

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a list of your test scores from the last five tests. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Dictionary: Collection storing pairs of key and value for quick lookup by name
Key: Unique name used to find its matching value inside dictionary quickly
Value: Data stored under a key that you retrieve when searching dictionary
Pair: Single key together with its value as one entry inside dictionary

Engage (5 min)

Teacher Students in a dictionary you look up a word to find its meaning. In Python a dictionary works similarly but instead of words and meanings it stores any key and its associated value. Think of it like a school register. The roll number is the key and the student name is the value. You look up by roll number to find the name. Today we learn Python dictionaries which are powerful data structures.

Students So a dictionary maps keys to values. I can look up any key instantly without searching through the entire collection. That is much faster than a list where I have to check each element one by one.

Explore (10 min)

Teacher Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me create a dictionary of KREIS houses: houses equals opening curly brace Arya colon green Bharati colon blue Chandrika colon red Drona colon yellow closing curly brace. Now I access a value using houses square bracket Arya which gives green. I can add a new house with houses Ekalavya equals purple. I can change a color with houses Drona equals orange. Try creating your own dictionary and performing these operations.

Students The curly braces create the dictionary. Keys go on the left of the colon and values on the right. Accessing by key is instant. Adding a new key value pair is simple. Changing a value is like updating an entry. Dictionaries feel more organized than lists.

Explain (10 min)

Teacher A dictionary stores data as key-value pairs. Keys must be unique and immutable like strings numbers or tuples. Values can be any data type. You create a dictionary using curly braces with key colon value pairs separated by commas. Access values using square brackets with the key name. Add new entries by assigning a value to a new key. Modify entries by reassigning. Delete using del statement or pop method. The keys method returns all keys. The values method returns all values. The items method returns key-value pairs as tuples. Dictionaries are unordered in Python versions before 3.7 and ordered by insertion in Python 3.7 and later.

Students Keys are unique identifiers and values are the data associated with them. Curly braces create dictionaries. Square brackets access values. del and pop remove entries. keys values and items methods help iterate through the data. Dictionaries are perfect for modeling real-world relationships like student records or product catalogs.

Elaborate (10 min)

Teacher Create a student profile dictionary with keys for name class roll_number subjects and marks. The subjects key should hold a list of subject names. The marks key should hold a list of corresponding marks. Access and print the student name. Access and print the third subject. Add a new key called attendance with a value of 95. Delete the roll_number key. Print all keys and values using the items method.

Students My dictionary has name Arya, class 9, roll 15, subjects list with 5 subjects and marks list with 5 marks. I accessed name directly. The third subject is at subjects list index 2. I added attendance 95 and deleted roll_number. The items method printed all key value pairs neatly.

Evaluate (5 min)

Teacher I will give you a scenario and ask you to design a dictionary that models it. You must choose appropriate keys and values and write code to access modify and iterate through the dictionary. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Choosing good keys is important. They should be descriptive and unique. Understanding when to use dictionaries versus lists is a key skill. Dictionaries are best when you need to look up data by a meaningful identifier.

Cross-Curricular Connections

Mathematics: Mathematics: Dictionaries map one set to another like mathematical functions. The domain is the set of keys and the range is the set of values.
Science: Science: Scientific databases use key-value pairs to store experimental data with labels and measurements. Dictionaries model this storage pattern.
Language: Language Arts: A dictionary is literally a word-to-definition mapping. The concept of looking up information by key is fundamental to reference works.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:

- Python IDLE
- Textbook Chapter 2 Page 5
- KREIS dictionary practice sheet

Differentiation

Approaching: Class 8 Foundation: Create simple dictionaries with string keys and values. Practice accessing values and adding entries. Use visual diagrams showing key-value mapping.
On Level: Class 9 Application: Create dictionaries with mixed value types including lists. Iterate using items method. Perform CRUD operations independently.
Advanced: Class 10 Mastery: Use dictionaries for complex data modeling. Explore nested dictionaries. Compare dictionary performance with list searching.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Dictionary creation and manipulation coding exercise
CT Skill Assessed: Pattern Recognition: Modeling real-world relationships as key-value pairs in dictionary structures

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a dictionary representing your daily schedule with time slots as keys and activities as values. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Lookup: Finding value by giving its key to dictionary quickly without scanning
Update: Changing value already stored under existing key to new value
Delete: Removing a key and its value from dictionary completely thereafter for cleanup
Iterate: Visiting each key-value pair one by one to process them all sequentially

Engage (5 min)

Teacher Students what happens if you try to access a key that does not exist in a dictionary. Python gives a KeyError. Today we learn safe dictionary operations that prevent errors and make our programs robust. Think of it like this: instead of just opening a box and hoping something is inside you check first whether the box contains what you need.

Students So I can check if a key exists before trying to access it. That prevents my program from crashing. The get method sounds useful because it provides a fallback value if the key is missing.

Explore (10 min)

Teacher Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me try accessing a missing key: student equals name colon Priya. If I try student square bracket age it gives KeyError. But if I use student dot get age 15 it returns 15 the default value. If I use student dot get age it returns None. I can also check with if age in student which returns True or False. Try dictionary comprehension: squares equals bracket key colon key times key for key in range 1, 6 bracket.

Students The get method prevented the KeyError by returning my default value 15. The in operator checked if age exists as a key. Dictionary comprehension created a new dictionary of squares in one line. These operations make dictionaries safer and more powerful.

Explain (10 min)

Teacher The get method provides safe dictionary access. It takes a key and an optional default value. If the key exists it returns the value. If not it returns the default or None. The in operator checks key existence and returns True or False. Always check before accessing to avoid KeyErrors. The update method merges another dictionary into the existing one. If keys overlap the new values replace old ones. Dictionary comprehension creates dictionaries from expressions using the pattern bracket key colon value for item in iterable bracket. These operations make dictionary handling safe and efficient.

Students Get is the safe way to access values. In checks existence. Update merges dictionaries. Comprehension builds them concisely. Together these tools make dictionaries robust data structures for any program.

Elaborate (10 min)

Teacher Write a program that tracks KREIS student attendance for a week. Use a dictionary with student names as keys and lists of attendance records as values. Use get to safely access attendance when a new student is added. Check if a student exists before recording attendance. Use update to merge attendance from two sections. Use dictionary comprehension to create a summary dictionary showing total present days for each student.

Students My attendance dictionary has student names and their daily records. get returns 0 for new students. I check with if name in attendance before recording. update merged Section A and Section B attendance. My comprehension creates total_present for each student by counting present entries in their list.

Evaluate (5 min)

Teacher I will give you a program that accesses dictionary keys without safety checks. You must add proper error handling using get and in operator. Then I will ask you to merge two dictionaries and explain what happens with duplicate keys. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Safe dictionary access prevents runtime errors. The get method with a default value is the most practical approach. update replaces old values with new ones when keys conflict.

Cross-Curricular Connections

Mathematics: Mathematics: Dictionary operations relate to set theory. Key existence checking is like set membership testing. Merging is like set union.
Science: Science: Merging data from multiple experiments is a common scientific task. Dictionary update models this data consolidation process.
Language: Language Arts: Safe dictionary access is like verifying a reference before citing it. You check that the source exists before using it in your work.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Python IDLE
- Textbook Chapter 2 Page 7
- KREIS dictionary operations guide

Differentiation

Approaching: Class 8 Foundation: Focus on get method and in operator for safe access. Use predefined dictionaries and practice one operation at a time with teacher guidance.
On Level: Class 9 Application: Use all dictionary operations independently. Create programs that safely handle

missing keys. Practice merging dictionaries and using comprehension.
Advanced: Class 10 Mastery: Combine dictionary operations with list operations for complex data processing. Analyze when to use each operation for optimal code clarity and performance.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Safe dictionary access and merging coding exercise
CT Skill Assessed: Evaluation: Assessing which dictionary operation to use for different scenarios and handling edge cases safely

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a dictionary of your monthly expenses with categories as keys and amounts as values. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Tuple: Ordered collection that cannot be changed after creation keeping data safe permanently
Immutable: Cannot be altered after creation protecting data from accidental change completely
Index: Position number starting zero used to fetch element from tuple quickly
Packing: Putting many values together into one tuple in single step conveniently

Engage (5 min)

Teacher Students imagine you write your exam paper in pen. Once written you cannot erase or change it. Tuples in Python work like that. They are like lists but you cannot change them after creation. Think of a tuple as a sealed container. You can look inside but you cannot add remove or change anything. This immutability is actually useful when you want to protect data from accidental changes.

Students So tuples are permanent while lists are changeable. If I have data that should never change like dates or coordinates a tuple is the right choice. It protects the data from accidental modification.

Explore (10 min)

Teacher Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me create a tuple: coordinates equals 10 comma 20. I can access with coordinates square bracket 0 which gives 10. I can slice with coordinates 0 colon 2. But if I try coordinates square bracket 0 equals 50 Python gives TypeError. Tuples also work as dictionary keys: location equals bracket coordinates colon school bracket. Try creating your own tuples and see what operations work and which do not.

Students Indexing and slicing work on tuples just like lists. But assignment does not work because tuples are immutable. I can use tuples as dictionary keys because they are immutable. Lists cannot be dictionary keys because they can change. This is an important difference between tuples and lists.

Explain (10 min)

Teacher Tuples are ordered collections enclosed in parentheses or just commas. They are immutable meaning once created they cannot be modified. You cannot add remove or change elements. However you can access elements using indexing and slicing. Tuples can be concatenated using the plus operator to create new tuples. The tuple function converts other iterables to tuples. Tuples are faster than lists and use less memory. They can be used as dictionary keys because they are hashable. Functions can return multiple values as tuples. Unpacking assigns tuple elements to multiple variables at once.

Students Tuples use parentheses and are immutable. They support indexing slicing and concatenation but not modification. They are faster than lists and can be dictionary keys. Multiple return values from functions come as tuples. Unpacking lets me assign multiple variables from a tuple in one line.

Elaborate (10 min)

Teacher Create a tuple representing the KREIS school coordinates with latitude and longitude. Use it as a dictionary key to store the school name. Write a function that returns the student name class and roll number as a tuple. Unpack the returned tuple into three variables. Create a list of tuples where each tuple contains a student name and their marks. Sort this list by marks using the second element of each tuple.

Students My coordinates tuple 12.9716 comma 77.5946 is the key for school name KREIS. My function returns the tuple with three values. Unpacking gives me name class and roll in separate variables. My student list has tuples sorted by marks using a lambda function. The immutable nature of tuples makes the data safe.

Evaluate (5 min)

Teacher I will give you scenarios and ask whether a tuple or list is more appropriate. You must justify your choice. I will also ask you to write code that uses tuple unpacking and returns multiple values from a function. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Choosing between tuple and list depends on whether the data needs to change. Fixed data like coordinates dates and configurations use tuples. Changeable data like student records and shopping lists use lists.

Cross-Curricular Connections

Mathematics: Mathematics: Tuples represent ordered pairs points in coordinate systems and mathematical relations. The immutability matches mathematical constants.
Science: Science: Experimental constants like physical properties are naturally immutable and suit tuple representation. Coordinates and measurements use tuples.
Language: Language Arts: A tuple is like a published quote that cannot be altered. Once a statement is recorded it remains in its original form forever.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Python IDLE
- Textbook Chapter 2 Page 9
- KREIS tuples and lists comparison chart

Differentiation

Approaching: Class 8 Foundation: Create simple tuples and practice indexing. Understand that tuples cannot be changed. Use provided scenarios to practice tuple creation and access.
On Level: Class 9 Application: Use tuples as dictionary keys and for multiple return values. Practice unpacking. Compare tuple and list performance.
Advanced: Class 10 Mastery: Use tuples in complex data structures. Explore named tuples and understand hashability. Analyze when tuples provide advantages over lists.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Tuple versus list decision making and coding exercise
CT Skill Assessed: Abstraction: Choosing the right data structure based on whether data needs to be mutable or immutable

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a tuple of your favorite foods. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Set: Collection holding only unique values with no duplicates allowed automatically for uniqueness
Unique: Appearing only once with no repetition inside the collection for clarity
Union: Combining all elements from two sets together without duplicating any value
Intersection: Keeping only elements present in both sets together as common part

Engage (5 min)

Teacher Students imagine you have a bag of marbles. A set in Python is like a bag that only allows one marble of each color. If you try to put two red marbles in the bag only one stays. Sets automatically remove duplicates. They are also useful for comparing groups like finding students who are in both the cricket team and the chess team or students who are in cricket but not chess.

Students So sets automatically remove duplicates. If I have a list with repeated elements converting it to a set gives me only unique elements. Set operations like union and intersection help me compare groups. This is useful for comparing attendance lists or team rosters.

Explore (10 min)

Teacher Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me create a set of cricket players: cricket equals curly brace Arjun Ravi Kavya Meena closing curly brace. And chess players: chess equals curly brace Ravi Priya Sanjay Arjun closing curly brace. Cricket union chess gives all players from both teams. Cricket intersection chess gives players in both teams like Arjun and Ravi. Cricket difference chess gives players only in cricket. Try creating your own sets and performing these operations.

Students Union combines both sets without duplicates. Intersection gives only common elements. Difference gives elements in the first set but not the second. These operations are like comparing lists of names to find overlaps and differences. Much faster than writing loops to compare.

Explain (10 min)

Teacher Sets are unordered collections of unique elements enclosed in curly braces. Duplicates are automatically removed. You cannot access elements by index because sets are unordered. The add method adds a single element. The remove method deletes an element. The union method combines two sets. The intersection method finds common elements. The difference method finds elements in one set but not the other. The symmetric_difference method finds elements in either set but not both. Sets are useful for removing duplicates and performing membership testing efficiently.

Students Sets are curly brace collections that enforce uniqueness. No indexing because they are unordered. Union intersection and difference are powerful comparison tools. Adding and removing elements modifies the set. Converting a list to a set removes duplicates. Sets are fast for membership testing.

Elaborate (10 min)

Teacher Write a program that finds duplicate entries in a student attendance list. Create a list with some repeated names. Convert it to a set to see unique names. Use set difference to find students who were absent today compared to yesterday. Use intersection to find students present on both days. Create a set of all subject names across three classes to find common subjects.

Students My attendance list had Arjun appearing three times. Converting to a set gave unique names. Set difference found 2 students absent today but present yesterday. Intersection found 8 students present both days. My subject set shows that Mathematics and English are common across all three classes.

Evaluate (5 min)

Teacher I will give you two lists of data and ask you to find common elements unique elements and elements only in one list. You must write the correct set operations and explain what each result represents. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Set operations are efficient for comparing groups. Union gives everyone. Intersection gives common members. Difference gives unique members. Symmetric difference gives members in either but not both.

Cross-Curricular Connections

Mathematics: Mathematics: Sets are a fundamental concept in mathematics. Union intersection and difference are core set operations in discrete mathematics.
Science: Science: Venn diagrams in science use set concepts to classify organisms and compare experimental groups. Set operations model these classifications.
Language: Language Arts: A set is like a collection of unique words in a text. Converting a word list to a set reveals the vocabulary diversity.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:

- Python IDLE
- Textbook Chapter 2 Page 11
- KREIS sets operations visual guide

Differentiation

Approaching: Class 8 Foundation: Create sets and practice add and remove. Understand uniqueness property. Use visual Venn diagrams to understand union and intersection before coding.
On Level: Class 9 Application: Perform all set operations independently. Apply sets to real problems like finding duplicates. Convert between lists and sets.
Advanced: Class 10 Mastery: Use sets for efficient data analysis. Explore set comprehensions. Analyze when sets provide advantages over lists for specific operations.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Set operations Venn diagram to code translation exercise
CT Skill Assessed: Pattern Recognition: Identifying which set operation matches a real-world comparison scenario

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a set of your hobbies and a set of your siblings hobbies. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

Teacher Students in mathematics you work with matrices which are grids of numbers. In Python we can represent matrices using nested lists. A list inside a list creates a 2D structure. Think of it like this: a classroom has rows of desks and each desk has items. The outer list is the rows and the inner lists are the desks. Nested structures let us model complex real-world data.

Students So I can put lists inside lists to create tables. Each inner list represents one row. I can also put lists inside dictionaries to create detailed records. This lets me model complex data like a class register with names and marks.

Explore (10 min)

Teacher Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me create a 3 by 3 matrix: matrix equals bracket bracket 1 comma 2 comma 3 bracket comma bracket 4 comma 5 comma 6 bracket comma bracket 7 comma 8 comma 9 bracket bracket. Access the center element with matrix square bracket 1 square bracket 1 which gives 5. Now create a student record: student equals bracket name colon Arya comma marks bracket colon bracket 85 comma 90 comma 78 bracket bracket. Access the second mark with student marks bracket 1.

Students Two indices access elements in nested lists. The first index selects the row and the second selects the column. For dictionaries with list values I access the key first then index into the list. Nested structures make it easy to organize complex related data.

Explain (10 min)

Teacher Nested structures are data structures that contain other data structures. A list of lists creates a 2D grid where each inner list is a row. Access uses multiple indices like matrix row column. Dictionaries can contain lists as values allowing one key to hold multiple items. Lists can contain dictionaries allowing each element to be a complex record. Nested dictionaries have dictionaries inside dictionaries. These structures model real-world data like tables student records and organizational hierarchies. Accessing deeply nested data requires multiple brackets or key lookups.

Students Nested structures combine different data types. Lists of lists create grids. Dictionaries with lists store multiple values per key. Lists of dictionaries store multiple records. Multiple indices or key lookups access deep elements. These structures are essential for modeling complex real-world data.

Elaborate (10 min)

Teacher Create a KREIS classroom register as a list of dictionaries. Each dictionary represents a student with keys for name roll_number and marks. The marks key holds a list of marks in different subjects. Write code to find the student with the highest total marks. Calculate the class average for each subject. Find students who scored below 50 in any subject. Add a new student record to the register.

Students My register has 10 student dictionaries each with name roll and marks list. I calculated totals by summing each student marks list and found the highest. Subject averages require accessing marks index for each student. Students below 50 are found by checking each mark. Adding a new student is appending a dictionary to the list.

Evaluate (5 min)

Teacher I will give you a nested data structure and ask you to access specific deep elements. You must also write code to create a nested structure from a description and modify specific elements within it. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Accessing nested structures requires understanding the depth. Each index or key peels one layer. Practice tracing through nested access to build confidence with complex data structures.

Cross-Curricular Connections

Mathematics: Mathematics: Nested lists directly represent matrices used in linear algebra. Matrix operations like addition and multiplication use nested iteration.
Science: Science: Experimental data tables have rows for trials and columns for measurements. Nested lists model these 2D scientific data structures.
Language: Language Arts: A nested structure is like a book with chapters paragraphs and sentences. Each level contains the next level of detail.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Python IDLE
- Textbook Chapter 2 Page 13
- KREIS nested structure examples

Differentiation

Approaching: Class 8 Foundation: Work with simple nested lists like 2D matrices. Use visual grid diagrams to understand row and column indexing. Access elements with guided step-by-step instructions.
On Level: Class 9 Application: Create and manipulate nested structures independently. Work with lists of dictionaries for student records. Perform calculations on nested data.
Advanced: Class 10 Mastery: Design complex nested structures for real applications. Explore deeply nested data. Combine nested structures with functions for data processing pipelines.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Nested data structure creation and access coding challenge
CT Skill Assessed: Decomposition: Breaking complex data into hierarchical nested structures that are organized and accessible

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a nested list representing the monthly attendance record for your class. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Sorting: Arranging items in order like smallest to largest for easy searching later
Bubble Sort: Repeatedly swapping neighbours that are out of order until list sorted completely
Comparison: Checking which of two items is smaller to decide swapping during sort
Complexity: Measure of how many steps sorting needs as list grows bigger gradually

Engage (5 min)

Teacher Students how does your teacher arrange you in order of height for the assembly photo. The teacher looks at the first two students and swaps them if they are in wrong order. Then looks at the next pair and so on. This is exactly how bubble sort works. Today we learn sorting algorithms which are step-by-step procedures for arranging data in order. These are fundamental algorithms in computer science.

Students Sorting algorithms are systematic ways to order data. Bubble sort compares neighbors and swaps them. Selection sort finds the smallest and puts it first. I can see how different approaches produce the same sorted result but with different amounts of work.

Explore (10 min)

Teacher Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me demonstrate bubble sort with marks 65 42 78 35 90. Pass 1: compare 65 and 42 swap them. Compare 65 and 78 no swap. Compare 78 and 35 swap. Compare 78 and 90 no swap. After pass 1 the largest value 90 is at the end. Pass 2 continues with remaining elements. Draw this on paper and count the number of comparisons and swaps in each pass.

Students I drew each pass on paper. Pass 1 needed 4 comparisons and 2 swaps. Pass 2 needed 3 comparisons and 1 swap. Pass 3 needed 2 comparisons and 1 swap. Pass 4 needed 1 comparison and 0 swaps. The list is now sorted. Bubble sort is simple but does many comparisons.

Explain (10 min)

Teacher Bubble sort works by repeatedly stepping through the list comparing adjacent elements and swapping them if they are in wrong order. The pass through the list is repeated until no swaps are needed. The largest unsorted element bubbles to its correct position after each pass. Selection sort works by dividing the list into sorted and unsorted portions. It finds the minimum from the unsorted portion and swaps it with the first unsorted element. Both algorithms have quadratic time complexity meaning they get much slower as the list grows. Understanding these algorithms builds the foundation for more efficient ones.

Students Bubble sort swaps neighbors repeatedly. Selection sort finds minimums and places them. Both are simple but slow for large datasets. Bubble sort does about n squared comparisons. Selection sort also does n squared but fewer swaps. More efficient algorithms exist but these build the foundation.

Elaborate (10 min)

Teacher Implement both bubble sort and selection sort in Python. Create a list of 10 random numbers and sort it using each algorithm. Count the total number of comparisons and swaps for each. Compare the results. Then use Python built-in sort method and note how much faster it is. Discuss why built-in sort is better for real applications.

Students My bubble sort did 45 comparisons and 12 swaps. My selection sort did 45 comparisons but only 8 swaps. Both produced the same sorted list. The built-in sort was nearly instant because it uses a highly optimized algorithm called Timsort. Learning the basic algorithms helps understand why efficient sorting matters.

Evaluate (5 min)

Teacher I will give you an unsorted list and ask you to trace through bubble sort or selection sort step by step. You must show each comparison and swap. Then I will ask you to count total operations and compare algorithm efficiency. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Tracing sorting algorithms requires patience. I write the list state after each comparison. Counting operations helps compare algorithms. The key insight is that more efficient algorithms do less work for the same result.

Cross-Curricular Connections

Mathematics: Mathematics: Sorting algorithms relate to permutation and comparison in discrete mathematics. Time complexity connects to mathematical growth functions.
Science: Science: Sorting algorithms model natural processes like sedimentation where heavier particles settle first. Crystal formation follows sorting-like patterns.
Language: Language Arts: Sorting is like organizing a library. Books can be sorted by title author or subject. Different sorting methods represent different organizational strategies.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Python IDLE
- Textbook Chapter 2 Page 15
- KREIS sorting algorithm tracing sheets

Differentiation

Approaching: Class 8 Foundation: Trace bubble sort with small lists of 5 elements. Use visual diagrams showing swaps. Focus on understanding the process not implementation. Compare with selection sort conceptually.
On Level: Class 9 Application: Implement both bubble sort and selection sort. Count operations and compare efficiency. Use Python to verify manual sorting results.
Advanced: Class 10 Mastery: Analyze time complexity of both algorithms. Explore optimizations like early

termination in bubble sort. Compare with built-in sort and discuss algorithm selection for different data sizes.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Algorithm tracing and implementation coding exercise
CT Skill Assessed: Algorithmic Thinking: Understanding step-by-step procedures for sorting and analyzing their efficiency

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write bubble sort to sort your test scores in ascending order. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Search: Looking through data to find item matching desired value quickly and efficiently
Linear Search: Checking each item one by one from start until found sequentially
Binary Search: Repeatedly halving sorted list to find item much faster than linear
Target: Value you are trying to locate inside the collection of data currently

Engage (5 min)

Teacher Students imagine you are looking for your friend in a crowded school ground. One approach is to check every person one by one. That is linear search. Another approach is to ask half the people if your friend is on the left or right side and then check only that half. That is binary search. Today we learn these two fundamental searching algorithms.

Students Linear search checks everything one by one. Binary search eliminates half the possibilities each step. Binary search sounds much faster but it requires the data to be sorted first. I can see how the sorted requirement is important.

Explore (10 min)

Teacher Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me demonstrate linear search. I have a list of 8 student roll numbers. I search for roll number 6 by checking each element from the beginning. It takes 6 comparisons. Now for binary search the list must be sorted. I check the middle element. If my target is smaller I search the left half. If larger I search the right half. In a sorted list of 8 elements binary search needs at most 3 comparisons. Try both methods on paper.

Students Linear search checked 6 elements to find 6. Binary search found it in 3 steps. The sorted list allowed binary search to eliminate half each time. For 8 elements log base 2 of 8 is 3 which matches the binary search steps. Much more efficient than checking all 8.

Explain (10 min)

Teacher Linear search checks each element one by one from the beginning. It works on any list whether sorted or not. It takes on average n divided by 2 comparisons and in the worst case n comparisons. Binary search works only on sorted lists. It checks the middle element and eliminates half the remaining elements each step. It takes at most log base 2 of n comparisons. For 1000 elements linear search needs about 500 comparisons while binary search needs about 10. Binary search is dramatically faster for large datasets but requires sorted data.

Students Linear search is simple but slow. Binary search is fast but needs sorted data. The choice depends on whether the data is sorted and how large the dataset is. Understanding both algorithms helps choose the right tool for each situation.

Elaborate (10 min)

Teacher Create a KREIS student database as a sorted list of roll numbers. Implement both linear and binary search to find a specific student. Count the number of comparisons for each algorithm. Test with different target values including the first element last element and middle element. Discuss which algorithm performs better in each case. Create an unsorted list and explain why binary search cannot be used directly.

Students My linear search found roll 25 in 5 comparisons. Binary search found it in 3 comparisons. For the first element linear search did 1 comparison while binary search still did 3. For unsorted data binary search gives wrong results because it assumes the data is ordered.

Evaluate (5 min)

Teacher I will give you a dataset and a search target. You must decide which algorithm to use and justify your choice. Then implement the chosen algorithm and count the comparisons made. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students The decision depends on two factors: is the data sorted and how large is it. Sorted data with binary search is always faster for large datasets. Unsorted data requires linear search unless we sort first.

Cross-Curricular Connections

Mathematics: Mathematics: Binary search relates to logarithmic functions. The number of steps is log base 2 of n which connects to logarithm concepts in mathematics.
Science: Science: Searching for a specific chemical element in the periodic table follows a binary-search-like pattern since elements are ordered by atomic number.
Language: Language Arts: Finding a word in a dictionary uses binary search naturally. You open to the middle and determine which half to search next.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:

- Python IDLE
- Textbook Chapter 2 Page 17
- KREIS searching algorithm comparison chart

Differentiation

Approaching: Class 8 Foundation: Focus on linear search implementation and tracing. Use small lists and count comparisons manually. Understand the sequential checking process step by step.
On Level: Class 9 Application: Implement both algorithms and compare their performance. Understand when binary search is applicable. Count and compare comparisons for different target values.
Advanced: Class 10 Mastery: Analyze time complexity of both algorithms. Explore how database indexing uses search concepts. Discuss trade-offs between sorting first then searching versus linear search.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Algorithm selection justification and implementation exercise
CT Skill Assessed: Evaluation: Choosing the most appropriate search algorithm based on data properties and performance requirements

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a sorted list of your classmates heights. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Stack: Pile where last added item is removed first like plates stacked
Queue: Line where first added item is removed first like school queue
Push: Adding new item to top of stack for storage immediately quickly
Pop: Removing top item from stack or front item from queue for use

Engage (5 min)

> **Teacher** Students imagine you are stacking plates in a canteen. You put the new plate on top and take the top plate when you need one. This is a stack. Now imagine students lining up at the water fountain. The first person

in line gets water first. This is a queue. Today we learn these two important data structures that model many real-world situations.

Students Stack is last-in-first-out like plates. Queue is first-in-first-out like a line. Both are used everywhere in computing. The undo button uses a stack and the print queue uses a queue.

Explore (10 min)

Teacher Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me implement a stack using a Python list. Push adds to the end with append. Pop removes from the end. Let me push three items: books. Then pop one. The last book pushed is the first one removed. Now a queue: enqueue adds to the end. Dequeue removes from the front using pop index 0. Let me add three names to a queue and remove them. The first name added is the first removed.

Students Stack push and pop both happen at the end. Queue enqueue happens at the end but dequeue happens at the front. Stack is like undoing actions in reverse order. Queue is like serving customers in the order they arrived.

Explain (10 min)

Teacher A stack follows last-in-first-out principle. The last element added is the first one removed. Push adds an element to the top. Pop removes the top element. Peek or top views the top element without removing it. A queue follows first-in-first-out principle. The first element added is the first one removed. Enqueue adds to the back. Dequeue removes from the front. Stacks are used for undo operations function calls and expression evaluation. Queues are used for print jobs task scheduling and breadth-first search.

Students Stack is LIFO push pop from top. Queue is FIFO enqueue from back dequeue from front. Stacks handle undo and function calls. Queues handle scheduling and processing order. Both are fundamental in computer science.

Elaborate (10 min)

Teacher Implement a stack-based undo system for a simple text editor. The stack stores each text change. Undo pops the last change. Then implement a printer queue that stores print jobs and processes them in order. Add a feature to view the queue without removing items. Discuss how your browser back button works like a stack and how a hospital waiting room works like a queue.

Students My text editor stack stores each version of the text. Undo pops and restores the previous version. The printer queue adds jobs at the back and prints from the front. My queue view function shows all pending jobs. The browser back button pops from a history stack. The hospital waiting room enqueues patients and dequeues when the doctor is ready.

Evaluate (5 min)

Teacher I will give you a real-world scenario and ask you to identify whether it models a stack or queue. You must justify your choice and implement the appropriate data structure. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Stacks are for last-first scenarios like undo and browsing history. Queues are for first-first scenarios like lines and print queues. The key question is whether the most recent or oldest item should be processed first.

Cross-Curricular Connections

Mathematics: Mathematics: Stacks and queues relate to sequences and ordering in mathematics. The LIFO and FIFO principles are logical ordering concepts.
Science: Science: Call stacks in computing mirror cause-and-effect chains in science. Queue processing models assembly line production in manufacturing.
Language: Language Arts: A stack is like a stack of papers on your desk. The top paper is processed first. A queue is like a library hold list where the first request is fulfilled first.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Python IDLE
- Textbook Chapter 2 Page 19
- KREIS stacks and queues visual guide

Differentiation

Approaching: Class 8 Foundation: Focus on understanding the concepts of LIFO and FIFO. Use physical demonstrations with plates and lines. Implement simple stack and queue operations with teacher guidance.
On Level: Class 9 Application: Implement stack and queue using lists. Apply them to real scenarios like undo and print queue. Understand performance implications of queue dequeue operation.
Advanced: Class 10 Mastery: Implement stack and queue with optimal performance. Explore applications in tree traversal and graph algorithms. Compare with Python collections deque for efficient queues.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Scenario identification and implementation coding exercise
CT Skill Assessed: Abstraction: Modeling real-world ordering scenarios using appropriate data structure abstractions

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Implement a stack to track your study tasks. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Data: Facts like numbers and words stored on a computer
Raw Data: Unprocessed facts collected before cleaning or organising properly
Information: Useful meaning produced after processing raw data carefully
Dataset: Collection of related data stored together in one place

Engage (5 min)

Teacher Students you have learned many data structures: lists dictionaries sets tuples stacks and queues. Each has strengths and weaknesses. Today we learn how to choose the right tool for the job. Think of it like a toolbox. You would not use a hammer to tighten a screw. Each tool has a specific purpose. Knowing which tool to use makes you an efficient programmer.

Students Choosing the right data structure is like choosing the right tool. Lists are flexible dictionaries are fast for lookups sets handle uniqueness and tuples protect data. I need to think about what my data needs before choosing.

Explore (10 min)

Teacher Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let me give you three scenarios. Scenario 1: store names of students in a class. Scenario 2: store student names and their marks. Scenario 3: store unique hobby names across the class. For each scenario discuss with your partner which data structure is best and why. Then I will show you the answer and explanation.

Students Scenario 1 needs a list of names since order matters and duplicates are allowed. Scenario 2 needs a dictionary mapping names to marks for fast lookup. Scenario 3 needs a set because we want unique hobby names without duplicates. Each scenario matches a different data structure.

Explain (10 min)

Teacher Lists are ordered and mutable. Use them when you need to maintain insertion order and modify elements. Dictionaries map keys to values. Use them when you need fast lookup by a unique identifier. Sets store unique elements. Use them when you need to remove duplicates or perform membership testing. Tuples are immutable. Use them when data should not change. The choice depends on: whether order matters whether you need key-value lookup whether uniqueness is required and whether mutability is needed. Consider performance for large datasets as different structures have different speeds for different operations.

Students Lists for ordered mutable data. Dictionaries for key-value mapping. Sets for unique data. Tuples for fixed data. The decision factors are order mutability lookup needs and uniqueness. Understanding these helps choose the most appropriate structure for any problem.

Elaborate (10 min)

Teacher Given these KREIS scenarios choose the best data structure and justify: storing the school timetable where each period has a subject, tracking unique visitors to the library today, maintaining a shopping list for the tuck shop, recording GPS coordinates of school buildings, and implementing an undo feature in a text editor. Write the justification and a small code example for each.

Students Timetable needs a dictionary with period as key and subject as value. Library visitors need a set for uniqueness. Shopping list needs a list for order and modification. GPS coordinates need a tuple because they are fixed. Undo feature needs a stack which is a list used as LIFO. Each choice matches the data requirements.

Evaluate (5 min)

Teacher I will give you a complex problem and ask you to identify all data structures needed and justify each choice. You must explain why alternatives would be less suitable. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students The key is matching data structure properties to problem requirements. If requirements conflict consider which property is most important. Sometimes combining multiple structures is the best approach.

Cross-Curricular Connections

Mathematics: Mathematics: Data structure selection relates to mathematical modeling where you choose the right mathematical representation for a problem.
Science: Science: Scientists choose instruments based on what they need to measure. Similarly programmers choose data structures based on what they need to store.
Language: Language Arts: Choosing a data structure is like choosing a writing format. An essay a list and a table serve different purposes for different content.

Digital Citizen Reminder: Choosing the right data structure affects how efficiently your program handles user data. Efficient structures protect user privacy by processing data faster and with fewer resources.

Resources Needed:

- Python IDLE
- Textbook Chapter 2 Page 21
- KREIS data structure comparison poster

Differentiation

Approaching: Class 8 Foundation: Focus on the basic differences between lists dictionaries and sets. Use matching activities where scenarios are matched to data structures. Practice one comparison at a time.
On Level: Class 9 Application: Analyze multiple scenarios and justify data structure choices. Implement each scenario with the chosen structure. Compare performance for small datasets.
Advanced: Class 10 Mastery: Analyze complex problems requiring multiple data structures. Consider performance implications for large datasets. Design optimal data structures for real applications.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Data structure selection and justification written exercise
CT Skill Assessed: Evaluation: Analyzing problem requirements and selecting the most appropriate data structure based on multiple criteria

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: For each of your school activities identify the best data structure to represent it. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Data: Facts like numbers and words stored on a computer
Raw Data: Unprocessed facts collected before cleaning or organising properly
Information: Useful meaning produced after processing raw data carefully
Dataset: Collection of related data stored together in one place

Engage (5 min)

Teacher Students this is our final project for the data structures chapter. You will combine everything you have learned into a complete application. This project requires you to think about which data structure best fits each part of your program. Like an architect who chooses different materials for different parts of a building you will choose different structures for different needs.

Students I am ready to combine all my knowledge of data structures into one project. I need to plan which parts need lists which need dictionaries and which need sets. The planning phase is as important as the coding phase.

Explore (10 min)

Teacher Concrete (CPA): Use index cards (one card per element or key-value pair) to physically model lists/dictionaries and sorting before coding abstractly. Let us design a KREIS Student Record Management System together. We need to store student profiles using dictionaries with name class and marks. We need to track attendance using a list of dictionaries. We need to find unique subjects using a set. We need to sort students by marks using list operations. Let me outline the functions and data structures needed for each feature.

Students The student profile is a dictionary for fast lookup by name. Attendance is a list because we add records chronologically. Subjects are a set to ensure uniqueness. Sorting uses the list sort method. Each feature uses the most appropriate data structure.

Explain (10 min)

Teacher A successful project requires careful planning before coding. Identify all features your program needs. For each feature determine which data structure best fits the requirements. Design functions that use these data structures effectively. Write pseudocode for each function. Implement one function at a time and test thoroughly. Use comments to explain your design decisions. Document why you chose each data structure. Test edge cases like empty inputs and duplicate entries. A well-designed project demonstrates mastery of both programming concepts and data structure selection.

Students Planning identifies features. Data structure selection matches tools to tasks. Pseudocode guides implementation. Incremental development catches errors early. Documentation explains design reasoning. Testing ensures reliability. The project brings all data structure knowledge together.

Elaborate (10 min)

Teacher Build the complete KREIS Student Record Management System. Include functions to add student records view all records search by name sort by marks calculate class statistics and generate a report. Use dictionaries for student profiles lists for attendance records and sets for unique subjects. Save all data to a file and load it when the program starts. Include error handling for all operations. Format the output neatly.

Students My project has seven functions using three data structure types. Student records are dictionaries for fast search. Attendance is a list for chronological records. Subjects are a set for uniqueness. The program saves to file and loads on startup. Error handling catches all invalid inputs. The output is formatted in a clean table.

Evaluate (5 min)

Teacher Your project will be evaluated on: correct use of multiple data structures with justified choices, proper implementation of all required functions, effective error handling, clean formatted output, and documentation of design decisions. Present your project explaining each data structure choice. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students My project uses all three main data structures with clear justification for each choice. All functions work correctly. Error handling prevents crashes. Output is professionally formatted. My documentation explains every design decision.

Cross-Curricular Connections

Mathematics: Mathematics: Project management involves tracking metrics and statistics which use mathematical operations on data structures.
Science: Science: Research projects require data collection storage and analysis. This project models the scientific data management process.
Language: Language Arts: Project documentation develops technical writing skills. Presenting the project develops oral communication abilities.

Digital Citizen Reminder: A student record system handles sensitive personal data. Implement proper data handling practices. Never hardcode personal information in the code. Ensure that data storage and retrieval follow privacy principles.

Resources Needed:
- Python IDLE
- Textbook Chapter 2 review pages
- KREIS project planning template

Differentiation

Approaching: Class 8 Foundation: Complete a simplified project with three functions using lists and dictionaries. Follow the teacher provided template and focus on understanding how data structures work together.
On Level: Class 9 Application: Build the complete project with all required functions. Use all three data structure types. Implement file handling and error handling independently.
Advanced: Class 10 Mastery: Extend the project with additional features like advanced search and statistical analysis. Add comprehensive documentation and present the project with design rationale to the class.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Project submission with documentation and class presentation
CT Skill Assessed: Creation: Designing and implementing a complete application using appropriate data structures for each component

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Build a School Inventory Management System. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 3 - Database Fundamentals

`(..) => ..`

Key Vocabulary

Data: Facts like numbers and words stored on a computer
Raw Data: Unprocessed facts collected before cleaning or organising properly
Information: Useful meaning produced after processing raw data carefully
Dataset: Collection of related data stored together in one place

Engage (5 min)

Teacher Students imagine you have 500 student records stored in separate text files. To find a specific student you would need to open each file and search. Now imagine all records in one organized system where you can search instantly. That is what a database does. A database is like a well-organized filing cabinet where everything has a specific place and you can find anything quickly.

Students So a database is a central place to store all related data. Instead of many separate files I have one organized system. Searching and updating data would be much faster and easier.

Explore (10 min)

Teacher Let me show you a real example. Our school library has a system that tracks which student borrowed which book and when it is due. Think about your phone contacts list. Each contact has a name phone number and email. These are examples of databases. Now let me show you the difference. A file-based system stores data in separate text files. A database stores data in tables that are linked together. Try to list five examples of databases you use in your daily life.

Students My examples are phone contacts school attendance system library catalog online shopping accounts and social media profiles. All of these store data in organized ways. A database is better than files because you can search update and connect data much more efficiently.

Explain (10 min)

Teacher A database is an organized collection of structured data stored electronically in a computer system. The word database comes from data meaning facts and base meaning foundation. A database management system or DBMS is software that creates and manages databases. Common DBMS include MySQL PostgreSQL and SQLite. Databases store data in tables with rows and columns. Tables can be linked to each other using relationships. Databases provide fast searching efficient storage data integrity and concurrent access. They are used in schools hospitals banks shops and virtually every organization.

Students Database is organized data. DBMS is the software that manages it. Tables store data in rows and columns. Relationships connect tables. Databases are faster more organized and more reliable than simple files. They are everywhere in modern life.

Elaborate (10 min)

Teacher Think about KREIS school systems. Identify at least four different databases the school might use: student records attendance library inventory and exam results. For each database describe what data it stores and who uses it. Then compare how this data would be stored in plain text files versus a database. Discuss the advantages of using a database for each case.

Students Student records database stores names addresses and parent info used by office staff. Attendance database tracks daily presence used by teachers. Library database tracks books and borrowers used by librarian. Exam results database stores marks used by exam department. In text files searching would be slow and updating would risk data loss. Databases provide instant search and safe updates.

Evaluate (5 min)

Teacher I will give you a scenario and ask you to design what data a database would store. You must identify the tables needed the columns in each table and who would use the database. You must also explain why a database is better than file storage for this scenario. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students A good database design identifies all the data that needs to be stored organizes it into logical tables and considers who will access the data. The key advantage of databases over files is the ability to search and connect related data efficiently.

Cross-Curricular Connections

Mathematics: Mathematics: Databases organize numerical data for analysis. Mathematical operations on database data enable statistics and reporting.
Science: Science: Scientific databases store experimental data. Research databases like PubMed organize scientific literature for efficient retrieval.
Language: Language Arts: A library catalog database organizes books by title author and subject. Database skills help students find resources efficiently.

Digital Citizen Reminder: Databases contain sensitive personal information. Understanding how databases work helps you appreciate why data protection laws exist. Always handle database data responsibly and ethically.

Resources Needed:
- SQLite or MySQL interface
- Textbook Chapter 3 Page 1
- KREIS database examples handout

Differentiation

Approaching: Class 8 Foundation: Focus on understanding what a database is through everyday examples. Identify databases in school systems. Compare with file storage using simple examples. Teacher guides discussion.
On Level: Class 9 Application: Analyze multiple database examples and their components. Design basic database schemas for KREIS scenarios. Work independently on database identification activities.
Advanced: Class 10 Mastery: Evaluate database design decisions and understand relationships between tables. Explore how databases support organizational decision-making. Connect to SQL in upcoming lessons.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Database identification and design scenario analysis
CT Skill Assessed: Abstraction: Identifying which real-world data needs database storage and how to organize it logically

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write down all the data your school stores about you as a student. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

> **Teacher** Students a database table is like a spreadsheet but more powerful. Each row represents one complete entry like one student. Each column represents a piece of information like name or marks. Think of your class

attendance sheet. Each row is a student and each column is a day of the week. Today we learn how to create and fill database tables.

Students A database table is like a structured spreadsheet. Rows are records and columns are fields. Each field has a name and a data type that determines what kind of data it can hold.

Explore (10 min)

Teacher Let me create a simple student table. The table has columns for id name class and marks. The id column holds numbers. The name column holds text. The class column holds text. The marks column holds numbers. I will insert five records one for each of you. Watch how each record fits into a row. Try to identify which column is the primary key and why.

Students The id column looks like the primary key because each student has a unique id number. No two students can have the same id. The name column holds text values like Arjun and Priya. The marks column holds numbers. Each record fills one complete row in the table.

Explain (10 min)

Teacher A table is the basic storage unit in a relational database. Each table has columns also called fields or attributes. Each column has a name and a data type. Common data types include INTEGER for whole numbers REAL for decimal numbers TEXT for strings and DATE for dates. Each row in a table is called a record or tuple. It represents one complete entry. A primary key is a column that uniquely identifies each record. No two records can have the same primary key value. Tables are related to each other through primary and foreign keys.

Students Tables have columns and rows. Columns define the data types. Rows contain the actual data. The primary key uniquely identifies each row. Different tables connect through keys. This structure organizes data efficiently and allows relationships between different data sets.

Elaborate (10 min)

Teacher Create a student table in the database with columns for student_id name class section and address. Insert records for five KREIS students with realistic data. Then create a subjects table with columns for subject_id subject_name and teacher_name. Insert records for five subjects. Observe how each table stores a different type of information. Identify the primary key for each table.

Students My student table has student_id as the primary key with unique values for each student. The subjects table has subject_id as the primary key. Each table is independent but stores related information about our school. The student table focuses on student information and the subjects table focuses on academic subjects.

Evaluate (5 min)

Teacher I will give you a scenario and ask you to design the table structure including column names data types and which column should be the primary key. You must also insert some sample records and explain your choices. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Good table design requires choosing appropriate column names selecting correct data types and identifying a suitable primary key. The primary key should be unique and never change for a given record.

Cross-Curricular Connections

Mathematics: Mathematics: Tables organize numerical data in rows and columns just like mathematical tables. Database tables enable calculations on structured data.
Science: Science: Experimental data tables in science follow the same row-column structure. Database tables formalize this familiar format for computers.
Language: Language Arts: A database table is like a structured paragraph with each row being a complete thought and each column a specific detail.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- SQLite or MySQL interface
- Textbook Chapter 3 Page 3
- KREIS database creation worksheet

Differentiation

Approaching: Class 8 Foundation: Focus on understanding the table structure concept. Use spreadsheet analogy to understand rows and columns. Create simple tables with provided column definitions and sample data.
On Level: Class 9 Application: Design and create tables with appropriate data types. Insert records independently. Identify primary keys and explain reasoning. Work on KREIS-specific table designs.
Advanced: Class 10 Mastery: Design complex table structures with multiple data types. Understand normalization concepts for table design. Connect tables through relationships in preparation for JOIN operations.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Table creation and data insertion practical exercise
CT Skill Assessed: Decomposition: Breaking real-world data into appropriate tables with well-defined columns and data types

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Design a table to store your personal library reading records. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

SELECT: Command to fetch chosen columns from a table for viewing results
Column: Vertical field showing one type of information for all records together
Row: Horizontal record showing all information about one item completely together
Query: Question you ask database to retrieve specific information quickly for answer

Engage (5 min)

Teacher Students SQL is the language used to talk to databases. SELECT is the most common SQL command. It lets you ask the database for specific information. Think of it like asking a librarian a question. You say I want to find books about science. The librarian searches and gives you the results. SELECT works the same way. You ask the database for data and it returns the results.

Students So SQL SELECT is like asking a question to the database. I can ask for all the data or just specific columns. I can also ask for unique values only. This is much faster than searching through files manually.

Explore (10 min)

Teacher Let me show you some SELECT statements. SELECT star from students retrieves all columns and all rows. SELECT name class from students retrieves only the name and class columns. SELECT distinct class from students shows only the unique class values without repetition. Try these statements on the student table we created and observe the results.

Students The star symbol means all columns. Choosing specific columns makes the output cleaner. DISTINCT removed duplicate class names so I see Class 8 Class 9 and Class 10 only once each. SELECT is like filtering what I want to see from the complete data.

Explain (10 min)

Teacher The SELECT statement retrieves data from a database table. SELECT star selects all columns. You can specify column names to select only certain columns. SELECT DISTINCT returns only unique values eliminating duplicates. You can use column aliases with the AS keyword to rename columns in the output. Expressions like marks plus 10 can create computed columns. The FROM clause specifies which table to query. SELECT is the foundation of SQL and is used in almost every database operation.

Students SELECT retrieves data. Star means all columns. Specific column names select those columns only. DISTINCT removes duplicates. AS renames columns. Expressions create new calculated columns. FROM specifies the source table. SELECT is the most fundamental SQL command.

Elaborate (10 min)

Teacher Write SELECT statements for the following KREIS scenarios: retrieve all student names and marks, retrieve only the unique class values, create a computed column showing marks plus 5 bonus points with an alias called adjusted_marks, and retrieve the first three columns of the student table. Test each statement and verify the output matches your expectations.

Students My first query returns all names and marks. The DISTINCT query shows three unique classes. The computed column adds 5 to each mark with the alias adjusted_marks. The third query specifies three column names. Each SELECT statement answers a different question about the data.

Evaluate (5 min)

Teacher I will give you a database table and ask you to write SELECT statements that retrieve specific information. You must use correct SQL syntax including proper column names table names and any needed keywords. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Writing SELECT statements requires knowing the table structure. I need to remember column names exactly and use correct SQL keywords. Practice helps me write queries faster and more accurately.

Cross-Curricular Connections

Mathematics: Mathematics: SELECT operations correspond to mathematical functions like selecting specific elements from a set. Computed columns use arithmetic operations.
Science: Science: Scientists query databases to extract specific experimental data. SELECT is the fundamental tool for data retrieval in research.
Language: Language Arts: SELECT is like asking a specific question in a library. The more precise your question the more useful the results.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- SQLite or MySQL interface
- Textbook Chapter 3 Page 5
- KREIS SQL SELECT practice sheet

Differentiation

Approaching: Class 8 Foundation: Focus on basic SELECT star and SELECT specific columns statements. Use predefined tables and practice writing simple queries with teacher guidance.
On Level: Class 9 Application: Write SELECT statements with DISTINCT and column aliases. Create computed columns. Write queries independently for various KREIS scenarios.
Advanced: Class 10 Mastery: Write complex SELECT statements with expressions and aliases. Combine with WHERE clause from next lesson. Understand how SELECT forms the basis of all SQL queries.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: SQL query writing exercise with specific retrieval requirements
CT Skill Assessed: Pattern Recognition: Translating data retrieval needs into correct SQL SELECT syntax

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write SELECT statements for a database of your classmates. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

WHERE: Clause filtering rows to show only those meeting a condition for query
Condition: Rule deciding which rows appear in result like age greater than 15
Filter: Keeping only matching rows and hiding others that do not satisfy condition
Operator: Symbol like equals or greater than used inside condition for comparison

Engage (5 min)

Teacher Students SELECT alone retrieves all records. But what if you only want students who scored above 80. The WHERE clause lets you filter records based on conditions. Think of it like this: SELECT is like opening a book and WHERE is like turning to a specific page. You only see what matches your criteria.

Students So WHERE filters the results. I can ask for students above a certain marks or students in a specific class. The database returns only the records that match my condition.

Explore (10 min)

Teacher Let me show you WHERE in action. SELECT star from students where marks greater than 80 returns only students with marks above 80. SELECT star from students where class equals Class 9 returns only Class 9 students. SELECT star from students where marks greater than 70 and class equals Class 8 returns Class 8 students with marks above 70. Try writing your own WHERE conditions.

Students WHERE with greater than 80 filtered out students below 80. WHERE with equals found students in a specific class. AND combined two conditions so both must be true. I can also use OR to match either condition. WHERE makes SELECT much more powerful.

Explain (10 min)

Teacher The WHERE clause filters records based on conditions. It follows the SELECT and FROM clauses. Comparison operators include equals not equals greater than less than greater than or equal and less than or equal. AND requires all conditions to be true. OR requires at least one condition to be true. NOT negates a condition. Conditions can use numbers text and dates. WHERE is essential for extracting specific information from large databases.

Students WHERE filters rows based on conditions. Comparison operators test values. AND needs all conditions true. OR needs one condition true. NOT reverses a condition. WHERE narrows down results to exactly what you need.

Elaborate (10 min)

Teacher Write WHERE queries for these KREIS scenarios: find all students who scored above 90, find all Class 10 students, find students who scored below 50 in Mathematics, find students who are in Class 9 OR scored above 85, and find students who are NOT in Class 8. Test each query and verify the results make sense.

Students My queries return specific subsets of the student data. The above 90 query finds high achievers. The Class 10 query finds senior students. The below 50 query identifies struggling students. The OR query finds either Class 9 or high scorers. The NOT query excludes Class 8 students.

Evaluate (5 min)

Teacher I will give you a filtering requirement in plain English and ask you to write the correct SQL WHERE clause. You must also predict how many records will match before running the query. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Translating English conditions to SQL requires understanding comparison operators and logical connectives. AND narrows results while OR widens them. NOT reverses the condition.

Cross-Curricular Connections

Mathematics: Mathematics: WHERE conditions use comparison operators from mathematics. AND and OR correspond to logical AND and OR in Boolean algebra.
Science: Science: Filtering experimental data by conditions is fundamental in scientific analysis. WHERE provides this filtering capability.
Language: Language Arts: WHERE is like applying criteria to a search. You specify exactly what you are looking for just like searching a library catalog with specific filters.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:

- SQLite or MySQL interface
- Textbook Chapter 3 Page 7
- KREIS SQL WHERE practice queries

Differentiation

Approaching: Class 8 Foundation: Focus on simple WHERE conditions with single comparisons like equals and greater than. Use provided tables and practice basic filtering with teacher guidance.
On Level: Class 9 Application: Write WHERE queries with AND and OR operators. Combine multiple conditions to filter complex scenarios. Work independently on various filtering tasks.
Advanced: Class 10 Mastery: Write complex WHERE queries with multiple AND and OR conditions. Use NOT to negate conditions. Understand operator precedence in complex WHERE clauses.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: SQL WHERE clause writing with English to SQL translation
CT Skill Assessed: Algorithmic Thinking: Translating natural language filtering requirements into SQL WHERE syntax

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write WHERE queries to find students from your dormitory who scored above 75, students who are in Class 10, and students who are NOT in your class. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

Teacher Students our school results are displayed in order from highest to lowest marks. SQL has a similar capability. The ORDER BY clause sorts your query results. Think of it like arranging books on a shelf. You can arrange them alphabetically by title or by author name or by publication year. ORDER BY lets you choose how to arrange your data.

Students So ORDER BY sorts the results. I can sort marks from high to low to find the class topper. I can sort names alphabetically for a neat display. I can even sort by multiple columns if needed.

Explore (10 min)

Teacher Let me show you ORDER BY. SELECT star from students order by marks descending shows students from highest marks to lowest. Adding ascending sorts from lowest to highest which is the default. ORDER BY class ascending marks descending first sorts by class then within each class sorts by marks from high to low. Try different sorting combinations and observe how the order changes.

Students Descending order shows highest marks first which is useful for finding toppers. Sorting by class then marks organizes students by class with each class showing highest marks first. Multiple sort columns handle cases where the first column has duplicate values.

Explain (10 min)

Teacher The ORDER BY clause sorts the result set. By default it sorts in ascending order. Use DESC for descending order and ASC for ascending order explicitly. You can sort by multiple columns separated by commas. The first column is the primary sort. If values tie the second column breaks the tie. ORDER BY comes after WHERE in the SQL statement. It sorts the filtered results if WHERE is used. ORDER BY is essential for presenting data in a meaningful order.

Students ORDER BY sorts results. DESC is descending. ASC is ascending which is default. Multiple columns create hierarchical sorting. ORDER BY comes after WHERE. Sorting organizes data for meaningful presentation and analysis.

Elaborate (10 min)

Teacher Write ORDER BY queries for these KREIS scenarios: display students sorted by marks from highest to lowest to find the class topper, display students sorted alphabetically by name, display students sorted by class then by marks within each class, and display only Class 9 students sorted by marks descending. Verify each result is correctly sorted.

Students My first query shows Arjun with 98 marks at the top. The alphabetical sort arranges names from A to Z. The multi-column sort groups by class then sorts within each class. The filtered and sorted query shows only Class 9 students in marks order. ORDER BY makes data presentation meaningful.

Evaluate (5 min)

Teacher I will give you a requirement like find the top 5 students and ask you to write the SQL query using ORDER BY. You must determine the correct sort order and any additional clauses needed. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students ORDER BY with DESC finds highest values. Combined with LIMIT it gives top results. Understanding sort order is essential for answering ranking questions.

Cross-Curricular Connections

Mathematics: Mathematics: ORDER BY sorts data like ordering numbers on a number line. Sorting is fundamental to finding median and quartiles.
Science: Science: Experimental results often need sorting to identify trends and outliers. ORDER BY provides this capability for database data.
Language: Language Arts: ORDER BY is like organizing a bibliography alphabetically or chronologically. Proper ordering makes references easy to find.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- SQLite or MySQL interface
- Textbook Chapter 3 Page 9
- KREIS SQL ORDER BY practice

Differentiation

Approaching: Class 8 Foundation: Focus on single column ORDER BY with ascending and descending order. Use simple sorting scenarios with teacher guided examples.
On Level: Class 9 Application: Write ORDER BY queries with multiple columns. Combine ORDER BY with WHERE for filtered sorted results. Work independently on various sorting tasks.
Advanced: Class 10 Mastery: Use ORDER BY in complex queries combining SELECT WHERE and ORDER BY. Understand how sorting affects query performance. Use ORDER BY to answer analytical questions about data.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: SQL ORDER BY query writing with sorting requirements
CT Skill Assessed: Pattern Recognition: Identifying the correct sort order and columns needed to answer specific data questions

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write ORDER BY queries to display your classmates sorted by marks from highest to lowest, by name alphabetically, and by roll number. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

Teacher Students when the teacher calculates class average marks or counts how many students passed the exam they are doing aggregate calculations. SQL has built-in functions for these tasks. COUNT counts records SUM adds up values AVG calculates averages MAX finds the highest and MIN finds the lowest. These functions process many records at once.

Students So I can calculate the class average total marks and highest score using simple SQL functions. COUNT tells me how many students. These functions save me from calculating manually.

Explore (10 min)

Teacher Let me demonstrate aggregate functions. SELECT count star from students counts all students. SELECT count star from students where marks greater than 80 counts students above 80. SELECT sum marks from students gives total marks. SELECT avg marks gives the average. SELECT max marks gives the highest and min marks gives the lowest. Try each function and verify the results manually.

Students COUNT gave 25 which is the total number of students. COUNT with WHERE gave 12 students above 80. SUM of all marks is 1875. Average is 75. Maximum is 98 and minimum is 35. These functions instantly calculate statistics that would take time to compute manually.

Explain (10 min)

Teacher Aggregate functions perform calculations on sets of records. COUNT returns the number of rows. COUNT star counts all rows. COUNT column counts non-null values in that column. SUM adds all numeric values. AVG calculates the arithmetic mean. MAX returns the highest value. MIN returns the lowest value. GROUP BY groups records by a column value before applying aggregate functions. HAVING filters groups after aggregation. These functions are essential for data analysis and reporting.

Students Aggregate functions summarize data. COUNT counts SUM adds AVG averages MAX finds highest and MIN finds lowest. GROUP BY creates groups for per-group aggregation. HAVING filters groups. Together they provide powerful data analysis capabilities.

Elaborate (10 min)

Teacher Write aggregate queries for KREIS scenarios: count total students in each class, find the average marks for each class, find the highest marks in each subject, count how many students scored above 80 in each class using GROUP BY and HAVING, and find the difference between highest and lowest marks in each class.

Students COUNT with GROUP BY shows Class 8 has 8 students Class 9 has 10 and Class 10 has 7. AVG by class shows Class 10 has the highest average. MAX by subject shows Mathematics has the highest individual score. HAVING filters classes with more than 2 students above 80. The range shows spread of marks in each class.

Evaluate (5 min)

Teacher I will give you a data analysis question like which class has the highest average and ask you to write the SQL query. You must choose the correct aggregate function and any needed GROUP BY or HAVING clauses. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Data analysis questions translate directly to aggregate function queries. The key is identifying which function answers the question and whether grouping is needed.

Cross-Curricular Connections

Mathematics: Mathematics: Aggregate functions are mathematical operations applied to datasets. SUM is addition AVG is mean MAX is maximum and MIN is minimum from statistics.
Science: Science: Data analysis in science uses aggregate functions to summarize experimental results. Averages and ranges are fundamental statistics.
Language: Language Arts: Aggregate functions summarize text data. COUNT can count word frequencies. These functions support data-driven writing.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- SQLite or MySQL interface
- Textbook Chapter 3 Page 11
- KREIS SQL aggregate functions guide

Differentiation

Approaching: Class 8 Foundation: Focus on basic aggregate functions COUNT SUM AVG MAX and MIN without GROUP BY. Use simple calculations on the entire table and verify results manually.
On Level: Class 9 Application: Use GROUP BY to aggregate by categories. Apply HAVING to filter groups. Write aggregate queries for various KREIS reporting scenarios independently.
Advanced: Class 10 Mastery: Write complex aggregate queries combining GROUP BY HAVING and ORDER BY. Use multiple aggregate functions in one query. Understand when to use each function for data analysis.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Aggregate function query writing with data analysis questions
CT Skill Assessed: Evaluation: Selecting the appropriate aggregate function to answer specific data analysis questions

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write aggregate queries to count how many books you have read this month, find your average rating, find your highest rated book, and find your lowest rated book. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

Teacher Students imagine you have one table with student names and another with their marks. To see a student name with their marks you need to connect the two tables. SQL JOIN does exactly this. Think of it like this: student table is like the left hand and marks table is like the right hand. JOIN brings both hands together to give a complete picture.

Students So JOIN connects related tables. I can combine student information with their marks even though they are stored in separate tables. This is why databases split data into multiple tables and connect them with keys.

Explore (10 min)

Teacher Let me show you INNER JOIN. I have a students table with id and name and a marks table with student_id and marks. SELECT students dot name marks dot marks from students INNER JOIN marks on students dot id equals marks dot student_id. This combines name and marks from both tables. Try a LEFT JOIN and see what additional records appear.

Students INNER JOIN showed only students who have marks records. LEFT JOIN showed all students including those without marks. The ON clause specifies how the tables are connected. The student_id in marks table links to id in students table.

Explain (10 min)

Teacher JOIN combines rows from two or more tables based on a related column. INNER JOIN returns only rows with matching values in both tables. LEFT JOIN returns all rows from the left table and matching rows from the right table. Non-matching right rows show NULL. RIGHT JOIN returns all rows from the right table and matching left rows. FULL JOIN returns all rows from both tables. The ON clause specifies the join condition using related columns. JOINs are essential for querying relational databases where data is split across tables.

Students INNER JOIN finds matches. LEFT JOIN keeps all left rows. RIGHT JOIN keeps all right rows. FULL JOIN keeps everything. ON specifies the matching condition. JOINs connect split data back together for complete queries.

Elaborate (10 min)

Teacher Create a students table and an enrollments table. The enrollments table has student_id and subject_id columns. Write an INNER JOIN to show which students are enrolled in which subjects. Write a LEFT JOIN to show all students even those not yet enrolled. Write a query to find students enrolled in Mathematics specifically.

Students My INNER JOIN shows 20 enrollment records connecting students to subjects. LEFT JOIN shows 25 students with 5 showing NULL for unenrolled students. The Mathematics filter shows 8 students enrolled in that subject. JOINs reveal relationships between students and their academic choices.

Evaluate (5 min)

Teacher I will give you two related tables and ask you to write JOIN queries that combine specific columns. You must choose the correct JOIN type based on whether you need only matches or all records from one table. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students INNER JOIN is for when both tables must have data. LEFT JOIN is for when you need all records from the primary table. The choice depends on what information you need to display.

Cross-Curricular Connections

Mathematics: Mathematics: JOINs relate to set operations. INNER JOIN is like intersection. LEFT JOIN preserves the entire left set while adding matching elements from the right set.
Science: Science: Scientific databases link experiments to results through JOIN operations. Connecting samples to measurements requires JOIN.
Language: Language Arts: A JOIN is like cross-referencing an index. You look up an entry and follow the reference to find related information in another section.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:

- SQLite or MySQL interface
- Textbook Chapter 3 Page 13
- KREIS SQL JOIN practice

Differentiation

Approaching: Class 8 Foundation: Focus on INNER JOIN with simple two-table relationships. Use visual diagrams to understand how tables connect. Practice with teacher guided step-by-step queries.
On Level: Class 9 Application: Write INNER JOIN and LEFT JOIN queries independently. Understand when to use each type. Work on KREIS-specific scenarios with multiple tables.
Advanced: Class 10 Mastery: Write complex JOIN queries with multiple tables. Combine JOIN with WHERE ORDER BY and aggregate functions. Understand JOIN performance implications.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: SQL JOIN query writing with table relationship diagrams
CT Skill Assessed: Abstraction: Understanding how separate tables represent different entities that need to be connected for complete information retrieval

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a database with a students table and a grades table. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Data: Facts like numbers and words stored on a computer
Raw Data: Unprocessed facts collected before cleaning or organising properly
Information: Useful meaning produced after processing raw data carefully
Dataset: Collection of related data stored together in one place

Engage (5 min)

Teacher Students before building a house an architect creates a blueprint. Similarly before creating a database you need a design plan. Database design determines what tables you need what columns each table has and how tables relate to each other. Good design prevents problems like duplicate data and inconsistent information.

Students Database design is the planning phase. I need to identify what data to store organize it into tables and plan how tables connect. Good design saves time and prevents data problems later.

Explore (10 min)

Teacher Let me design a KREIS library database. First identify the main entities: books students and loans. Books have title author and availability. Students have name class and contact. Loans connect a student to a book with borrow and return dates. Now create tables for each entity with appropriate columns. Draw an Entity-Relationship diagram showing how the tables connect.

Students My library database has three tables. The books table stores book details with book_id as primary key. The students table stores student info with student_id as primary key. The loans table connects them using both IDs as foreign keys. The ER diagram shows rectangles for tables and lines for relationships.

Explain (10 min)

Teacher Database design follows a systematic process. First identify the entities which are the main objects to store data about. Then identify the attributes which are the properties of each entity. Next normalize the data to eliminate redundancy. Then create the schema defining tables columns and data types. Finally establish relationships between tables using primary and foreign keys. An Entity-Relationship diagram visualizes the design with rectangles for entities ovals for attributes and diamonds for relationships. Good design ensures data integrity and efficient queries.

Students Design steps are identify entities determine attributes normalize data create schema and establish relationships. ER diagrams visualize the design. Good design prevents redundancy and ensures data integrity. Planning saves time during implementation.

Elaborate (10 min)

Teacher Design a KREIS examination database. Identify entities like students subjects exams and results. Determine attributes for each. Create the database schema with appropriate tables and columns. Draw an ER diagram showing relationships. Implement the design by creating tables in the database and inserting sample data. Test your design by writing queries that answer common questions.

Students My exam database has four tables. Students store student info. Subjects store subject details. Exams store exam information. Results connect students exams and subjects with marks. The ER diagram shows all relationships clearly. My queries successfully retrieve student results and class statistics.

Evaluate (5 min)

Teacher I will give you a real-world scenario and ask you to design a complete database. You must identify entities attributes and relationships create an ER diagram and implement the schema. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Good database design requires understanding the problem thoroughly. Entities come from nouns in the description. Attributes come from properties. Relationships come from how entities interact.

Cross-Curricular Connections

Mathematics: Mathematics: Database design uses logical thinking similar to mathematical proof construction. Each design decision must be justified logically.

Science: Science: Scientific database design must consider how experiments generate data. The schema must capture the experimental process.
Language: Language Arts: Database design is like outlining an essay. You plan the structure before writing. Good planning produces better results.

Digital Citizen Reminder: Database design must consider privacy from the start. Only store necessary data. Design access controls into the schema. Consider data retention policies during the design phase.

Resources Needed:
- SQLite or MySQL interface
- Textbook Chapter 3 Page 15
- KREIS database design template

Differentiation

Approaching: Class 8 Foundation: Design simple two-table databases with basic relationships. Use provided scenarios and follow the design steps with teacher guidance. Create simple ER diagrams.
On Level: Class 9 Application: Design multi-table databases independently. Create ER diagrams and implement schemas. Test designs with queries to verify they work correctly.
Advanced: Class 10 Mastery: Design complex normalized databases with multiple relationships. Consider performance and scalability. Analyze existing database designs and suggest improvements.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Database design project with ER diagram and implementation
CT Skill Assessed: Decomposition: Breaking a complex information need into organized tables with clear relationships

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Design a database for your personal study planner. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations

Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

Teacher Students every student in KREIS has a unique roll number. No two students share the same roll number. This roll number is like a primary key in a database. Now when the library records which student borrowed which book it uses the roll number to identify the student. The roll number in the library table is a foreign key because it refers to the roll number in the student table. Keys connect tables together.

Students So a primary key uniquely identifies each record. A foreign key connects one table to another by referencing the primary key. This is how databases link related data across different tables.

Explore (10 min)

Teacher Let me show you keys in action. The students table has student_id as the primary key. Each student has a unique student_id. The enrollments table has student_id and subject_id. Both are foreign keys. student_id refers to students table and subject_id refers to subjects table. Try inserting a record with a duplicate primary key and see the error. Then insert with a foreign key that does not exist in the referenced table.

Students The duplicate primary key was rejected because each student_id must be unique. The foreign key error happened because I tried to reference a student_id that does not exist in the students table. Keys enforce data integrity by preventing invalid references.

Explain (10 min)

Teacher A primary key uniquely identifies each record in a table. It must be unique and cannot be NULL. A table can have only one primary key. A foreign key is a column that references the primary key of another table. It creates a link between the two tables. Foreign keys can have duplicate values and can be NULL. Keys enforce referential integrity meaning you cannot reference non-existent records. They also enable JOIN operations to combine related data. Keys are the foundation of relational databases.

Students Primary key is unique identifier for each row. Foreign key links to another table primary key. Keys enforce data integrity. They prevent orphan records and enable table relationships. Without keys databases would be disconnected collections of data.

Elaborate (10 min)

Teacher Create a school database with a teachers table using teacher_id as primary key and a classes table using class_id as primary key. Create a teaches table linking teachers to classes using both IDs as foreign keys. Insert sample data. Test that you cannot insert a foreign key that references a non-existent teacher. Write a JOIN query to show which teacher teaches which class.

Students My teachers table has three teachers with unique teacher_id. My classes table has five classes. The teaches table links them with foreign keys. I tried inserting a teaches record with teacher_id 99 which does not exist and got a foreign key error. The JOIN query shows Mr. Sharma teaches Mathematics to Class 10.

Evaluate (5 min)

Teacher I will give you a scenario and ask you to identify the primary key for each table and the foreign keys needed to connect them. You must also explain why each key choice is appropriate. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Primary keys should be unique identifiers that never change. Foreign keys should reference appropriate primary keys. The key choices determine how tables can be connected and queried.

Cross-Curricular Connections

Mathematics: Mathematics: Primary keys relate to unique elements in a set. Foreign keys model mathematical functions mapping between sets.
Science: Science: Primary keys are like unique sample identifiers in laboratory notebooks. Foreign keys connect samples to results.
Language: Language Arts: A primary key is like a unique page number in a book. A foreign key is like a citation referencing another page.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- SQLite or MySQL interface
- Textbook Chapter 3 Page 17
- KREIS keys and relationships diagram

Differentiation

Approaching: Class 8 Foundation: Focus on understanding primary key uniqueness concept. Identify primary keys in provided tables. Understand foreign key as a reference with visual diagrams.
On Level: Class 9 Application: Create tables with appropriate primary and foreign keys. Insert data and test referential integrity. Write queries using JOINs with foreign keys.
Advanced: Class 10 Mastery: Design databases with complex key relationships. Understand composite primary keys and many-to-many relationships. Analyze key design choices for efficiency.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Key identification and creation exercise with relationship testing
CT Skill Assessed: Pattern Recognition: Identifying how entities relate to each other through primary and foreign key connections

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a database with a books table and an authors table. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Data: Facts like numbers and words stored on a computer
Raw Data: Unprocessed facts collected before cleaning or organising properly
Information: Useful meaning produced after processing raw data carefully
Dataset: Collection of related data stored together in one place

Engage (5 min)

Teacher Students imagine a table that stores student name class teacher name and teacher phone for every student. If ten students have the same teacher the teacher name and phone are repeated ten times. This wastes space and causes problems if the teacher phone changes. Normalization fixes this by splitting data into separate tables. It is like organizing a messy room into separate drawers.

Students Normalization removes redundant data. Instead of repeating teacher information for every student I store it once in a teachers table and reference it with a foreign key. This saves space and prevents inconsistencies.

Explore (10 min)

Teacher Let me show you a denormalized table with columns student_name class teacher_name teacher_phone subject and marks. Notice how teacher_name and teacher_phone repeat for every student in the same class. Now normalize it by creating a students table a teachers table and a marks table. Compare the before and after. Count how many duplicate values were removed.

Students The original table had 25 rows with teacher info repeated 25 times. After normalization the teachers table has only 3 rows one for each teacher. The students table has 25 rows with just student data. The marks table connects them. We removed 22 duplicate teacher entries saving significant space.

Explain (10 min)

Teacher Normalization is the process of organizing a database to reduce data redundancy and improve data integrity. Redundancy means storing the same data in multiple places. It wastes space and causes update anomalies where changing data in one place but not another creates inconsistency. First Normal Form requires each column to contain atomic values. Second Normal Form requires non-key columns to depend on the entire primary key. Third Normal Form requires non-key columns to depend only on the primary key not on other non-key columns. Normalization splits large tables into smaller focused tables connected by keys.

Students Normalization eliminates redundancy and prevents anomalies. 1NF requires atomic values. 2NF removes partial dependencies. 3NF removes transitive dependencies. The result is smaller focused tables connected by keys. This saves space and improves data integrity.

Elaborate (10 min)

Teacher Take a denormalized KREIS table containing student name class subject teacher name and teacher qualification. Normalize it into separate tables. Identify the primary keys for each new table. Draw the relationships. Insert sample data into the normalized tables. Verify that queries on the normalized tables produce the same results as the original table.

Students My normalized design has three tables. Students with student_id and class. Subjects with subject_id and teacher_id. Teachers with teacher_id and qualification. Queries using JOINs produce identical results to the original table but with no duplicate data. The design is cleaner and more maintainable.

Evaluate (5 min)

Teacher I will give you a table with redundant data and ask you to normalize it. You must identify the redundancy create appropriate tables and explain how normalization improves the design. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Normalization requires identifying what data repeats and what depends on what. The key insight is that repeating groups of data should be in separate tables connected by foreign keys.

Cross-Curricular Connections

Mathematics: Mathematics: Normalization relates to set theory where data is partitioned into disjoint sets with minimal redundancy.
Science: Science: Scientific data normalization ensures that measurements are recorded without unnecessary repetition improving accuracy.
Language: Language Arts: Normalization is like editing a draft to remove repeated information. The final version is cleaner and more efficient.

Digital Citizen Reminder: Normalization reduces data exposure by storing sensitive information in fewer places. This improves security because there are fewer copies to protect.

Resources Needed:
- SQLite or MySQL interface
- Textbook Chapter 3 Page 19
- KREIS normalization examples

Differentiation

Approaching: Class 8 Foundation: Focus on identifying redundancy in tables. Understand why repeating data is problematic. Practice splitting one table into two with teacher guidance.
On Level: Class 9 Application: Normalize tables independently applying basic normalization rules. Create normalized schemas and verify they produce equivalent query results.
Advanced: Class 10 Mastery: Apply 1NF 2NF and 3NF systematically. Analyze complex tables for normalization needs. Understand when denormalization might be acceptable for performance.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Table normalization exercise with redundancy identification
CT Skill Assessed: Evaluation: Analyzing table structures to identify redundancy and determining the best normalization approach

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Look at a school timetable and identify any repeated data. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

> **Teacher** Students CRUD stands for Create Read Update and Delete. These are the four fundamental operations for managing data in any database. Think of it like managing your diary. Create means writing a new entry. Read means reviewing what you wrote. Update means changing an entry. Delete means removing an entry. Every database application uses these four operations.

> **Students** CRUD covers all data management tasks. INSERT creates new records. SELECT reads data. UPDATE modifies records. DELETE removes records. These four operations handle everything I need to do with data.

Explore (10 min)

> **Teacher** Let me demonstrate all four CRUD operations. CREATE: INSERT INTO students name class marks VALUES Arjun Class 10 95. READ: SELECT star from students. UPDATE: UPDATE students set marks equals 98 where name equals Arjun. DELETE: DELETE from students where name equals Arjun. Try each operation and verify the results.

> **Students** INSERT added a new student record. SELECT displayed all records. UPDATE changed Arjun marks from 95 to 98. DELETE removed the record. Each operation affects the table differently. WHERE is important in UPDATE and DELETE to target specific records.

Explain (10 min)

> **Teacher** CRUD represents the four basic database operations. CREATE uses INSERT INTO to add new records. Specify the table columns and values to insert. READ uses SELECT to retrieve data. Use WHERE to filter specific records. UPDATE modifies existing records using UPDATE table SET column equals value WHERE condition.

DELETE removes records using DELETE FROM table WHERE condition. Always use WHERE in UPDATE and DELETE to avoid affecting all records. These operations are the foundation of all database applications.

Students INSERT adds new data. SELECT retrieves data. UPDATE modifies existing data. DELETE removes data. WHERE clause targets specific records in UPDATE and DELETE. Without WHERE all records are affected. CRUD operations manage all database interactions.

Elaborate (10 min)

Teacher Build a complete CRUD application for KREIS student records. Implement functions to add a new student read all students update a student marks and delete a student record. Include error handling for each operation. Test all four operations and verify the database state after each change.

Students My CRUD application has four functions. add_student inserts a new record. view_students displays all records. update_marks modifies a specific student. delete_student removes a record. Each function includes error handling. After all operations the database contains the correct records.

Evaluate (5 min)

Teacher I will give you a database scenario and ask you to write CRUD operations for specific requirements. You must use correct SQL syntax and include WHERE clauses where needed. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students CRUD operations are straightforward but require attention to syntax and WHERE clauses. Always verify the number of records affected by UPDATE and DELETE operations.

Cross-Curricular Connections

Mathematics: Mathematics: CRUD operations correspond to set operations. INSERT adds elements. SELECT retrieves subsets. UPDATE modifies elements. DELETE removes elements.
Science: Science: Data collection is CREATE. Data retrieval is READ. Data correction is UPDATE. Data removal is DELETE. The scientific method uses all four operations.
Language: Language Arts: CRUD mirrors the writing process. Create is drafting. Read is reviewing. Update is editing. Delete is revising.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- SQLite or MySQL interface
- Textbook Chapter 3 Page 21
- KREIS CRUD operations reference

Differentiation

Approaching: Class 8 Foundation: Focus on basic INSERT and SELECT operations. Practice adding and viewing records with simple syntax. Understand what each CRUD operation does conceptually.
On Level: Class 9 Application: Implement all four CRUD operations. Write UPDATE and DELETE with WHERE clauses. Build a simple menu-driven CRUD application independently.
Advanced: Class 10 Mastery: Implement complete CRUD applications with error handling and validation. Combine CRUD with JOINs for complex operations. Understand transaction concepts.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: CRUD operation implementation with verification
CT Skill Assessed: Algorithmic Thinking: Planning and executing the correct sequence of CRUD operations to achieve a desired database state

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a CRUD application for managing your personal contact list. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Data: Facts like numbers and words stored on a computer
Raw Data: Unprocessed facts collected before cleaning or organising properly
Information: Useful meaning produced after processing raw data carefully
Dataset: Collection of related data stored together in one place

Engage (5 min)

Teacher Students this is the final project for our database chapter. You will apply everything you learned about databases from design to implementation to querying. This project requires you to think like a database administrator who designs the schema creates the tables populates them with data and writes queries to extract useful information.

Students I am ready to combine all my database knowledge into one complete project. I need to plan the database design implement it with SQL and write queries to answer real questions about the data.

Explore (10 min)

Teacher Let us plan a KREIS Student Information System together. First identify the main data needs: student profiles attendance records exam results and extracurricular activities. Design tables for each. Consider relationships between tables. Plan the queries needed to answer common questions like who is the class topper and what is the attendance percentage.

Students My system needs four main tables. Students for basic info. Attendance for daily records. Results for exam marks. Activities for clubs and sports. The queries need JOINs to connect these tables. Aggregate functions will answer statistical questions.

Explain (10 min)

Teacher A database project follows the complete database lifecycle. Start with requirements analysis to understand what data to store. Design the schema including tables columns keys and relationships. Implement by creating tables with SQL DDL statements. Populate with sample data using INSERT statements. Query the database to extract information. Test thoroughly to ensure correctness. Document the design and queries for future maintenance. The project demonstrates mastery of all database concepts.

Students Project lifecycle is design implement populate query test and document. Each phase builds on the previous one. Good design makes implementation easier. Thorough testing ensures correctness. Documentation helps maintenance and future development.

Elaborate (10 min)

Teacher Build the complete KREIS Student Information System. Create at least four related tables with appropriate primary and foreign keys. Insert at least 20 records across all tables. Write queries using SELECT WHERE JOIN GROUP BY and ORDER BY to answer ten different questions about the data. Include questions about student performance attendance and activity participation.

Students My system has four tables with 25 student records 50 attendance records and 30 result records. My queries find class averages attendance percentages top performers in each subject and students who participate in activities. The database provides complete information about our school.

Evaluate (5 min)

Teacher Your project will be evaluated on: database design quality with appropriate normalization, correct SQL syntax for all operations, comprehensive queries answering multiple questions, proper use of keys and relationships, and clear documentation of design decisions. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students My project meets all criteria. The database is well normalized. All SQL operations work correctly. My queries answer diverse questions about the data. Keys properly connect the tables. My documentation explains every design decision.

Cross-Curricular Connections

Mathematics: Mathematics: Database projects involve statistical queries using aggregate functions. Mathematical analysis of data supports decision making.
Science: Science: Scientific databases follow the same lifecycle of design implementation and analysis. This project models scientific data management.
Language: Language Arts: Project documentation develops technical writing skills. Presenting the project develops communication abilities.

Digital Citizen Reminder: A student information system contains sensitive data. Implement proper access controls. Never share database credentials. Ensure data backup procedures are in place.

Resources Needed:
- SQLite or MySQL interface
- Textbook Chapter 3 review pages
- KREIS project requirements document

Differentiation

Approaching: Class 8 Foundation: Create a simplified two-table database with basic queries. Follow the teacher provided template and focus on understanding the database lifecycle.
On Level: Class 9 Application: Build the complete four-table database independently. Write diverse queries combining multiple SQL concepts. Test and verify all operations.
Advanced: Class 10 Mastery: Extend the database with additional tables and complex queries. Add data validation and error handling. Present the project with comprehensive documentation.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Complete database project with design documentation and presentation
CT Skill Assessed: Creation: Designing and implementing a complete database system from requirements to queries

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Build a Library Management System database. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 4 - Web Technologies

`( . . ) => . .`

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

Teacher Students when you type a website address in your browser and press Enter something amazing happens in seconds. Your browser sends a request to a computer called a server far away and the server sends back the web page. Today we learn how this client-server communication works. Think of it like ordering food by phone. You are the client calling the restaurant the server. You place an order the request and they deliver food the response.

Students So my browser is the client and the website is stored on a server. When I visit a website my browser asks the server for the page and the server sends it back. This happens every time I visit any website.

Explore (10 min)

Teacher Let me trace what happens when you visit kreis.edu. First your browser needs the servers address. It asks DNS which is like a phone book for websites. DNS translates kreis.edu to an IP address like 192.168.1.1. Then your browser sends an HTTP request to that address. The server receives the request and sends back an HTTP response containing the HTML code. Your browser reads the HTML and displays the page. Try to draw this process as a flowchart.

Students I drew the flowchart: browser asks DNS for IP address DNS returns the address browser sends HTTP request to server server processes request server sends HTTP response with HTML browser renders the HTML and displays the page. The whole process takes less than a second.

Explain (10 min)

Teacher The web operates on a client-server model. The client is your web browser like Chrome or Firefox. The server is a computer that stores websites. Communication happens through HTTP or HTTPS which is the language of the web. A URL is the address of a web resource. It contains the protocol domain name and path. DNS translates domain names to IP addresses. The browser sends an HTTP request to the server. The server processes the request and sends an HTTP response containing the web page content. The browser then renders the content for you to see.

Students Client-server model means browsers request and servers respond. HTTP is the communication language. URLs are web addresses. DNS translates names to numbers. Requests go out and responses come back. This cycle happens for every web page you visit.

Elaborate (10 min)

Teacher Create a diagram showing the complete web communication process including DNS resolution HTTP request and response and page rendering. Label each component and describe what happens at each step. Then research and explain the difference between HTTP and HTTPS. Why should you always look for HTTPS when entering personal information online.

Students My diagram shows six steps from typing the URL to seeing the page. DNS is step one translating the name. HTTP request is step two. Server processing is step three. HTTP response is step four. Browser rendering is step five. Display is step six. HTTPS adds encryption making the communication secure.

Evaluate (5 min)

Teacher I will ask you to explain the complete process of visiting a website from typing the URL to seeing the page. You must identify each component and its role in the process. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Understanding the web process requires knowing each component. DNS translates names. HTTP carries requests and responses. Servers store and serve content. Browsers render and display pages.

Cross-Curricular Connections

Mathematics: Mathematics: IP addresses use numerical notation. DNS involves mathematical mapping between names and numbers.
Science: Science: Network communication follows protocols similar to scientific measurement standards. Consistency enables global communication.

Language: Language Arts: HTTP is like a postal system with addresses envelopes containing messages and delivery confirmation.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Web browser
- Textbook Chapter 4 Page 1
- KREIS web communication diagram handout

Differentiation

Approaching: Class 8 Foundation: Focus on the basic client-server concept using the restaurant analogy. Understand that browsers request and servers respond. Trace simple web visits with teacher guidance.
On Level: Class 9 Application: Understand DNS HTTP and the request-response cycle in detail. Create diagrams showing the complete process. Research HTTPS security benefits independently.
Advanced: Class 10 Mastery: Analyze the complete web communication process including HTTP methods headers and status codes. Understand how caching and proxies optimize web delivery.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Web communication process diagram and explanation
CT Skill Assessed: Abstraction: Understanding the layered communication model that enables global web access

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Draw a diagram showing how a web page travels from the server to your computer. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

HTML: Language using tags to build structure of web pages for browsers
Tag: Word inside angle brackets marking start and end of element like <p>
Element: Complete part made of opening tag content and closing tag together
Attribute: Extra information inside tag like colour or link address for control

Engage (5 min)

Teacher Students every website you visit is built using HTML which stands for HyperText Markup Language. HTML is the skeleton of a web page. Think of it like this: if a web page were a human body HTML would be the bones that give it structure. CSS would be the skin and clothes that make it look good. JavaScript would be the muscles that make it move. Today we learn HTML which provides the structure.

Students So HTML is the foundation of every website. It defines what content appears on the page like headings paragraphs and images. Without HTML there would be no web pages to see.

Explore (10 min)

Teacher Let me create your first HTML page. I will write a simple document with a heading and a paragraph. The document starts with doctype html then html tags. Inside html we have head and body sections. The head contains the title. The body contains the visible content. Save this as index dot html and open it in a browser. What do you see.

Students I see my first web page with a heading Hello KREIS and a paragraph Welcome to our school website. The browser displayed the heading larger than the paragraph. The title appears in the browser tab. HTML tags tell the browser how to display the content.

Explain (10 min)

Teacher HTML uses tags to mark up content. Tags come in pairs with an opening tag and a closing tag. The closing tag has a forward slash. The head section contains metadata like the title and links to stylesheets. The body section contains the visible page content. Heading tags h1 through h6 create headings with h1 being the largest. The p tag creates paragraphs. The br tag creates line breaks. HTML is not a programming language but a markup language that describes the structure of content.

Students HTML tags come in pairs. Head holds metadata. Body holds content. Headings use h1 to h6. Paragraphs use p tags. HTML describes structure not behavior. The browser reads HTML and decides how to display the content.

Elaborate (10 min)

Teacher Create an HTML page about KREIS school. Include a main heading h1 with the school name. Add a subheading h2 with About Our School. Write two paragraphs describing the school. Add another subheading h2 with Our Values. List three values in separate paragraphs. Open the page in a browser and verify the structure displays correctly.

Students My HTML page has the school name as the main heading. About Our School section describes the school in two paragraphs. Our Values section lists three values each in its own paragraph. The browser shows headings in decreasing size and paragraphs with spacing.

Evaluate (5 min)

Teacher I will give you a content description and ask you to write the correct HTML structure. You must use appropriate heading levels paragraph tags and proper document structure. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students HTML structure follows a hierarchy. h1 is the main heading h2 is section heading and so on. Paragraphs use p tags. The document must have proper html head and body structure.

Cross-Curricular Connections

Mathematics: Mathematics: HTML structure follows hierarchical organization similar to mathematical proof structure with main theorem and lemmas.
Science: Science: Scientific reports follow a structure similar to HTML with title abstract body and conclusion sections.
Language: Language Arts: HTML organizes content hierarchically just like an essay has a title topic sentences and paragraphs.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Text editor and web browser
- Textbook Chapter 4 Page 3
- KREIS HTML basics cheat sheet

Differentiation

Approaching: Class 8 Foundation: Create a simple HTML page with one heading and one paragraph. Use provided template and fill in content. Focus on understanding tag pairs and document structure.
On Level: Class 9 Application: Create HTML pages with multiple heading levels and paragraphs. Structure content logically. Work independently on page creation.
Advanced: Class 10 Mastery: Create well-structured HTML documents with semantic organization. Understand heading hierarchy importance. Begin connecting HTML to CSS for styling.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: HTML page creation with correct structure and content
CT Skill Assessed: Pattern Recognition: Identifying how content maps to appropriate HTML tags and structure

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create an HTML page about your dormitory. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

Teacher Students yesterday we created basic text pages. Today we learn HTML elements that make pages more useful and interesting. Lists organize information. Images make pages visual. Links connect pages together. Think of these elements as the furniture in a room. Without furniture a room is empty. Without these elements a web page is just plain text.

Students Lists images and links are essential web page elements. Lists organize content images add visual interest and links connect pages. Together they make web pages functional and engaging.

Explore (10 min)

Teacher Let me show you lists first. ul creates an unordered bullet list and ol creates an ordered numbered list. Each item uses the li tag. Now images: img src equals photo dot jpg alt equals description creates an image. The alt text appears if the image cannot load. Links: a href equals URL creates a clickable link. Try creating each.

Students My bullet list shows school subjects with dots. My numbered list shows steps in a process with numbers. My image displays a photo with the alt text showing when I removed the image file. My link clicks through to the school website. These elements bring pages to life.

Explain (10 min)

Teacher HTML elements extend basic text with rich content. The ul element creates unordered lists with bullet points. The ol element creates ordered lists with numbers. The li element defines each list item. The img element embeds images using the src attribute for the image path and alt attribute for alternative text. The a element creates hyperlinks using the href attribute for the destination URL. These elements are fundamental for creating functional web pages.

Students Lists organize content with ul for bullets and ol for numbers. Images use img with src and alt. Links use a with href. These three element types transform plain text into a rich web page.

Elaborate (10 min)

Teacher Create an HTML page about KREIS with at least three different element types. Include an unordered list of school facilities an ordered list of daily schedule and an image of the school. Add at least two links to other pages or external websites. Use proper alt text for the image and meaningful link text.

Students My page has a bullet list of facilities including library lab and playground. My numbered list shows the daily schedule. The image shows the school building with alt text. Links connect to the admission page and education department website.

Evaluate (5 min)

Teacher I will give you a content description and ask you to write the correct HTML using appropriate elements. You must choose between ordered and unordered lists and provide proper attributes. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Choosing between ul and ol depends on whether order matters. Images always need alt text for accessibility. Links need meaningful text that describes the destination.

Cross-Curricular Connections

Mathematics: Mathematics: Ordered lists correspond to sequences in mathematics. The numbering represents mathematical ordering.
Science: Science: Scientific reports use figures with captions similar to HTML images with alt text. References use hyperlinks.
Language: Language Arts: Lists organize information like bullet points in an essay. Links are like citations and references.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:

- Text editor and web browser
- Textbook Chapter 4 Page 5
- KREIS HTML elements reference

Differentiation

Approaching: Class 8 Foundation: Create simple lists and add one image to a page. Use provided image files and URLs. Focus on understanding element syntax with teacher guidance.
On Level: Class 9 Application: Create pages with multiple element types including lists images and links. Use proper attributes and write meaningful alt text independently.
Advanced: Class 10 Mastery: Create complex pages with nested lists multiple images and internal page links. Understand semantic HTML and accessibility best practices.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: HTML page with multiple element types and proper attributes
CT Skill Assessed: Pattern Recognition: Mapping content requirements to appropriate HTML elements and attributes

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create an HTML page about your favorite book. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

Teacher Students when you fill out an online form to create an account or submit homework you are interacting with HTML forms. Forms are how websites collect information from users. Today we learn to create forms. Think of a form like a paper application form but digital. It has fields for you to fill in and a submit button to send your answers.

Students HTML forms collect user input. I see forms everywhere for login registration and search. Learning to create forms lets me build interactive web pages that can gather information.

Explore (10 min)

Teacher Let me create a student feedback form. The form element wraps all input fields. Input type text creates a text field. Input type email validates email format. Input type password masks the input. Input type submit creates a submit button. A label element connects to an input using the for attribute. Try creating a form with at least three different input types.

Students My form has a name field with type text an email field with type email and a feedback textarea for longer messages. The submit button says Submit Feedback. When I click submit the form tries to send the data. The email field shows an error if I enter invalid email format.

Explain (10 min)

Teacher HTML forms collect user input. The form element is the container for all form controls. Input elements create different field types: text for single-line text email for email validation password for masked input number for numeric values and checkbox for multiple selections. The textarea element creates multi-line text areas. Select and option elements create dropdown menus. The button element creates clickable buttons. Labels improve accessibility by connecting text descriptions to inputs.

Students Forms contain input controls. Text email password number and checkbox are input types. Textarea handles multi-line input. Select creates dropdowns. Labels improve accessibility. Submit sends the form data. Forms enable user interaction.

Elaborate (10 min)

Teacher Create a KREIS admission inquiry form with fields for student name parent name email phone number class applying for and a textarea for additional questions. Add a dropdown for preferred section and checkboxes for facilities interested in. Include proper labels for each field and a submit button.

Students My admission form collects all necessary information. Name and parent name use text inputs. Email and phone use appropriate types. Class uses a dropdown with Class 8 9 and 10 options. Facilities uses checkboxes for library lab and sports. The textarea collects additional questions.

Evaluate (5 min)

Teacher I will give you a form description and ask you to write the HTML code. You must choose appropriate input types and include labels for accessibility. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Choosing input types depends on the data. Text for names email for email addresses password for passwords and number for numeric values. Labels are essential for accessibility.

Cross-Curricular Connections

Mathematics: Mathematics: Form inputs collect numerical data that can be validated using mathematical rules like range checking.
Science: Science: Research forms collect experimental data. HTML forms model this data collection process digitally.
Language: Language Arts: Forms are like structured questionnaires. The labels guide respondents just like questions guide survey takers.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Text editor and web browser
- Textbook Chapter 4 Page 7
- KREIS HTML forms reference

Differentiation

Approaching: Class 8 Foundation: Create a simple form with text and email inputs only. Use provided templates and focus on understanding the form element and basic input types.
On Level: Class 9 Application: Create forms with multiple input types including text email password and dropdown. Add labels and buttons. Work independently on form creation.
Advanced: Class 10 Mastery: Create complex forms with validation attributes. Understand form submission methods. Begin connecting forms to JavaScript for client-side validation.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: HTML form creation with appropriate input types and labels
CT Skill Assessed: Decomposition: Breaking a data collection need into appropriate form fields and controls

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create an HTML form for a library book request. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

CSS: Language styling HTML by setting colours fonts and layout beautifully together
Selector: Pattern choosing which HTML elements to style together as group
Property: Aspect like colour or margin you want to change for selected elements
Rule: Complete instruction pairing selector with property and value for styling

Engage (5 min)

Teacher Students HTML provides the structure of a web page but CSS makes it beautiful. CSS stands for Cascading Style Sheets. Think of HTML as the skeleton and CSS as the skin and clothes. Without CSS every web page would look plain black text on white background. CSS adds color style and visual appeal. Today we learn how to make our web pages look professional.

Students So CSS is like the makeup and clothing of a web page. HTML defines what content appears and CSS defines how it looks. I can change colors fonts sizes and backgrounds using CSS.

Explore (10 min)

Teacher Let me show you three ways to add CSS. Inline CSS goes directly on an element using the style attribute. Internal CSS goes in the head section using a style tag. External CSS goes in a separate dot css file linked with a link tag. Try changing the text color using each method. Which is easiest for quick changes and which is best for many pages.

Students Inline is easy for one element. Internal works for one page. External is best for many pages because one CSS file styles all linked pages. External is the most professional approach.

Explain (10 min)

Teacher CSS controls the visual presentation of HTML elements. Inline CSS uses the style attribute on individual elements. Internal CSS uses a style tag in the document head. External CSS uses a separate file linked with the link tag. CSS properties include color for text color background-color for background color font-size for text size font-family for typeface and text-align for horizontal alignment. The selector determines which elements the styles apply to. CSS separates content from presentation.

Students Three CSS methods: inline on elements internal in head external in separate file. Properties control color font size and alignment. Selectors target elements. External CSS is the best practice for maintainability.

Elaborate (10 min)

Teacher Create an external CSS file to style your KREIS HTML page. Set the body background to light blue. Make h1 headings dark green and centered. Set paragraph text to dark gray with font size 16 pixels. Style links with underline and hover color change. Link the CSS file to your HTML page and verify.

Students My CSS file sets the background to light blue. Headings are dark green and centered. Paragraphs are dark gray at 16 pixels. Links are blue with underline and change to red on hover. The HTML page loads the CSS file and displays all styles.

Evaluate (5 min)

Teacher I will give you an HTML page and ask you to write CSS to achieve a specific visual appearance. You must choose appropriate properties and values. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students CSS properties map to visual attributes. Color changes text color. Background-color changes the background. Font-size controls text size. The selector determines which elements receive the styles.

Cross-Curricular Connections

Mathematics: Mathematics: CSS uses numerical values for sizes and colors. Percentage-based layouts relate to proportional reasoning.
Science: Science: CSS styling follows cause-and-effect relationships. Changing a property produces a predictable visual result.
Language: Language Arts: CSS is like choosing fonts and formatting for a document. The same content looks different with different styles.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Text editor and web browser
- Textbook Chapter 4 Page 9
- KREIS CSS introduction guide

Differentiation

Approaching: Class 8 Foundation: Apply inline CSS to change text color and background. Focus on understanding what CSS does versus what HTML does. Use provided CSS snippets.
On Level: Class 9 Application: Create internal and external CSS files. Apply multiple properties to style a page. Work independently on creating styled pages.
Advanced: Class 10 Mastery: Create comprehensive CSS stylesheets with multiple selectors and properties. Understand the cascade and specificity concepts.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: CSS styling exercise with property application
CT Skill Assessed: Pattern Recognition: Identifying which CSS properties produce desired visual effects

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a CSS file to style your HTML dormitory page. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

CSS: Language styling HTML by setting colours fonts and layout beautifully together
Selector: Pattern choosing which HTML elements to style together as group
Property: Aspect like colour or margin you want to change for selected elements
Rule: Complete instruction pairing selector with property and value for styling

Engage (5 min)

Teacher Students what if you want to style only some paragraphs differently. CSS selectors tell CSS exactly which elements to style. Think of calling students by name versus calling everyone.

Students Selectors target specific elements. Element selectors style all of a type. Class selectors style groups. ID selectors style one unique element.

Explore (10 min)

Teacher The element selector p styles all paragraphs. The class selector dot highlight styles elements with that class. The ID selector hash header styles one element. Try adding a class to one paragraph and styling it differently.

Students My element selector changed all paragraphs. My class selector made only the highlighted paragraph yellow. ID selector made only the header red.

Explain (10 min)

Teacher CSS selectors determine which elements receive styles. Element targets all of a type. Class uses a dot prefix. ID uses a hash prefix. Specificity determines which style wins: ID is more specific than class which is more specific than element. Pseudo-class selectors like hover style elements based on state.

Students Element targets types. Class targets groups. ID targets one element. Specificity order is ID then class then element. Pseudo-classes target states like hover.

Elaborate (10 min)

Teacher Create a styled KREIS page using all selector types. Use element selector for body text. Create a class called special for important paragraphs. Use an ID for the main heading. Add hover effects on links.

Students My page uses element selector for paragraphs the special class for highlighted content the ID for the heading and hover effects on links.

Evaluate (5 min)

Teacher Write CSS using correct selectors to target specific elements without affecting others. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students IDs for unique elements classes for groups and elements for all of a type.

Cross-Curricular Connections

Mathematics: Mathematics: Selector specificity follows mathematical ordering.
Science: Science: Selectors are like filters isolating specific variables.
Language: Language Arts: Selectors are like addressing specific people versus everyone.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Text editor and web browser
- Textbook Chapter 4 Page 11
- KREIS CSS selectors reference

Differentiation

Approaching: Class 8 Foundation: Focus on element and class selectors with teacher guidance.
On Level: Class 9 Application: Use all selector types including ID and pseudo-class.
Advanced: Class 10 Mastery: Apply advanced selectors and understand specificity rules.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Selector identification and CSS writing exercise
CT Skill Assessed: Pattern Recognition: Identifying the most appropriate selector type for each styling requirement

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Style your dormitory page using element class and ID selectors. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

CSS: Language styling HTML by setting colours fonts and layout beautifully together
Selector: Pattern choosing which HTML elements to style together as group
Property: Aspect like colour or margin you want to change for selected elements
Rule: Complete instruction pairing selector with property and value for styling

Engage (5 min)

Teacher Students every HTML element is like a box. Content is inside then padding then border then margin. This is the CSS box model. Think of a gift box. The gift is content wrapping is padding box edge is border and space between boxes is margin.

Students The box model explains how elements take up space. Padding adds space inside border adds a line and margin adds space outside.

Explore (10 min)

Teacher Create two boxes with different padding and margin. Box one has padding 20 and margin 10. Box two has padding 5 and margin 30. Now try flexbox. A container with display flex arranges children in a row. Use justify-content to center items.

Students Box one has more internal space. Box two has more external space. Flexbox lined up items and justify-content center moved them to the middle.

Explain (10 min)

Teacher The box model has content padding border and margin. Content is actual text. Padding is space inside the border. Margin is space outside. Display property controls flow: block takes full width inline flows with text and flex creates flexible layouts. Flexbox provides alignment and distribution tools.

Students Box model is content plus padding plus border plus margin. Display block is full width. Inline is flowing. Flex creates responsive layouts.

Elaborate (10 min)

Teacher Create a KREIS website layout with header sidebar and main content using flexbox. Set header to span full width. Use flex for sidebar and content side by side. Apply padding and margin. Add a footer spanning full width.

Students My layout has a green header spanning full width. Sidebar takes 25 percent and content takes 75 percent. Footer sits at the bottom.

Evaluate (5 min)

Teacher Write CSS to achieve a described page layout using box model and flexbox. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Layout requires the box model for spacing and flexbox for arrangement.

Cross-Curricular Connections

Mathematics: Mathematics: The box model involves adding dimensions.
Science: Science: Layout design follows principles of visual hierarchy.
Language: Language Arts: Page layout is like formatting a newspaper.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Text editor and web browser
- Textbook Chapter 4 Page 13
- KREIS CSS layout examples

Differentiation

Approaching: Class 8 Foundation: Focus on box model. Practice adjusting padding and margin.
On Level: Class 9 Application: Use flexbox to create multi-column layouts independently.
Advanced: Class 10 Mastery: Create complex responsive layouts using flexbox.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: CSS layout implementation with box model and flexbox
CT Skill Assessed: Algorithmic Thinking: Planning layout structure and applying CSS properties correctly

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a layout for your dormitory page with header and two columns using flexbox. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

Teacher Students you browse websites on phones tablets and laptops. Each has a different screen size. Responsive design makes pages adapt to any size. Think of water taking the shape of its container.

Students Responsive design means one website works on all devices. One adaptive site adjusts its layout automatically.

Explore (10 min)

Teacher The viewport meta tag tells the browser to use actual device width. Media queries apply CSS rules based on screen width. Try resizing your browser window and watch the layout change at breakpoints.

Students The viewport tag made text readable on phone. The media query changed the layout when I made the window narrow. Layout switched from two columns to one.

Explain (10 min)

Teacher Responsive design ensures pages work on all devices. Viewport meta tag tells the browser to use actual width. Media queries use syntax like at media max-width 600px. Relative units like percentages create flexible layouts. Mobile-first starts with smallest screen. Breakpoints are widths where layout changes occur.

Students Viewport enables mobile rendering. Media queries detect screen size. Relative units create flexible sizing. These techniques create adaptive pages.

Elaborate (10 min)

Teacher Make your KREIS website responsive. Add viewport meta tag. Use media queries to stack columns on screens smaller than 600 pixels. Change font sizes for small screens. Test by resizing browser.

Students My website now has the viewport tag. Media query stacks sidebar below content on small screens. Font sizes decrease for mobile.

Evaluate (5 min)

Teacher Make a desktop layout responsive by adding viewport tag media queries and adjusting layouts. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Responsive design requires viewport tag media queries and relative units.

Cross-Curricular Connections

Mathematics: Mathematics: Responsive design uses breakpoints and percentage calculations.
Science: Science: Responsive design parallels how organisms adapt to environments.
Language: Language Arts: Responsive design is like formatting differently for poster vs book.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:

- Text editor and web browser
- Textbook Chapter 4 Page 15
- KREIS responsive design

Differentiation

Approaching: Class 8 Foundation: Add viewport meta tag and observe the difference.
On Level: Class 9 Application: Create responsive layouts with media queries independently.
Advanced: Class 10 Mastery: Design mobile-first responsive websites with breakpoints.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Responsive layout creation with media query testing
CT Skill Assessed: Evaluation: Assessing which layout works best for different screens

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Make your dormitory page responsive with a viewport tag and media query. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

JavaScript: Language adding interactivity like clicks and animations to web pages dynamically
Script: List of instructions running inside browser to make page interactive for users
Event: Action like click or key press that triggers JavaScript code to run
DOM: Tree structure representing webpage that JavaScript can change live instantly

Engage (5 min)

Teacher Students HTML gives structure CSS gives style and JavaScript gives life to web pages. JavaScript makes pages interactive. If HTML is the body and CSS is the clothes then JavaScript is the brain.

Students JavaScript makes web pages interactive. I can create programs that run in the browser and respond to user actions.

Explore (10 min)

Teacher Add JavaScript using a script tag. Alert shows a popup box. Document dot write adds text to the page. Console dot log sends messages to the developer console. Try these and see what happens.

Students Alert showed a popup. Document dot write added text. Console dot log sent a message I could see by pressing F12.

Explain (10 min)

Teacher JavaScript is a programming language that runs in browsers. Write in script tags or external js files. Variables store data using let and const. Conditionals use if-else. Loops repeat using for and while. Alert shows popups. Console dot log helps debugging.

Students JavaScript is the behavior layer. Variables store data. Conditionals decide. Loops repeat. Alert shows popups. Console helps debugging.

Elaborate (10 min)

Teacher Create a JavaScript program that greets the user by name. Use a variable for the name. Use if-else to check if empty and show default greeting. Add a for loop printing 1 to 5 using console dot log.

Students My program stores the name in a variable. If-else checks for empty name. The for loop prints 1 through 5.

Evaluate (5 min)

Teacher Write JavaScript using variables conditionals and loops to solve a problem. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Variables store data. Conditionals decide. Loops repeat.

Cross-Curricular Connections

Mathematics: Mathematics: JavaScript variables store numerical values for calculations.
Science: Science: JavaScript can simulate experiments in the browser.
Language: Language Arts: JavaScript programs follow logical sequences.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Text editor and web browser
- Textbook Chapter 4 Page 17
- KREIS JavaScript basics

Differentiation

Approaching: Class 8 Foundation: Write simple JavaScript with alert and document write.
On Level: Class 9 Application: Write programs with variables conditionals and loops.
Advanced: Class 10 Mastery: Create interactive programs with complex logic.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: JavaScript code writing with variables and control structures
CT Skill Assessed: Algorithmic Thinking: Writing step-by-step JavaScript instructions

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a JavaScript program that calculates your dormitory room area. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

> **Teacher** Students the browser builds a tree from HTML called the Document Object Model or DOM. JavaScript can find any element and change it. Think of a family tree where JavaScript can find any member.

> **Students** The DOM is a map of the web page. JavaScript uses it to find and change elements without reloading the page.

Explore (10 min)

Teacher Select an element with document dot getElementById and change its text content. Use querySelector to select by CSS selector. Change style using element dot style dot color equals red. Watch the page change without reloading.

Students I selected the heading and changed its text and color. The page updated instantly.

Explain (10 min)

Teacher The DOM is a tree of HTML elements the browser creates. getElementById finds by ID. querySelector finds by CSS selector. textContent changes text. Style properties change appearance. addEventListener responds to events. The DOM bridges HTML and JavaScript.

Students DOM is the browser tree. getElementById and querySelector find elements. textContent and style change them. addEventListener makes them respond.

Elaborate (10 min)

Teacher Create a KREIS page with a button that changes background color when clicked. Use getElementById to select the button. Add a click event listener. Change document dot body dot style dot backgroundColor on click.

Students My button changes background from white to blue on click. Each click cycles through colors.

Evaluate (5 min)

Teacher Write JavaScript that selects HTML elements and modifies their content or style. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students DOM methods find elements. Properties change them. Events trigger changes.

Cross-Curricular Connections

Mathematics: Mathematics: The DOM tree is a hierarchical data structure.
Science: Science: DOM manipulation models real-time data visualization.
Language: Language Arts: DOM changes are like editing a document in real time.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Text editor and web browser
- Textbook Chapter 4 Page 19
- KREIS DOM guide

Differentiation

Approaching: Class 8 Foundation: Use getElementById to select and change one element.
On Level: Class 9 Application: Use querySelector and event listeners for interaction.

Advanced: Class 10 Mastery: Create complex DOM manipulation with multiple events.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: DOM manipulation exercise with selection and modification
CT Skill Assessed: Algorithmic Thinking: Planning DOM operations for interactive behavior

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Add a button that shows and hides your daily routine paragraph. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

Teacher Students the web has dangers. Hackers try to steal passwords and personal data. Today we learn about web security threats and protection. Think of it like locking your house door.

Students Web security means protecting yourself online. I need to recognize threats and know how to stay safe.

Explore (10 min)

Teacher Let me show you what a phishing email looks like. It tries to trick you into clicking fake links. HTTPS encrypts data between browser and server. Check for the lock icon in the address bar.

Students The phishing email had suspicious links. HTTPS sites show a lock icon. HTTP sites show not secure warning.

Explain (10 min)

Teacher Web security threats include phishing which tricks users into revealing data and malware which damages devices. HTTPS encrypts communication. SSL certificates verify website identity. Strong passwords protect accounts. Two-factor authentication adds extra security.

Students Phishing tricks users. Malware damages devices. HTTPS encrypts data. SSL verifies identity. Strong passwords protect accounts.

Elaborate (10 min)

Teacher Create a web security awareness poster for KREIS. Include tips for creating strong passwords recognizing phishing emails and checking HTTPS. Add a section about social media privacy.

Students My poster includes password tips recognizing phishing by checking sender addresses and looking for HTTPS lock icon.

Evaluate (5 min)

Teacher Analyze and compare security threats in given scenarios and recommend protection measures. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Recognizing threats requires awareness. Protection involves technical and behavioral measures.

Cross-Curricular Connections

Mathematics: Mathematics: Encryption uses mathematical algorithms to protect data.
Science: Science: Cybersecurity is like biological defense against threats.
Language: Language Arts: Phishing uses persuasive language to trick readers.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Web browser
- Textbook Chapter 4 Page 21
- KREIS web security guide

Differentiation

Approaching: Class 8 Foundation: Focus on recognizing phishing and using HTTPS.
On Level: Class 9 Application: Understand encryption and implement password best practices.
Advanced: Class 10 Mastery: Analyze security vulnerabilities and design protection strategies.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Security scenario analysis and protection recommendation
CT Skill Assessed: Evaluation: Assessing threats and selecting appropriate protections

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a checklist of web security practices to follow every day. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

Teacher Students this is our final web project. You will combine everything learned about HTML CSS and JavaScript into a personal portfolio website. This is like building your digital identity.

Students I am ready to combine all web technologies into one complete project. I need to plan structure style and add interactive elements.

Explore (10 min)

Teacher Let us plan a KREIS student portfolio. We need pages for home about skills and contact. HTML provides structure. CSS provides styling. JavaScript adds interactivity like a dark mode toggle.

Students My portfolio has four sections. HTML structures the content. CSS styles everything. JavaScript toggles dark mode. Media queries make it responsive.

Explain (10 min)

Teacher A portfolio website demonstrates all web technologies working together. HTML creates structure with semantic elements. CSS styles with selectors box model and flexbox. JavaScript adds interactivity with DOM manipulation. Responsive design uses media queries and relative units.

Students Portfolio combines HTML structure CSS style and JavaScript behavior. Responsive design ensures mobile compatibility.

Elaborate (10 min)

Teacher Build your personal portfolio with at least four sections. Include home with photo and introduction about section with education skills section with progress bars and a contact form. Style with external CSS. Add one JavaScript feature.

Students My portfolio has home about skills and contact sections. CSS flexbox creates layout. JavaScript toggles dark mode.

Evaluate (5 min)

Teacher Your project will be evaluated on proper HTML structure correct CSS styling JavaScript functionality responsive design and documentation. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students My project uses semantic HTML organized CSS and functional JavaScript. It works on all screen sizes.

Cross-Curricular Connections

Mathematics: Mathematics: Portfolio layout uses mathematical proportions and grid systems.
Science: Science: The development process mirrors scientific methodology.
Language: Language Arts: Portfolio content requires clear written communication.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Text editor and web browser
- Textbook Chapter 4 review pages
- KREIS portfolio template

Differentiation

Approaching: Class 8 Foundation: Build a simplified portfolio with HTML and CSS only using the teacher template.
On Level: Class 9 Application: Build the complete portfolio with HTML CSS and basic JavaScript independently.
Advanced: Class 10 Mastery: Build an advanced portfolio with multiple JavaScript features and comprehensive responsive design.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Complete portfolio website with presentation and documentation
CT Skill Assessed: Creation: Combining HTML CSS and JavaScript into a cohesive personal website

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Build a personal portfolio page about yourself with hobbies favorite subjects and goals. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 5 - Algorithms and Computational Thinking

`(..) => ..`

Key Vocabulary

Algorithm: Step-by-step instructions solving a problem clearly and completely in order
Flowchart: Diagram using boxes and arrows showing algorithm steps visually for understanding
Pseudocode: Simple English-like description of algorithm before writing real code for planning
Efficiency: How quickly algorithm finishes and how little memory it uses together

Engage (5 min)

Teacher Students when you make tea you follow specific steps. That recipe is an algorithm. An algorithm is a step-by-step procedure to solve a problem. Today we learn to think algorithmically.

Students So an algorithm is like a recipe. If the steps are clear anyone can follow them and get the same result.

Explore (10 min)

Teacher Write down every step for making chai. Include every detail. Compare with your partner. What happens if steps are missing or in wrong order.

Students My steps were take water add sugar add tea boil strain and serve. My partner missed boiling. Order matters.

Explain (10 min)

Teacher An algorithm is a finite set of unambiguous steps that solve a specific problem. Key properties are finite unambiguous and effective. Algorithms exist everywhere from recipes to traffic rules. A good algorithm produces correct output for all valid inputs.

Students Algorithms are finite clear and doable. They exist everywhere. Good algorithms give correct output and run efficiently.

Elaborate (10 min)

Teacher Write algorithms for: tie shoelaces pack school bag and find a book in the library. Each must have at least 5 steps. Test by having a partner follow exactly.

Students My shoelace algorithm has 7 steps. My bag packing checks the timetable first. My partner followed each successfully.

Evaluate (5 min)

Teacher Write a complete algorithm for a given task. Check if steps are finite clear and produce correct result. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Good algorithms are specific enough that anyone can follow them.

Cross-Curricular Connections

Mathematics: Mathematics: Mathematical proofs follow algorithmic thinking with clear logical steps.
Science: Science: The scientific method is an algorithm: observe hypothesize experiment analyze conclude.
Language: Language Arts: Following a recipe is following an algorithm.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Paper and pen
- Textbook Chapter 5 Page 1
- KREIS algorithm examples handout

Differentiation

Approaching: Class 8 Foundation: Write algorithms for simple tasks like brushing teeth.
On Level: Class 9 Application: Write algorithms for complex tasks with decision points.
Advanced: Class 10 Mastery: Write algorithms with conditions and loops and analyze efficiency.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Algorithm writing exercise with peer testing
CT Skill Assessed: Algorithmic Thinking: Breaking tasks into clear sequential steps

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write an algorithm for your evening study routine from sitting down to going to bed. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

`(..) => ..`

Key Vocabulary

Chart: Visual picture showing numbers as bars lines or pies for easy comparison
Bar Chart: Rectangles whose height shows value of each category clearly for viewer
Pie Chart: Circle divided into slices showing share of each part
Legend: Key matching colours to names so chart is understandable quickly

Engage (5 min)

Teacher Algorithms can be drawn as diagrams called flowcharts. Think of a map with directions. The flowchart shows the path your program takes from start to finish using different shapes.

Students Flowcharts are like maps for programs. The shapes tell me what type of action happens at each step.

Explore (10 min)

Teacher Standard symbols: oval for start end, rectangle for process, diamond for decision with yes/no branches, arrows for flow. Draw a flowchart for making tea.

Students I used oval for start and end, rectangles for boiling and straining, diamond for is it ready. Arrows connect everything.

Explain (10 min)

Teacher Flowcharts use standard symbols. Oval represents start and end. Rectangle represents process. Diamond represents decision with two exits. Parallelogram represents input or output. Arrows show flow direction. Flowcharts make algorithms visual and help identify errors before coding.

Students Oval is start end. Rectangle is process. Diamond is decision. Parallelogram is input output. Flowcharts visualize algorithms.

Elaborate (10 min)

Teacher Convert your algorithm for finding the largest of three numbers into a flowchart. Use proper symbols. Trace through with test values 5 8 3 to verify.

Students My flowchart starts with input three numbers. First diamond compares A and B. Second compares larger with C. Tracing confirms it works.

Evaluate (5 min)

Teacher Draw a flowchart for a given algorithm and describe what a given flowchart does. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Converting between algorithm and flowchart requires understanding the logic.

Cross-Curricular Connections

Mathematics: Mathematics: Flowcharts represent decision trees and logical sequences.
Science: Science: Experimental procedures follow flowchart-like paths.
Language: Language Arts: Flowcharts are like story maps with decision points.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Paper and pen
- Textbook Chapter 5 Page 3
- KREIS flowchart symbols poster

Differentiation

Approaching: Class 8 Foundation: Draw simple flowcharts with linear steps and one decision.
On Level: Class 9 Application: Draw flowcharts with multiple decisions and loops.
Advanced: Class 10 Mastery: Create complex flowcharts and analyze efficiency.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Flowchart drawing and interpretation exercise
CT Skill Assessed: Abstraction: Converting text algorithms into visual representations

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Draw a flowchart for making your bed from waking up to finishing. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

Teacher Students pseudocode is like writing an algorithm in a format that looks like code but uses plain English. It is halfway between a flowchart and actual programming.

Students Pseudocode is structured English that describes algorithms. It uses keywords like IF and WHILE but does not follow strict syntax rules.

Explore (10 min)

Teacher Let me write pseudocode for finding the largest of three numbers. BEGIN input A B C. IF A greater than B AND A greater than C THEN output A. ELSE IF B greater than C THEN output B. ELSE output C. END.

Students My pseudocode clearly shows the decision structure. IF-THEN-ELSE handles different cases. BEGIN and END mark the boundaries.

Explain (10 min)

Teacher Pseudocode is an informal high-level description of an algorithm. It uses keywords like BEGIN END IF THEN ELSE WHILE DO FOR and RETURN. It does not follow strict syntax so it focuses on logic not syntax. Pseudocode bridges the gap between human language and programming.

Students Pseudocode uses keywords in plain English. It bridges human language and programming.

Elaborate (10 min)

Teacher Write pseudocode for: calculate average of three marks, find if a number is even or odd, and sort three numbers.

Students My average algorithm computes sum then divides by 3. My even-odd uses MOD operator. My sorting uses nested IF statements.

Evaluate (5 min)

Teacher Convert a given flowchart to pseudocode and verify both represent the same logic. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Pseudocode and flowcharts represent the same algorithms in different formats.

Cross-Curricular Connections

Mathematics: Mathematics: Pseudocode expresses mathematical algorithms in structured format.
Science: Science: Scientific procedures can be written as pseudocode for reproducibility.
Language: Language Arts: Pseudocode is a structured form of technical writing.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Paper and pen
- Textbook Chapter 5 Page 5
- KREIS pseudocode reference card

Differentiation

Approaching: Class 8 Foundation: Write pseudocode for linear algorithms using BEGIN END and simple steps.
On Level: Class 9 Application: Write pseudocode with IF THEN ELSE and loops.
Advanced: Class 10 Mastery: Write complex pseudocode with nested structures and functions.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Pseudocode writing and conversion exercise
CT Skill Assessed: Algorithmic Thinking: Expressing algorithms in structured pseudocode format

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write pseudocode for your morning routine from alarm to leaving for school. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Algorithm: Step-by-step instructions solving a problem clearly and completely in order
Flowchart: Diagram using boxes and arrows showing algorithm steps visually for understanding
Pseudocode: Simple English-like description of algorithm before writing real code for planning
Efficiency: How quickly algorithm finishes and how little memory it uses together

Engage (5 min)

Teacher Today we focus on sequential algorithms where steps happen in a fixed order. Think of boarding a bus: show ticket find seat sit down. Each step follows the previous one.

Students Sequential algorithms are the simplest type. Steps follow in order.

Explore (10 min)

Teacher Write a sequential algorithm to calculate total marks for 5 subjects. Input each mark then add them all and output total. Trace with sample values.

Students I input marks 85 90 78 92 88. I added step by step. The total is 433.

Explain (10 min)

Teacher Sequential algorithms have steps that execute in order from first to last. There are no decisions or repetitions. They are the foundation for all other algorithm types.

Students Sequential steps run in fixed order. No decisions or loops. They form the foundation.

Elaborate (10 min)

Teacher Write sequential algorithms for: calculate area of rectangle, convert Celsius to Fahrenheit, and calculate distance given speed and time.

Students My area algorithm multiplies length times width. My temperature uses F equals C times 9/5 plus 32.

Evaluate (5 min)

Teacher Given a problem write a sequential algorithm and trace it with test values. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Sequential algorithms are verified by tracing each step in order.

Cross-Curricular Connections

Mathematics: Mathematics: Sequential algorithms implement mathematical formulas step by step.
Science: Science: Following experimental procedures step by step is sequential algorithmic thinking.
Language: Language Arts: Following assembly instructions is executing a sequential algorithm.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Paper and pen
- Textbook Chapter 5 Page 7
- KREIS sequential algorithm worksheets

Differentiation

Approaching: Class 8 Foundation: Trace pre-written sequential algorithms with sample values.
On Level: Class 9 Application: Write sequential algorithms for mathematical calculations.
Advanced: Class 10 Mastery: Write complex sequential algorithms and analyze their steps.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Sequential algorithm writing and tracing exercise
CT Skill Assessed: Algorithmic Thinking: Organizing steps in correct sequential order

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a sequential algorithm to calculate how many hours you studied today. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Algorithm: Step-by-step instructions solving a problem clearly and completely in order
Flowchart: Diagram using boxes and arrows showing algorithm steps visually for understanding
Pseudocode: Simple English-like description of algorithm before writing real code for planning
Efficiency: How quickly algorithm finishes and how little memory it uses together

Engage (5 min)

Teacher Sometimes algorithms need to make decisions. If it rains take umbrella else wear sunglasses. Selection algorithms choose between paths based on conditions.

Students Selection algorithms make choices. IF a condition is true one path is taken. ELSE another path is taken.

Explore (10 min)

Teacher Write an algorithm: IF temperature greater than 35 THEN output hot ELSE IF temperature less than 10 THEN output cold ELSE output pleasant. Trace with values 40, 5, and 20.

Students For 40 it outputs hot. For 5 it outputs cold. For 20 it outputs pleasant. Different inputs take different paths.

Explain (10 min)

Teacher Selection algorithms use conditions to choose between paths. IF condition THEN action ELSE alternative. Nested selection adds more conditions. AND requires both true. OR requires one true.

Students IF-THEN-ELSE chooses paths. Nested IF handles multiple conditions.

Elaborate (10 min)

Teacher Write selection algorithms for: grade a student mark, determine if a year is a leap year, and classify a number as positive negative or zero.

Students My grading uses ranges. Leap year checks divisibility by 4 and not by 100 or by 400.

Evaluate (5 min)

Teacher Write a selection algorithm and trace it with different inputs to verify all paths. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Tracing with multiple inputs ensures all paths produce correct results.

Cross-Curricular Connections

Mathematics: Mathematics: Selection implements conditional logic like absolute value functions.
Science: Science: Scientific classification uses selection logic.
Language: Language Arts: Decision points in stories mirror selection algorithms.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Paper and pen
- Textbook Chapter 5 Page 9
- KREIS selection algorithm examples

Differentiation

Approaching: Class 8 Foundation: Write simple IF-THEN-ELSE algorithms with one condition.
On Level: Class 9 Application: Write algorithms with nested selections and multiple conditions.
Advanced: Class 10 Mastery: Write complex selection algorithms and analyze all possible paths.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Selection algorithm writing with multi-input tracing
CT Skill Assessed: Algorithmic Thinking: Designing conditional paths for different inputs

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a selection algorithm that decides what to wear based on weather temperature. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Algorithm: Step-by-step instructions solving a problem clearly and completely in order
Flowchart: Diagram using boxes and arrows showing algorithm steps visually for understanding
Pseudocode: Simple English-like description of algorithm before writing real code for planning
Efficiency: How quickly algorithm finishes and how little memory it uses together

Engage (5 min)

Teacher Sometimes you need to repeat a step multiple times. Do 10 jumping jacks. Sing the prayer three times. Iteration is repetition using loops.

Students Iteration means repetition. FOR loops repeat a known number of times. WHILE loops repeat until a condition changes.

Explore (10 min)

Teacher Write: FOR i from 1 to 5 DO output i times 2. Trace: outputs 2 4 6 8 10. Now WHILE: set count=1 WHILE count less than or equal to 5 DO output count count equals count plus 1.

Students FOR loop runs exactly 5 times. WHILE loop also runs 5 times but stops when count exceeds 5.

Explain (10 min)

Teacher FOR loops repeat a known number of times. WHILE loops repeat until a condition becomes false. The loop variable must change toward the exit condition to avoid infinite loops. Counter-controlled FOR and condition-controlled WHILE are the two main types.

Students FOR is for known counts. WHILE is for unknown counts. The variable must change to avoid infinite loops.

Elaborate (10 min)

Teacher Write iteration algorithms for: print multiplication table of a number, sum numbers from 1 to N, and find the largest number in a list of 10 numbers.

Students My multiplication table uses FOR loop from 1 to 10. My sum uses WHILE adding each number. My largest compares each element.

Evaluate (5 min)

Teacher Write an iteration algorithm and trace it counting each loop execution. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Tracing loops requires tracking the counter and output at each iteration.

Cross-Curricular Connections

Mathematics: Mathematics: Iteration implements sequences and series calculations.
Science: Science: Repeated measurements use iterative processes.
Language: Language Arts: Repeated practice like writing paragraphs is iterative.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Paper and pen
- Textbook Chapter 5 Page 11
- KREIS iteration algorithm examples

Differentiation

Approaching: Class 8 Foundation: Write simple FOR loops with known repetition counts.
On Level: Class 9 Application: Write both FOR and WHILE loops for different scenarios.
Advanced: Class 10 Mastery: Write nested loops and analyze loop efficiency.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Iteration algorithm writing with loop tracing
CT Skill Assessed: Algorithmic Thinking: Using loops to efficiently repeat operations

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write an algorithm that counts from 1 to 20 and prints only even numbers. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Big-O: Notation describing how runtime grows when input size increases dramatically for comparison
Complexity: Amount of time or memory algorithm needs as data grows larger
Linear: Time growing directly with input size like checking each item once
Constant: Time staying same no matter how large input becomes for operation

Engage (5 min)

Teacher Students not all algorithms are equally fast. Some handle 100 items quickly but struggle with 10000. Big-O notation measures how an algorithm grows as input gets bigger. Think of it like measuring how long a journey takes as the distance increases.

Students Big-O tells us how fast an algorithm runs. O(1) is constant time. O(n) grows with input. O(n squared) grows much faster.

Explore (10 min)

Teacher Compare two search methods on a list of 100 items. Linear search checks up to 100 items. Binary search checks about 7 items. For 1000 items linear checks 1000 but binary checks about 10. The difference grows as input grows.

Students Linear search is O(n). Binary search is O(log n). For large inputs binary is much faster.

Explain (10 min)

Teacher Big-O describes the worst-case growth rate. O(1) constant means same time regardless of input. O(n) linear means time grows with input. O(n squared) quadratic means time grows with input squared. O(log n) logarithmic means time grows very slowly.

Students O(1) is fastest. O(log n) is fast. O(n) is moderate. O(n squared) is slow for large inputs.

Elaborate (10 min)

Teacher Analyze the time complexity of: checking every element in a list, using binary search on a sorted list, and comparing every pair in a list.

Students Checking every element is O(n). Binary search is O(log n). Comparing every pair is O(n squared).

Evaluate (5 min)

Teacher Given an algorithm determine its Big-O notation and explain why. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Big-O depends on how many operations the algorithm performs relative to input size.

Cross-Curricular Connections

Mathematics: Mathematics: Big-O notation is related to mathematical limits and growth functions.
Science: Science: Efficiency matters in scientific computing where datasets are huge.
Language: Language Arts: Big-O is like estimating how long a book takes to read based on page count.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:

- Paper and pen
- Textbook Chapter 5 Page 13
- KREIS Big-O comparison chart

Differentiation

Approaching: Class 8 Foundation: Understand that some algorithms are faster than others for large inputs.
On Level: Class 9 Application: Identify O(1) O(n) and O(n squared) for simple algorithms.
Advanced: Class 10 Mastery: Analyze time complexity of algorithms and compare efficiency.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Algorithm efficiency analysis exercise
CT Skill Assessed: Evaluation: Comparing algorithms based on their computational efficiency

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Compare the efficiency of finding a word in a dictionary by checking every page versus opening to the middle. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

Teacher Students divide and conquer means breaking a big problem into smaller easier pieces solve each piece and combine the results. Think of dividing class work among groups. Each group handles a part and you combine the answers.

Students Divide and conquer breaks hard problems into easy pieces. Each piece is solved independently. Results are combined.

Explore (10 min)

Teacher Binary search divides the search range in half each time. Merge sort divides the list in half sorts each half then merges. Both are faster than checking every element. Draw the steps for merge sort on 8 numbers.

Students Merge sort divides 8 into two 4s, each into two 2s, each into two 1s. Then merges back sorted. It takes about 24 operations instead of 64 for bubble sort.

Explain (10 min)

Teacher Divide and conquer algorithms break a problem into subproblems solve each recursively and combine results. They are often more efficient than brute force. Binary search merge sort and quicksort all use this strategy.

Students Divide into subproblems, solve each, combine results. More efficient than brute force.

Elaborate (10 min)

Teacher Apply divide and conquer to find the largest element in a list: split the list in half find the largest in each half then compare the two largest. Trace with 8 numbers.

Students Split 8 into two 4s. Find largest in each. Compare the two winners. Much fewer comparisons than checking all 8.

Evaluate (5 min)

Teacher Analyze and compare whether a given algorithm uses divide and conquer and explain the subproblem structure. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Divide and conquer algorithms have clear divide solve and combine steps.

Cross-Curricular Connections

Mathematics: Mathematics: Merge sort relates to mathematical induction and recursive definitions.
Science: Science: Complex experiments are divided into smaller controlled sub-experiments.
Language: Language Arts: Writing a book chapter by chapter is divide and conquer.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Paper and pen
- Textbook Chapter 5 Page 15
- KREIS divide and conquer examples

Differentiation

Approaching: Class 8 Foundation: Understand the concept of dividing a problem into smaller parts.
On Level: Class 9 Application: Trace merge sort and binary search step by step.
Advanced: Class 10 Mastery: Analyze divide and conquer efficiency and compare with other approaches.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Divide and conquer tracing exercise
CT Skill Assessed: Decomposition: Breaking complex problems into manageable subproblems

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Use divide and conquer to find the median of a list of 6 numbers. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Algorithm: Step-by-step instructions solving a problem clearly and completely in order
Flowchart: Diagram using boxes and arrows showing algorithm steps visually for understanding
Pseudocode: Simple English-like description of algorithm before writing real code for planning
Efficiency: How quickly algorithm finishes and how little memory it uses together

Engage (5 min)

Teacher Students greedy algorithms make the best choice right now without worrying about the future. Like getting the best deal at each shop while shopping. Sometimes this gives the best overall result but not always.

Students Greedy algorithms pick the best local option. Sometimes this gives the global best. Sometimes it does not.

Explore (10 min)

Teacher Make change for 37 rupees using the fewest coins. Greedy approach: use the largest coin first. 37 = 25 + 10 + 1 + 1 = 4 coins. Is this always the best? Try making change for a currency where greedy fails.

Students For Indian currency greedy works well. But for some coin systems it fails. Greedy is simple but not always optimal.

Explain (10 min)

Teacher Greedy algorithms make the locally optimal choice at each step hoping for a globally optimal result. They work well for some problems like activity selection and Huffman coding. They fail for problems like the traveling salesman.

Students Greedy picks the best local choice. Works for some problems. Fails for others.

Elaborate (10 min)

Teacher Apply greedy to: schedule maximum activities from a list with start and end times, and make change for 63 rupees with minimum coins.

Students For activities I sort by end time and pick non-overlapping ones. For change 63 = 50 + 10 + 1 + 1 + 1 = 5 coins.

Evaluate (5 min)

Teacher Determine if a greedy approach works for a given problem and justify your answer. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Greedy works when local optimal choices lead to global optimal solution.

Cross-Curricular Connections

Mathematics: Mathematics: Greedy relates to optimization problems in mathematics.
Science: Science: Resource allocation in experiments often uses greedy principles.
Language: Language Arts: Greedy is like choosing the best sentence in each paragraph.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Paper and pen
- Textbook Chapter 5 Page 17
- KREIS greedy algorithm examples

Differentiation

Approaching: Class 8 Foundation: Understand the concept of making the best local choice.
On Level: Class 9 Application: Apply greedy to simple problems like making change.

Advanced: Class 10 Mastery: Analyze when greedy works and when it fails with proof.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Greedy algorithm application exercise
CT Skill Assessed: Evaluation: Assessing whether greedy approach produces optimal solution

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Use a greedy approach to pack your bag placing the most important items first. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Algorithm: Step-by-step instructions solving a problem clearly and completely in order
Flowchart: Diagram using boxes and arrows showing algorithm steps visually for understanding
Pseudocode: Simple English-like description of algorithm before writing real code for planning
Efficiency: How quickly algorithm finishes and how little memory it uses together

Engage (5 min)

Teacher Students efficiency matters when working with large data. An efficient algorithm on a slow computer beats an inefficient algorithm on a fast computer. Today we compare algorithm efficiencies.

Students Efficiency determines how well an algorithm scales. Choosing the right algorithm is more important than faster hardware.

Explore (10 min)

Teacher Compare bubble sort O(n squared) with binary search O(log n) on a list of 10000 items. Bubble sort does about 100 million operations. Binary search does about 14. The difference is enormous.

Students Bubble sort: 10000 squared equals 100 million. Binary search: log of 10000 is about 14. Binary search is millions of times faster.

Explain (10 min)

Teacher Efficiency is measured by time complexity and space complexity. Time complexity measures how operations grow. Space complexity measures memory usage. The best algorithm depends on data size structure and available resources.

Students Time and space complexity measure efficiency. Best algorithm depends on the specific problem.

Elaborate (10 min)

Teacher Compare three algorithms for finding duplicates in a list: check every pair O(n squared) sort first O(n log n) and use a set O(n). Discuss when each is appropriate.

Students Check every pair is simple but slow. Sort is faster. Set is fastest but uses more memory.

Evaluate (5 min)

Teacher For a given problem compare different algorithm approaches and recommend the most efficient one with justification. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students The best algorithm depends on data size, memory availability, and code complexity.

Cross-Curricular Connections

Mathematics: Mathematics: Efficiency relates to optimization and mathematical complexity theory.
Science: Science: Scientific computing requires efficient algorithms for large datasets.
Language: Language Arts: Efficient writing communicates clearly with fewer words.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Paper and pen
- Textbook Chapter 5 Page 19
- KREIS algorithm efficiency comparison

Differentiation

Approaching: Class 8 Foundation: Understand that some algorithms are faster than others.
On Level: Class 9 Application: Compare algorithm efficiencies for small and large inputs.
Advanced: Class 10 Mastery: Analyze time and space complexity and choose optimal algorithms.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Algorithm efficiency comparison exercise
CT Skill Assessed: Evaluation: Selecting the most efficient algorithm for given constraints

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Compare two ways to find your name in a phone directory: page by page versus binary search. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

Teacher Students complex problems are overwhelming when viewed as a whole. Decomposition breaks them into smaller pieces. Like eating an elephant one bite at a time. Each subproblem is easier to solve.

Students Decomposition makes hard problems manageable. Each small piece is solved separately then combined.

Explore (10 min)

Teacher Design a student report card system. Decompose into: input student data, calculate averages, determine grades, generate report. Each subproblem is simpler than the whole.

Students My system has four subproblems. Each can be solved independently. Together they produce the complete report card.

Explain (10 min)

Teacher Decomposition is breaking a complex problem into simpler subproblems. Each subproblem can be solved independently. Subproblems are then combined to solve the original problem. This is a fundamental problem-solving strategy in computing.

Students Break complex into simple. Solve each independently. Combine solutions.

Elaborate (10 min)

Teacher Decompose these KREIS problems: manage library system, plan school event, and create online quiz. Identify at least 3 subproblems for each.

Students Library system needs catalog, borrowing, and returns. Event needs venue, schedule, and invitations.

Evaluate (5 min)

Teacher Given a complex problem decompose it into subproblems and describe how solving each contributes to the whole. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Good decomposition creates independent subproblems that cover the entire problem.

Cross-Curricular Connections

Mathematics: Mathematics: Complex proofs are decomposed into lemmas and supporting statements.
Science: Science: Research projects decompose into experiments analysis and reporting.
Language: Language Arts: Writing decomposes into planning drafting and editing.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:

- Paper and pen
- Textbook Chapter 5 Page 21
- KREIS decomposition worksheet

Differentiation

Approaching: Class 8 Foundation: Decompose simple problems into 2-3 subproblems.
On Level: Class 9 Application: Decompose complex problems and solve each subproblem.
Advanced: Class 10 Mastery: Apply decomposition to large projects with dependencies.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Problem decomposition exercise
CT Skill Assessed: Decomposition: Breaking complex problems into manageable subproblems

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Decompose the problem of organizing a class picnic into subproblems. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Algorithm: Step-by-step instructions solving a problem clearly and completely in order
Flowchart: Diagram using boxes and arrows showing algorithm steps visually for understanding
Pseudocode: Simple English-like description of algorithm before writing real code for planning
Efficiency: How quickly algorithm finishes and how little memory it uses together

Engage (5 min)

Teacher Students this is our final algorithm project. You will use everything learned: algorithms flowcharts pseudocode selection iteration decomposition and efficiency analysis to solve a real problem.

Students I need to plan my solution using algorithms choose efficient approaches and document everything clearly.

Explore (10 min)

Teacher Design a KREIS canteen queue management system. Identify the problems to solve: take orders, manage queue, calculate bills, track inventory. Design algorithms for each. Create flowcharts. Analyze efficiency.

Students My system has four algorithms. The queue uses FIFO. Bill calculation uses selection for discounts. Inventory uses iteration to check stock levels.

Explain (10 min)

Teacher A complete algorithm design project follows these steps: understand the problem, decompose into subproblems, design algorithms for each, represent with flowcharts and pseudocode, analyze efficiency, and document the process.

Students Project steps: understand decompose design represent analyze document.

Elaborate (10 min)

Teacher Design a complete solution for: online homework submission system or digital attendance tracker. Include all design steps from decomposition to efficiency analysis.

Students My attendance system decomposes into: mark attendance view records generate report. Each has efficient algorithms.

Evaluate (5 min)

Teacher Present your design explaining each algorithm choice and efficiency analysis. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Good design documentation explains why choices were made not just what was chosen.

Cross-Curricular Connections

Mathematics: Mathematics: Algorithm design applies mathematical thinking to problem solving.
Science: Science: Research methodology follows a structured design process.
Language: Language Arts: Design documentation requires clear technical writing.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Paper and pen
- Textbook Chapter 5 review pages
- KREIS project planning template

Differentiation

Approaching: Class 8 Foundation: Design a simplified system with 2-3 algorithms.
On Level: Class 9 Application: Design a complete system with flowcharts and pseudocode.
Advanced: Class 10 Mastery: Design with efficiency analysis and comprehensive documentation.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Complete algorithm design project with documentation
CT Skill Assessed: Creation: Applying all algorithmic concepts to design a complete solution

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Design an algorithm for a simple calculator that handles + - * / operations. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 6 - Emerging Technologies

`(..) => ..`

Key Vocabulary

AI: Machines mimicking human thinking to solve problems without explicit instructions always
Model: Trained system that makes predictions based on data it learned from examples
Data: Examples used to teach AI system patterns for making decisions later
Ethics: Rules ensuring fair and safe use of AI for everyone

Engage (5 min)

Teacher Students you have seen robots in movies that think and act like humans. Today we learn about artificial intelligence or AI which is making machines smart. AI is already around you in your phone assistant and video recommendations.

Students AI is making computers smart. My phone recognizes my face and my voice assistant answers questions. AI is already part of my daily life.

Explore (10 min)

Teacher List five examples of AI you use: voice assistants like Alexa face unlock on phones recommendation videos on YouTube spell check in documents and auto-complete in messages. Discuss how each uses intelligence.

Students My examples are Siri face recognition YouTube recommendations Google spell check and WhatsApp auto-complete. Each mimics a human ability like seeing hearing or predicting.

Explain (10 min)

Teacher Artificial intelligence is machines performing tasks that normally require human intelligence. This includes learning from data recognizing patterns making decisions and understanding language. AI works by training on large datasets to find patterns. Narrow AI handles specific tasks like voice recognition. General AI would handle any intellectual task like a human.

Students AI mimics human intelligence. It learns from data and finds patterns. Narrow AI handles specific tasks. General AI does not yet exist.

Elaborate (10 min)

Teacher Research one AI application used in India. Explain how it works and who benefits from it. Consider agriculture healthcare education or banking.

Students I researched AI in Indian agriculture. It analyzes soil and weather data to advise farmers. This helps farmers grow better crops with less waste.

Evaluate (5 min)

Teacher For each AI example identify the human intelligence it mimics and discuss benefits and risks. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students AI mimics human abilities like seeing hearing and deciding. Benefits include convenience. Risks include privacy concerns.

Cross-Curricular Connections

Mathematics: Mathematics: AI uses statistical models and mathematical optimization.
Science: Science: AI assists in scientific research by analyzing large datasets quickly.
Language: Language Arts: Natural language processing enables AI to understand human language.

Digital Citizen Reminder: AI systems collect data about your behavior. Be aware of how your data is used to train AI models.

Resources Needed:

- Internet access and device with AI features
- Textbook Chapter 6 Page 1
- KREIS AI examples handout

Differentiation

Approaching: Class 8 Foundation: Focus on recognizing AI in everyday life and understanding what AI can and cannot do.
On Level: Class 9 Application: Analyze how AI works in specific applications and discuss benefits and risks.
Advanced: Class 10 Mastery: Compare narrow and general AI and evaluate the ethical implications of AI adoption.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: AI identification and analysis exercise
CT Skill Assessed: Abstraction: Understanding how AI abstracts human intelligence into computational processes

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: List five AI tools you use this week and describe what each does for you. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations

Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

Teacher Students machine learning is a type of AI where computers learn from examples instead of being told every rule. Like how you learn to recognize animals by seeing many animals. Today we learn how computers learn.

Students Machine learning means computers learn from data. Instead of writing rules for every situation I give the computer examples and it learns the patterns.

Explore (10 min)

Teacher Show a simple example: sorting fruits by weight and color. Give the computer 100 examples of apples and oranges with their weight and color. The computer finds the pattern: apples tend to be smaller and redder oranges are larger and orange. Now it can sort new fruits.

Students The computer learned the pattern from examples. Weight and color help distinguish apples from oranges. More examples make the computer more accurate.

Explain (10 min)

Teacher Machine learning is AI that improves through experience. Supervised learning uses labeled training data with correct answers. Unsupervised learning finds hidden patterns in unlabeled data. Reinforcement learning learns through trial and error with rewards.

Students Supervised uses labeled data. Unsupervised finds hidden patterns. Reinforcement learns through rewards.

Elaborate (10 min)

Teacher Describe a machine learning scenario for KREIS: predicting which students might need extra help based on past performance. What data would you use and what type of learning is this.

Students This is supervised learning. I would use past marks attendance and homework data. The model learns which patterns predict low performance.

Evaluate (5 min)

Teacher For a given scenario identify whether it uses supervised unsupervised or reinforcement learning and justify your choice. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students The type depends on whether data has labels and whether there is a reward system.

Cross-Curricular Connections

Mathematics: Mathematics: Machine learning uses statistical models and probability theory.
Science: Science: Machine learning analyzes experimental data to find patterns.
Language: Language Arts: Natural language processing is machine learning applied to text.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Internet access
- Textbook Chapter 6 Page 3
- KREIS machine learning examples

Differentiation

Approaching: Class 8 Foundation: Understand that computers can learn from examples and data.
On Level: Class 9 Application: Identify the three types of machine learning and give examples of each.
Advanced: Class 10 Mastery: Analyze how machine learning models are trained and discuss bias in training data.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Machine learning type identification exercise
CT Skill Assessed: Pattern Recognition: Understanding how machines identify patterns in data

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Give three examples of machine learning from your daily life and classify each type. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Neural Network: System inspired by brain with layers of connected nodes learning patterns together
Neuron: Single unit receiving inputs and passing result to next layer for processing
Layer: Group of neurons processing together at same stage before passing onward
Training: Adjusting connections by showing many examples until accuracy improves gradually step

Engage (5 min)

Teacher Students your brain has billions of neurons connected to each other. A neural network is a computer system inspired by this. Each artificial neuron receives inputs processes them and sends outputs to other neurons.

Students Neural networks are like digital brains. They have layers of connected nodes. Each node processes information and passes it forward.

Explore (10 min)

Teacher Draw a simple neural network with 3 input nodes 2 hidden nodes and 1 output node. Input values flow through connections with weights. Each hidden node calculates a weighted sum. The output node produces the final result.

Students My network has inputs feeding into hidden nodes which feed into output. Weights determine how much each input matters. The output changes as weights change.

Explain (10 min)

Teacher Neural networks are layers of interconnected nodes inspired by biological neurons. Input layer receives data. Hidden layers process it. Output layer produces results. Networks learn by adjusting connection weights during training.

Students Input-hidden-output layers. Weights adjust during training. Networks learn patterns from data.

Elaborate (10 min)

Teacher Research how neural networks are used in handwriting recognition. Explain the layers involved and how the network learns to identify different letters.

Students Handwriting recognition uses input nodes for pixel values. Hidden layers detect features like curves and lines. Output nodes identify the letter. Training adjusts weights until recognition is accurate.

Evaluate (5 min)

Teacher Explain how a neural network processes information from input to output for a given scenario. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Data flows through layers. Weights determine importance. Output represents the prediction.

Cross-Curricular Connections

Mathematics: Mathematics: Neural networks use matrix multiplication and calculus for weight adjustment.
Science: Science: Neural networks are inspired by biological neuroscience.
Language: Language Arts: Natural language processing uses neural networks to understand text.

Digital Citizen Reminder: Neural networks power many AI applications. Understanding them helps you evaluate AI capabilities and limitations.

Resources Needed:
- Internet access and drawing materials
- Textbook Chapter 6 Page 5
- KREIS neural network diagram

Differentiation

Approaching: Class 8 Foundation: Understand the concept of layers and how information flows through a network.
On Level: Class 9 Application: Draw simple neural networks and trace data flow through the layers.
Advanced: Class 10 Mastery: Explain how neural networks learn through weight adjustment and backpropagation.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Neural network diagram and explanation exercise
CT Skill Assessed: Abstraction: Modeling brain-inspired computation using layers of connected nodes

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Draw a neural network that decides if a student will pass or fail based on marks and attendance inputs. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

IoT: Everyday objects like bulbs connected to internet to be controlled remotely easily
Sensor: Device detecting temperature light or motion and sending data to system
Actuator: Device that moves or switches something after receiving command from system
Smart Device: Object that senses acts and connects to internet automatically without human help

Engage (5 min)

Teacher Students imagine your fan automatically turns on when the room gets hot. Your water bottle tells you how much water you drank. This is IoT the Internet of Things where everyday objects connect to the internet.

Students IoT means connecting everyday objects to the internet. Smart fans water bottles and watches can all collect data and communicate.

Explore (10 min)

Teacher List IoT devices in a smart home: smart bulbs that adjust brightness smart fans that detect temperature smart meters that track electricity and security cameras that send alerts. Each device has a sensor and internet connection.

Students Smart bulbs have light sensors. Smart fans have temperature sensors. Smart meters track usage. Cameras have motion sensors. All send data to the internet.

Explain (10 min)

Teacher IoT connects physical objects to the internet. Devices have sensors to collect data processors to analyze it and network connections to share it. Data is often processed in the cloud. IoT enables smart homes cities and industries.

Students Sensors collect data. Processors analyze it. Network shares it. Cloud stores and processes it.

Elaborate (10 min)

Teacher Design a smart school system using IoT. Identify what devices would be connected what sensors they would have and what data they would collect.

Students My smart school has attendance sensors at the door temperature sensors in classrooms and energy sensors on lights. All data goes to a central dashboard.

Evaluate (5 min)

Teacher For a given scenario identify the IoT devices sensors data collected and benefits. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students IoT requires sensors connectivity and data processing to provide benefits.

Cross-Curricular Connections

Mathematics: Mathematics: IoT generates data that requires statistical analysis.
Science: Science: IoT sensors measure physical quantities like temperature and humidity.
Language: Language Arts: IoT data can be used to write automated reports and summaries.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- IoT device examples and internet access
- Textbook Chapter 6 Page 7
- KREIS IoT examples poster

Differentiation

Approaching: Class 8 Foundation: Identify IoT devices and understand they collect and share data.
On Level: Class 9 Application: Design simple IoT systems with sensors data collection and benefits.
Advanced: Class 10 Mastery: Analyze IoT security challenges and design comprehensive IoT solutions.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: IoT device identification and design exercise
CT Skill Assessed: Pattern Recognition: Identifying how everyday objects can be enhanced with sensors and connectivity

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: List five IoT devices that could improve life at KREIS and explain what data each would collect. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

Teacher Students smart devices are like regular devices with a brain. A regular fan just spins. A smart fan senses temperature and adjusts speed. Today we explore how smart devices work and their impact.

Students Smart devices add intelligence to regular devices. They sense their environment make decisions and act automatically.

Explore (10 min)

Teacher Compare a regular alarm clock with a smart alarm. Regular just rings at set time. Smart analyzes sleep patterns wakes you during light sleep and adjusts based on your schedule. The smart version adapts to your needs.

Students Regular alarm is simple and fixed. Smart alarm adapts uses data and provides personalized experience.

Explain (10 min)

Teacher Smart devices have four components: sensors to collect data processors to analyze it connectivity to share data and actuators to take action. This creates a cycle: sense decide act communicate. Smart homes schools and cities use networks of these devices.

Students Sensors process data. Processors decide. Actuators act. Connectivity communicates. Sense-decide-act-communicate cycle.

Elaborate (10 min)

Teacher Analyze the impact of smart devices at KREIS: smart attendance smart classroom and smart library. For each identify the technology used benefits and potential concerns.

Students Smart attendance uses facial recognition benefits include accuracy concerns include privacy. Smart classroom uses interactive boards benefits include engagement.

Evaluate (5 min)

Teacher Evaluate the benefits and drawbacks of adopting smart devices in a school environment. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Benefits include efficiency and personalization. Drawbacks include cost privacy and dependency.

Cross-Curricular Connections

Mathematics: Mathematics: Smart devices generate numerical data for analysis.
Science: Science: Smart sensors measure physical phenomena and convert to digital data.
Language: Language Arts: Smart devices with voice assistants process natural language.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Smart device examples
- Textbook Chapter 6 Page 9
- KREIS smart device comparison chart

Differentiation

Approaching: Class 8 Foundation: Understand what makes a device smart and identify examples.
On Level: Class 9 Application: Analyze smart device components and their functions.
Advanced: Class 10 Mastery: Evaluate the societal impact of smart device adoption.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Smart device analysis and comparison exercise
CT Skill Assessed: Evaluation: Assessing the benefits and risks of smart technology adoption

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a report on how three smart devices could improve life at KREIS. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Cloud: Servers on internet that store your files safely always
Server: Powerful computer storing many files for many users together
Upload: Sending your file from device to cloud storage
Download: Bringing a file from cloud to your own device

Engage (5 min)

Teacher Students instead of storing everything on your computer you can store it on the internet called the cloud. Google Drive YouTube and Gmail all use cloud computing. Today we learn how the cloud works.

Students Cloud computing means using internet servers instead of your own device. I can access my files from any device anywhere.

Explore (10 min)

Teacher Compare cloud services: Gmail is SaaS you use it directly. Google App Engine is PaaS you build apps on it. AWS is IaaS you rent virtual computers. Each gives you different levels of control.

Students SaaS gives ready-to-use apps. PaaS gives development platforms. IaaS gives raw computing resources. Each level adds more control and responsibility.

Explain (10 min)

Teacher Cloud computing delivers computing services over the internet. IaaS provides virtual machines and storage. PaaS provides development platforms. SaaS provides ready-to-use applications. Benefits include cost savings scalability and accessibility.

Students Cloud is computing over internet. IaaS PaaS SaaS are service levels. Benefits are cost scalability accessibility.

Elaborate (10 min)

Teacher Analyze how KREIS could use cloud computing: Google Classroom for LMS Google Drive for file storage and Zoom for online classes. For each identify the service model and benefits.

Students Google Classroom is SaaS for learning. Google Drive is SaaS for storage. Zoom is SaaS for communication. Benefits include accessibility and collaboration.

Evaluate (5 min)

Teacher Compare cloud computing with local computing and recommend when each is appropriate. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Cloud is better for collaboration and accessibility. Local is better for speed and privacy.

Cross-Curricular Connections

Mathematics: Mathematics: Cloud computing uses mathematical models for resource allocation.
Science: Science: Cloud computing enables scientific collaboration across institutions.
Language: Language Arts: Cloud-based tools enable collaborative writing and editing.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:

- Internet access
- Textbook Chapter 6 Page 11
- KREIS cloud computing comparison chart

Differentiation

Approaching: Class 8 Foundation: Understand that cloud computing stores data on internet servers.
On Level: Class 9 Application: Compare IaaS PaaS and SaaS with real-world examples.
Advanced: Class 10 Mastery: Analyze cloud computing costs benefits and security considerations.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Cloud computing service model comparison exercise
CT Skill Assessed: Evaluation: Assessing when cloud computing is appropriate versus local computing

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: List five cloud services you use and identify whether each is IaaS PaaS or SaaS. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

AI: Machines mimicking human thinking to solve problems without explicit instructions always
Model: Trained system that makes predictions based on data it learned from examples
Data: Examples used to teach AI system patterns for making decisions later
Ethics: Rules ensuring fair and safe use of AI for everyone

Engage (5 min)

Teacher Students blockchain is like a digital ledger that everyone can see but no one can change. Originally designed for Bitcoin it now has many uses. Think of it as a chain of blocks where each block contains records.

Students Blockchain is a permanent digital record. Once something is recorded it cannot be changed. This makes it trustworthy.

Explore (10 min)

Teacher Demonstrate blockchain: write a transaction on a block connect it to the previous block with a hash. Try to change one block and observe that all following blocks become invalid. This shows immutability.

Students Each block has a unique hash. Changing one block changes its hash which breaks the chain. This makes blockchain tamper-proof.

Explain (10 min)

Teacher Blockchain is a distributed ledger across many computers. Each block contains transactions and a hash linking to the previous block. Once recorded data cannot be altered. This creates trust without a central authority.

Students Distributed ledger across computers. Blocks contain transactions and hashes. Immutable once recorded. Trust without central authority.

Elaborate (10 min)

Teacher Research three non-cryptocurrency applications of blockchain: supply chain tracking medical records and digital identity. Explain how blockchain benefits each application.

Students Supply chain tracks products from farm to table. Medical records ensure data integrity. Digital identity prevents fraud.

Evaluate (5 min)

Teacher Explain how blockchain achieves trust and security without a central authority. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Distributed storage consensus mechanisms and cryptographic hashing create trust.

Cross-Curricular Connections

Mathematics: Mathematics: Blockchain uses cryptographic hashing and mathematical proofs.
Science: Science: Blockchain ensures scientific data integrity and reproducibility.
Language: Language Arts: Blockchain can verify authorship and prevent plagiarism.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:

- Internet access
- Textbook Chapter 6 Page 13
- KREIS blockchain concept diagram

Differentiation

Approaching: Class 8 Foundation: Understand blockchain as a permanent digital record that cannot be changed.
On Level: Class 9 Application: Explain how blocks are linked and why changing one breaks the chain.
Advanced: Class 10 Mastery: Analyze blockchain applications and evaluate their benefits and limitations.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Blockchain concept explanation and application analysis
CT Skill Assessed: Abstraction: Understanding distributed trust systems without central authorities

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Draw a simple blockchain with 5 blocks and show what happens when someone tries to change block 3. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Quantum: Computing using tiny particles that hold many states at once
Qubit: Unit holding zero one or both together enabling powerful calculations beyond normal
Superposition: Being in multiple states simultaneously until measured for quantum advantage greatly
Entanglement: Link where two qubits affect each other instantly even when far apart

Engage (5 min)

Teacher Students classical computers use bits that are 0 or 1. Quantum computers use qubits that can be 0 and 1 at the same time called superposition. This makes them potentially much more powerful for certain problems.

Students Quantum computers use qubits instead of bits. Qubits can be 0 and 1 simultaneously. This allows exploring many solutions at once.

Explore (10 min)

Teacher Compare classical and quantum approaches to finding a word in a dictionary. Classical checks one page at a time. Quantum could theoretically check all pages simultaneously using superposition.

Students Classical checks pages one by one. Quantum explores many pages at once. Quantum is exponentially faster for some problems.

Explain (10 min)

Teacher Quantum computing uses qubits which exploit superposition being in multiple states at once and entanglement where qubits are connected. This enables solving certain problems exponentially faster than classical computers.

Students Qubits use superposition and entanglement. Quantum solves some problems exponentially faster than classical.

Elaborate (10 min)

Teacher Research one real-world problem that quantum computing could solve: drug discovery climate modeling or cryptography. Explain why classical computers struggle with this problem.

Students Drug discovery requires simulating molecular interactions. Classical computers cannot handle the complexity. Quantum computers could simulate molecules accurately.

Evaluate (5 min)

Teacher Explain the potential advantages and current limitations of quantum computing. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Quantum is faster for specific problems but currently expensive fragile and limited in availability.

Cross-Curricular Connections

Mathematics: Mathematics: Quantum computing uses linear algebra and probability theory.
Science: Science: Quantum computing applies quantum mechanics principles to computation.
Language: Language Arts: Quantum computing research requires clear technical communication.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Internet access
- Textbook Chapter 6 Page 15
- KREIS quantum computing concept illustration

Differentiation

Approaching: Class 8 Foundation: Understand that quantum computers work differently from classical computers.
On Level: Class 9 Application: Explain superposition and entanglement concepts in simple terms.
Advanced: Class 10 Mastery: Compare quantum and classical computing capabilities and limitations.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Quantum computing concept explanation exercise
CT Skill Assessed: Abstraction: Understanding how quantum phenomena enable new computing paradigms

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Explain in your own words why quantum computers could be faster than classical computers for certain tasks. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

Teacher Students technology is powerful but must be used responsibly. Today we discuss ethics in technology. Questions like who owns your data should AI make decisions about people and is technology fair to everyone.

Students Technology ethics means using technology responsibly. I need to think about fairness privacy and the impact of technology on people.

Explore (10 min)

Teacher Discuss these scenarios: an AI hiring system that discriminates against women a social media app that sells user data and a facial recognition system that works better for some races. Identify the ethical issues.

Students The hiring AI shows bias. The social media app violates privacy. The facial recognition is unfair. All involve ethical issues that affect real people.

Explain (10 min)

Teacher Technology ethics covers privacy fair access algorithmic bias intellectual property and environmental impact. Developers have responsibility to create fair transparent and safe technologies. Users have responsibility to use technology ethically.

Students Privacy fair access bias transparency responsibility. Both developers and users have ethical obligations.

Elaborate (10 min)

Teacher Analyze an ethical dilemma: Should schools use facial recognition for attendance? Consider privacy accuracy fairness and alternatives. Present both sides of the argument.

Students Benefits include accuracy and automation. Risks include privacy violation and data security. Alternatives include ID cards.

Evaluate (5 min)

Teacher Propose ethical guidelines for technology use at KREIS considering privacy fairness and responsibility. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Good guidelines address data collection consent fairness and accountability.

Cross-Curricular Connections

Mathematics: Mathematics: Algorithmic fairness requires mathematical definitions of bias.
Science: Science: Research ethics ensure responsible conduct of scientific technology.
Language: Language Arts: Ethical arguments require clear reasoning and evidence.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Internet access
- Textbook Chapter 6 Page 17
- KREIS technology ethics discussion cards

Differentiation

Approaching: Class 8 Foundation: Identify basic ethical issues in technology use.
On Level: Class 9 Application: Analyze ethical scenarios and discuss both sides.
Advanced: Class 10 Mastery: Evaluate technology ethics and propose guidelines for responsible use.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Technology ethics scenario analysis exercise
CT Skill Assessed: Evaluation: Assessing ethical implications of technology decisions

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write three ethical guidelines for using technology at KREIS. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

> **Teacher** Students not everyone has equal access to technology. Some have smartphones and internet. Others have neither. This gap is called the digital divide. Today we explore why it exists and how to bridge it.

Students The digital divide means unequal access to technology. Some people have much more technology than others. This creates inequality.

Explore (10 min)

Teacher Research the digital divide in Karnataka: urban schools have computer labs while rural schools may not. Urban students have internet at home while rural students may not. List the causes and effects of this gap.

Students Causes include poverty infrastructure and education. Effects include unequal learning opportunities and job prospects.

Explain (10 min)

Teacher The digital divide has three aspects: access to devices availability of internet and digital literacy. It exists between urban and rural areas wealthy and poor nations and different age groups. Bridging it requires infrastructure education and affordable technology.

Students Three aspects: devices internet literacy. Exists between urban-rural rich-poor. Requires infrastructure education affordability.

Elaborate (10 min)

Teacher Propose solutions for bridging the digital divide at KREIS: how can the school ensure all students have equal access to technology for learning.

Students Solutions include shared computer labs mobile device lending libraries and digital literacy programs.

Evaluate (5 min)

Teacher Evaluate the impact of the digital divide on education and propose policy recommendations. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Digital divide creates educational inequality. Policies should address access literacy and affordability.

Cross-Curricular Connections

Mathematics: Mathematics: Digital divide statistics require data analysis and interpretation.
Science: Science: Technology access affects scientific education and research opportunities.
Language: Language Arts: Digital literacy requires reading and writing skills for online communication.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Internet access
- Textbook Chapter 6 Page 19
- KREIS digital divide statistics

Differentiation

Approaching: Class 8 Foundation: Understand what the digital divide is and identify examples.
On Level: Class 9 Application: Analyze causes and effects of the digital divide.
Advanced: Class 10 Mastery: Evaluate solutions and propose policies to bridge the divide.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Digital divide analysis and solution proposal exercise
CT Skill Assessed: Evaluation: Assessing social impact of technology inequality

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a proposal for bridging the digital divide in your local community. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

Teacher Students technology is creating new careers that did not exist before. AI specialist data scientist cybersecurity expert and app developer are jobs today that were not around 20 years ago. Let us explore future careers.

Students Technology creates new careers. AI specialist data scientist and cybersecurity are growing fields. I need to prepare for these opportunities.

Explore (10 min)

Teacher Research five technology careers: their role required skills and education path. Compare with traditional careers like doctor engineer and teacher. How do technology careers differ.

Students Technology careers require coding data analysis and continuous learning. They often offer remote work and high demand.

Explain (10 min)

Teacher Technology is transforming all careers. Traditional jobs now require digital skills. New careers emerge from AI data science cybersecurity and renewable technology. Key skills include coding data analysis problem-solving and adaptability.

Students Technology transforms all careers. New roles emerge. Key skills are coding data analysis problem-solving adaptability.

Elaborate (10 min)

Teacher Create a career exploration plan for yourself: identify three technology careers that interest you the skills required and the steps to prepare for them during and after school.

Students My plan includes learning Python for AI studying data analysis for data science and understanding networks for cybersecurity.

Evaluate (5 min)

Teacher Compare a traditional career path with a technology career path and identify the transferable skills. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Both require communication problem-solving and continuous learning. Technology careers add coding and data skills.

Cross-Curricular Connections

Mathematics: Mathematics: Data science careers require strong mathematical foundations.
Science: Science: Research careers increasingly require computational skills.
Language: Language Arts: Technical communication is essential for all technology careers.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Internet access
- Textbook Chapter 6 Page 21
- KREIS career exploration worksheet

Differentiation

Approaching: Class 8 Foundation: Identify technology careers and the basic skills they require.
On Level: Class 9 Application: Research career paths and plan skill development.
Advanced: Class 10 Mastery: Create comprehensive career plans with educational pathways.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Career exploration and planning exercise
CT Skill Assessed: Creation: Designing a personal career development plan for technology fields

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Interview a technology professional and write about their career journey. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

```
(..) => ..
```

Key Vocabulary

Concept: Main idea that helps you understand the lesson topic clearly and correctly
Example: Specific case showing how the concept works in real life situations
Practice: Repeating task many times to improve skill and remember better always
Apply: Using what you learned to solve new problem or daily task usefully

Engage (5 min)

Teacher Students this is our final project. You will write an essay analyzing technology impact using everything learned about AI IoT cloud computing ethics and the digital divide. This brings all chapter topics together.

Students I need to organize my knowledge from all chapter topics into a well-structured essay with clear arguments and evidence.

Explore (10 min)

Teacher Plan your essay: introduction body paragraphs on AI IoT ethics and digital divide and conclusion. Use evidence from lessons to support each point. Consider both benefits and risks.

Students My essay has five paragraphs. Introduction sets context. Body paragraphs cover AI benefits IoT transformation ethics challenges and digital divide concerns. Conclusion balances both perspectives.

Explain (10 min)

Teacher A technology impact essay has a clear thesis statement evidence-based body paragraphs and a thoughtful conclusion. It presents balanced perspectives acknowledging both benefits and risks. Good essays use specific examples and data.

Students Clear thesis evidence-based paragraphs balanced perspectives specific examples.

Elaborate (10 min)

Teacher Write a 500-word essay on technology impact covering at least four chapter topics. Include a thesis statement evidence for each point and a balanced conclusion.

Students My essay argues that technology brings great benefits but requires responsible use. I covered AI IoT ethics and digital divide with specific examples.

Evaluate (5 min)

Teacher Evaluate peer essays for thesis clarity evidence use and balanced perspective. Hinge Q: [question with 3 options A/B/C, correct in brackets]. Students answer on mini-boards; if <80% correct, reteach 3 min. Self-assessment: tick Success Criteria.

Students Good essays have clear arguments supported by evidence from multiple topics.

Cross-Curricular Connections

Mathematics: Mathematics: Essay structure follows logical mathematical argumentation.
Science: Science: Technology impact requires evidence-based analysis like scientific research.
Language: Language Arts: Essay writing develops critical thinking and communication skills.

Digital Citizen Reminder: No specific citizenship link this lesson - reinforce general respectful use.

Resources Needed:
- Writing materials and internet access
- Textbook Chapter 6 review pages
- KREIS essay writing guide

Differentiation

Approaching: Class 8 Foundation: Write a simple essay with introduction body and conclusion on one technology topic.
On Level: Class 9 Application: Write an essay covering three topics with evidence and balanced perspective.
Advanced: Class 10 Mastery: Write a comprehensive essay covering all chapter topics with strong arguments and evidence.
SEND: Trace code with finger and use visual block cards with teacher support

Formative Assessment

Method: Technology impact essay writing and peer review
CT Skill Assessed: Creation: Synthesizing all chapter knowledge into a coherent analytical essay

Dormitory — Retrieval (Paper)

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write an essay about how technology has changed education in India. Spiral: also recall one fact from last week.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Colophon

Author: Hoysala T, Computer Science Teacher, MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga — 577519
Contact: hoy.sala@email.com · GitHub: hoy-sala

Version: August 2026 · ICT Levels 1 (Class 6), 2 (Class 7), 3 (Classes 8–10) · 216 lessons
Attribution: Created for residential school learners. No official affiliation. Use and adapt freely.

Set in Atkinson Hyperlegible / Noto Sans for accessibility. Built with Typst.